THE GROUND

Yang–Mills Theory after the Alpöge–Sussman Manuscripts

What Is Constructed, What Is Claimed, and the Object That Remains

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Abstract

A proof that quantum Yang–Mills theory has a mass gap must stand on a construction of the theory, and the most complete public construction now in view is the corpus of manuscripts that Levent Alpöge and Ethan Sussman have posted since 6 October 2026, carrying Bałaban's renormalisation group from the lattice to the continuum for the gauge group SU(2). This paper fixes that ground. It reads the corpus from its primary sources and proves, clause by clause and with exact references, what the corpus asserts and at what strength. In three dimensions an 854-page manuscript asserts continuum Schwinger functions of the plaquette-energy density, the lattice form of tr F^2, which satisfy the Osterwalder–Schrader axioms in the clustering components of infinite-volume limits taken along subsequences, together with a short-distance law that makes the theory non-trivial; it asserts no mass gap. In four dimensions a 153-page draft deduces, from inputs it declares clause by clause and from supporting statements two of which are not yet written, continuum limits on $\mathbb{R}^4$ along subsequences with regularity, reflection positivity, symmetry and hypercubic invariance; its authors call the underlying argument extremely long and not understandable as it stands, and state that the corpus has not been satisfactorily vetted. The paper then proves that, for a reflection-positive theory, a mass gap is the same property as exponential clustering of the reflection-positive correlations, and that a lower bound on the lattice mass gap, uniform in physical units along the lattice sequence of such a construction, supplies at once the three properties the corpus withholds: a unique vacuum, the cluster property and a mass gap. It

then reduces that bound to two finite statements: if, at a fixed physical scale, the blocked measure satisfies Dobrushin's condition with an exponential weight and the field inside the blocks clusters at the block scale, both uniformly in the cutoff, then every limit point has a unique vacuum and a mass gap of at least min(κ, log(1/ρ′))/a_c; the open mathematics of the problem is thereby two statements about one block at the crossover scale, stated in the form that is uniform in the cutoff, and the paper says why the form on the fine measure's blocks is not. Finally it proves a theorem at the strong end: for the SU(2) Wilson theory on $\mathbb{Z}^4$ at $\beta \le 0.241$, an exact decimation of one link in every plaquette and Dobrushin's criterion, with constants that Turán's inequality for the modified Bessel functions makes exact, give a unique infinite-volume Gibbs measure, exponential clustering and a mass gap of the lattice theory (Theorem 8.7), and the same construction on $\mathbb{Z}^3$, along a cover by three families of parallel lines, gives them at $\beta \le 0.3739$ (Theorem 8.8); on $\mathbb{Z}^4$ the flux-surface gas, which prices the surfaces of the polymer expansion by the representation theory of SU(2), extends them to $\beta \le 0.4764$, conditional on one stated lemma on contact leaves (Theorem 8.9). On the hierarchical model, where the Migdal–Kadanoff recursion is the exact renormalisation group, the crossover is crossed with certified numbers: from $\beta = 2.2$ and from $\beta = 2.3$ the eighth iterate of the Wilson weight satisfies Dobrushin's condition on $\mathbb{Z}^4$ with row sums at most 0.0336 and 0.2820, the first step at which the certified budget closes; the inequality that would carry that crossing to $\mathbb{Z}^4$ must depend on the gauge group, since the same recursion carries the U(1) weight into the disk while four-dimensional U(1) has a massless phase. These theorems lie at strong coupling and on the hierarchical model; the interval between them and the crossover of $\mathbb{Z}^4$ is the open part of the problem, and the object on which the proof that follows must stand.

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1. Introduction

The quantum Yang–Mills problem of the Clay Mathematics Institute is a single sentence. For every compact simple gauge group G it asks for a proof that a non-trivial quantum Yang–Mills theory exists on $\mathbb{R}^4$ and has a mass gap $\Delta > 0$.[1] The sentence joins two demands of different kinds. The first is a construction: a system of correlation functions, free of every cutoff, defined on the whole of space-time and obeying axioms at least as strong as those of Wightman and of Osterwalder and Schrader. The second is an inequality: the spectrum of the Hamiltonian of the theory so constructed must leave an interval above zero empty. The inequality cannot be stated before the construction exists, and when the construction proceeds by compactness, the problem's own text, in a footnote examined in Section 3, does not count it as complete until properties of this kind are proved for the limit.

In two dimensions the continuum Yang–Mills measure was constructed in 1989, by Driver and by Gross, King and Sengupta.[2] In three dimensions Bałaban controlled Wilson's lattice theory on a torus by a renormalisation group whose bounds are uniform in the lattice spacing; in four dimensions he proved the same bounds under the hypothesis that the running couplings stay below a constant at every scale; the discharge of that hypothesis, his Theorem 2, was announced in 1987 and never published, and it is a theorem on this work's path (§8.3).[3] When the problem was posed, its authors wrote that a detailed construction on a compact space would be a major breakthrough, and that even then "no present ideas point the direction to establish the existence of a mass gap that is uniform in the volume. Nor do present methods suggest how to obtain the existence of the infinite volume limit $T^4 \to \mathbb{R}^4$."[4]

On 6 October 2026 Ethan Sussman opened a public repository for the joint work of Levent Alpöge and himself, whose stated aim is "to induce LLMs to develop, push, and extend Balaban's program to construct Yang--Mills_4 (YM4) and other quantum gauge theories".[5] It now holds an 854-page manuscript on three-dimensional SU(2) Yang–Mills theory, a revised edition of 856 pages, a 1,044-page sequel on curvature bilinears, a 263-page manuscript on three-dimensional quantum electrodynamics, a 30-page manuscript on rotation invariance from reflection positivity, and a 153-page zeroth draft on four-dimensional SU(2) Yang–Mills theory.[6] Peter Woit discussed the repository

[1] A. Jaffe and E. Witten, "Quantum Yang–Mills theory", official problem description of the Clay Mathematics Institute, https://www.claymath.org/wp-content/uploads/2022/06/yangmills.pdf, §4, p. 6.

[2] B. K. Driver, "YM_2: continuum expectations, lattice convergence, and lassos", *Comm. Math. Phys.* 123 (1989); L. Gross, C. King and A. Sengupta, "Two dimensional Yang–Mills theory via stochastic differential equations", *Ann. Physics* 194 (1989).

[3] T. Bałaban, "Ultraviolet stability of three-dimensional lattice pure gauge field theories", *Comm. Math. Phys.* 102 (1985), 255–275; "Renormalization group approach to lattice gauge field theories. I", *Comm. Math. Phys.* 109 (1987), 249–301; and the further papers of the series, *Comm. Math. Phys.* 95–122 (1984–1989).

[4] Jaffe and Witten, *op. cit.*, §6, p. 7.

[5] L. Alpöge and E. Sussman, README of the repository *Yang--Mills-LLM-experiment*, https://github.com/ethanwsussman/Yang--Mills-LLM-experiment (repository created 6 October 2026, 09:51:17 UTC; README last changed 26 October 2026, 16:18:10 UTC; read 28 October 2026).

[6] Measured by counting the page breaks of the text extracted from each file of https://github.com/ethanwsussman/Yang--Mills-LLM-experiment: YM3/YMM1.pdf, 854 pages (posted 26 October 2026, 02:10 UTC); YM3/Revisions/YMM1-2026-10-26.pdf, 856; YM3/YMM2.pdf, 1,044 (posted 27 October 2026, 01:14 UTC); QED3/qed3.pdf, 263; RPM1.pdf, 30; ym4_manuscript_zeroth_draft.pdf, 153.

on 26 October 2026.[7] Its standing has since been compressed into a single sentence: a public three-dimensional manuscript proving the Osterwalder–Schrader axioms for tr F^2, about 854 pages, "LLM-approved", unverified, with subsequential infinite-volume limits and no gap; and a four-dimensional existence argument that its authors call extremely long and not currently understandable, being rewritten slowly.

This paper replaces that sentence by a proof. Each of its clauses is unfolded into a precise mathematical statement, each statement is traced to the sentence of the primary source that carries it, and the consequences for the Millennium problem are deduced. The outcome is the Ground Theorem of Section 2. Its first three parts establish what the corpus asserts in three dimensions, what it asserts in four, and on what standing it asserts either. Its fourth part establishes what the corpus leaves undone and proves that one inequality completes it: a lower bound on the lattice mass gap, uniform in physical units along the construction's own sequence of lattices. That inequality is the object on which the proof that follows will stand.

Quotations are verbatim, including the authors' spelling. Page counts were measured by counting the page breaks in the text extracted from each file. Every source was read on 28 October 2026; the repository's most recent commit is dated 27 October 2026, 10:45:28 UTC, and its README was last changed on 26 October 2026, 16:18:10 UTC. The deductions of Sections 7 and 8 assume nothing about the correctness of the corpus. They hold for every construction of the shape the corpus describes.

2. The Ground Theorem

By the *corpus* we mean the content of the repository *Yang--Mills-LLM-experiment* of L. Alpöge and E. Sussman at its commit of 27 October 2026. By the *problem statement* we mean the official description of Jaffe and Witten. Throughout, a denotes a lattice spacing and $b > 0$ the physical inverse coupling of the three-dimensional theory, with bare inverse coupling $\beta = b/a$.

Theorem 2.1 (The Ground). *(i) Three dimensions. For the SU(2) Wilson lattice theory on the lattices $a_K\mathbb{Z}^3$, with $a_K = L^{-K}$ and bare inverse coupling $L^K b$, the manuscript YMM1 asserts four things: on each fixed torus the connected correlations of the centred plaquette-energy density converge along the full sequence of cutoffs; infinite-volume limits exist along subsequences; in each canonical clustering component of such a limit the Schwinger functions satisfy the Osterwalder–Schrader axioms (E0)–(E4) and reconstruct a Hermitian scalar Wightman field; and a short-distance two-point law, at sufficiently small reference coupling, determines b and proves the theory non-trivial. YMM1 asserts no mass gap. For the original limit it asserts neither the uniqueness of the vacuum nor clustering, and the clustering bound it proves in the components is the inverse first power of the separation.*

(ii) Four dimensions. The zeroth draft asserts, under proof inputs that it declares clause by clause, continuum limits on $\mathbb{R}^4$ along subsequences of cutoffs and volumes whose Schwinger functions satisfy (E0) in a stated form, invariance under translations and the hypercubic group, reflection positivity and symmetry, whence a Hilbert space with a vacuum and a non-negative Hamiltonian;

[7] P. Woit, "Secret Unified Physics Slop", *Not Even Wrong*, 26 October 2026, with its updates and comments, https://www.math.columbia.edu/~woit/wordpress/?p=15948 (read 28 October 2026).

non-triviality uniformly in the volume, through a two-point law in the running coupling; and invariance under O(4) under a displayed hypothesis on the flux of the lattice rotation current. It asserts neither the cluster property, nor the uniqueness of the vacuum, nor a mass gap, nor the uniqueness of the limit. Two of its supporting statements are not yet written. Its authors describe the four-dimensional argument behind it as extremely long and not understandable as it stands, and state that it is being written out methodically and slowly.

(iii) Standing. The authors state that the corpus has not been satisfactorily vetted and that all its claims are to be understood as tentative, pending verification.

(iv) Residue and completion. Measured against the problem statement, the corpus supplies a mass gap in no dimension. In four dimensions it supplies neither the cluster property, nor the uniqueness of the vacuum, nor unconditional invariance under O(4), nor the relativistic spectral condition. In every dimension it treats the single gauge group SU(2) and a restricted class of curvature observables. For a reflection-positive theory a mass gap is equivalent to exponential clustering of the reflection-positive correlations. Consequently, uniform exponential clustering of the lattice correlations at a rate fixed in physical units, which a lower bound on the lattice mass gap uniform in physical units supplies together with the bounds at separated points that such a construction provides, gives every limit point of the construction's sequence the uniqueness of the vacuum, the cluster property and a mass gap.

The rest of the paper is the proof. Sections 3 and 4 set down the definitions on which every clause depends. Section 5 proves part (i), Section 6 proves parts (ii) and (iii), and Section 7 proves part (iv). Section 8 states the object that remains, the instruments with which the proof that follows will meet it, and the reduction of that object to two statements about one block at a fixed physical scale, Theorem 8.5, and proves at the strong end a mass gap of the lattice theory in infinite volume, Theorem 8.7 on $\mathbb{Z}^4$ and Theorem 8.8 on $\mathbb{Z}^3$, extends the four-dimensional theorem by the flux-surface gas under one stated lemma, Theorem 8.9, and certifies the crossing of the crossover on the hierarchical model.

3. What the Problem Demands

3.1 Euclidean fields and the axioms of Osterwalder and Schrader

A Euclidean quantum field theory in d dimensions is given by its Schwinger functions S_n, $n \geq 1$: continuous linear functionals on test functions of n points of $\mathbb{R}^d$. For the composite fields from which a gauge theory is built, the natural test functions vanish, with all their derivatives, wherever two points coincide, and the corpus works with exactly these.[8] Let θ be the reflection $x_d \mapsto -x_d$, let τ_v be translation by v, and let $\mathcal{S}_+$ consist of the test functions supported where every point has positive last coordinate. A *finite family* is a sequence $f = (f_0, f_1, \ldots, f_N)$ with $f_n \in \mathcal{S}_+(\mathbb{R}^{dn})$; it stands for a polynomial in the smeared field, and we write

[8] *YMM1*, "Scalar fields from three-dimensional lattice Yang–Mills theory", YM3/YMM1.pdf and its source YM3/YMM1.tex in the repository, p. 4; *The zeroth draft*, "SU(2) Yang–Mills theory in four dimensions via Balaban's renormalisation group: construction, axioms and non-triviality", dated 6 October 2026, ym4_manuscript_zeroth_draft.pdf in the repository, Remark 4.11.

$$S(\theta f \cdot \tau_t f) = \sum_{n,n'} S_{n+n'}(\theta f_n \otimes \tau_t f_{n'}) \tag{3.1}$$

for the corresponding correlation, with τ_t the translation by t in the last coordinate. The axioms of Osterwalder and Schrader are five.[9]

(E0) *Regularity.* Each S_n is a distribution on its test space; in the strengthened form of the second paper, $|S_n(f)|$ is bounded by one Schwartz seminorm of degree proportional to n, with constants growing at most as a power of $n!$.

(E1) *Euclidean invariance.* $S_n(h \cdot f) = S_n(f)$ for every Euclidean motion h.

(E2) *Reflection positivity.* $S(\theta \bar{f} \cdot f) \geq 0$ for every finite family f.

(E3) *Symmetry.* Each S_n is invariant under permutations of its arguments.

(E4) *Cluster property.* For finite families f, g and every spatial vector $v \neq 0$, $S(\theta \bar{f} \cdot \tau_{sv} g) \to S(\theta \bar{f})\, S(g)$ as $s \to \infty$.

Theorem 3.1 (Osterwalder–Schrader reconstruction). *Schwinger functions satisfying (E0) in its strengthened form and (E1)–(E3) determine a Hilbert space H, a unit vector Ω, a positive self-adjoint Hamiltonian H with $H\Omega = 0$, a unitary representation of the Poincaré group and Wightman fields, from which the S_n are recovered by analytic continuation. Each finite family f defines a vector $[f] \in H$; these vectors are dense in H, $\langle \Omega, [f] \rangle = S(f)$, and $\langle [f], e^{-tH}[f] \rangle = S(\theta \bar{f} \cdot \tau_t f)$ for $t \geq 0$. Given (E0)–(E3), the cluster property (E4) holds exactly when Ω is, up to a phase, the only vector invariant under the translations.*

This is the theorem of Osterwalder and Schrader.[10] Reflection positivity is the axiom that manufactures the Hilbert space, and with it the Hamiltonian whose spectrum the mass gap concerns.

3.2 The mass gap

The problem statement defines the gap and states the problem in these words:[11]

> "A quantum field theory has a mass gap Δ if H has no spectrum in the interval $(0, \Delta)$ for some $\Delta > 0$. The supremum of such Δ is the mass m, and we require $m < \infty$.
>
> Yang–Mills Existence and Mass Gap. Prove that for any compact simple gauge group G, a non-trivial quantum Yang–Mills theory exists on $\mathbb{R}^4$ and has a mass gap $\Delta > 0$. Existence includes establishing axiomatic properties at least as strong as those cited in [45, 35]."

The two references are Streater and Wightman's axioms and those of Osterwalder and Schrader. The text adds that clustering is "an important consequence of the existence of a mass gap": connected vacuum correlations of local operators decay exponentially in the distance. It also records that "the existence of a uniform gap for finite-volume approximations may play a fundamental role in the proof of existence of the infinite-volume limit."[12] Section 7 proves the precise form of both remarks that the present ground requires.

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[9] K. Osterwalder and R. Schrader, "Axioms for Euclidean Green's functions", *Comm. Math. Phys.* 31 (1973), 83–112, and "Axioms for Euclidean Green's functions. II", *Comm. Math. Phys.* 42 (1975), 281–305.

[10] Osterwalder and Schrader, *op. cit.*; see also J. Glimm and A. Jaffe, *Quantum Physics: A Functional Integral Point of View*, 2nd ed., Springer, 1987, Chapter 6.

[11] Jaffe and Witten, *op. cit.*, §4, p. 6.

[12] Jaffe and Witten, *op. cit.*, §5, p. 6.

3.3 Non-triviality

The problem fixes what it means for the constructed theory to be Yang–Mills theory and not some other theory:[13]

> "To establish existence of four-dimensional quantum gauge theory with gauge group G, one should define a quantum field theory (in the above sense) with local quantum field operators in correspondence with the gauge-invariant local polynomials in the curvature F and its covariant derivatives, such as Tr $F_{ij}F_{kl}(x)$. Correlation functions of the quantum field operators should agree at short distances with the predictions of asymptotic freedom and perturbative renormalization theory, as described in textbooks."

Two features of this demand govern the reading of the corpus: the theory must carry all the gauge-invariant local polynomials in the curvature, and its short-distance behaviour must be that of asymptotic freedom.

3.4 Limits obtained by compactness

A construction often proves bounds uniform in the cutoffs, extracts a convergent subsequence by compactness, and calls the limit the theory. The problem statement addresses this practice in its second footnote:[14]

> "We specifically exclude weak-existence (compactness) as the solution to the existence part of the Millennium problem, unless one also uses other techniques to establish properties of the limit (such as the existence of a mass gap and the axioms)."

Proposition 3.2 (The standard for a limit by compactness). *A system of Schwinger functions obtained as a limit point of cutoff theories by compactness establishes the existence part of the problem only together with a proof of properties of that limit, and the problem statement names the mass gap and the axioms as such properties.*

Proof. This is the content of the footnote quoted above, read at its literal strength. The footnote does not require the limit to be unique; it requires the properties to be proved for the limit that has been obtained. ■

4. The Lattice, Its Positivity and Its Gap

4.1 Wilson's theory

Let Λ be the periodic lattice of spacing a and side L. To each oriented bond b of Λ Wilson's theory assigns $U_b \in G$, with $U_{-b} = U_b^{-1}$, and to each plaquette p the ordered product U_p of the four bond variables around it. For $G = \mathrm{SU}(N)$ Wilson's measure is[15]

$$d\mu_\beta(U) \;=\; Z^{-1}\, exp\big(\,-\beta \textstyle\sum_p \big(1 \,-\, \frac{1}{N}\, Re\; tr\; U_p\big)\,\big) \prod_b dU_b, \tag{4.1}$$

[13] Jaffe and Witten, *op. cit.*, §4, p. 6.

[14] Jaffe and Witten, *op. cit.*, §6.5, footnote 2, p. 12.

[15] K. G. Wilson, "Confinement of quarks", *Phys. Rev. D* 10 (1974), 2445–2459.

with dU_{b} the normalised Haar measure. It is invariant under the gauge transformations $U_{\mathrm{xy}} \mapsto g_{\mathrm{x}} U_{\mathrm{xy}} g_{\mathrm{y}}^{-1}$, and the observables of the theory are the gauge-invariant functions of the bond variables. For SU(2) the plaquette energy $a_{\mathrm{p}}(U) = 1 - \tfrac{1}{2} \operatorname{tr} U_{\mathrm{p}}$ is the gauge-invariant scalar whose smeared and centred sums are the corpus's basic observable, the lattice form of the action density $\operatorname{tr} \mathrm{F}^2$.[16]

4.2 Reflection positivity and the transfer matrix

Osterwalder and Seiler proved that Wilson's measure is reflection positive: if ϑ is the reflection of Λ in a lattice hyperplane orthogonal to a coordinate axis and F is a function of the bond variables on one side of it, then $\int \vartheta \bar{F} \cdot F \, \mathrm{d}\mu_\beta \geq 0$.[17] The problem statement calls this "a major advantage" and adds that "establishing a quantum mechanical Hilbert space is part of the solution to this Millennium problem."[18] Reflection positivity makes the lattice theory a quantum mechanics. Lüscher constructed for Wilson's action a self-adjoint, strictly positive transfer matrix T acting on the gauge-invariant functions of the bond variables of one time slice, so that time-ordered correlations on a torus of time extent Na are traces of products of T with the observables.[19] On a torus of finite spatial volume T is an integral operator with a continuous, strictly positive kernel on a compact space. By Jentzsch's theorem its largest eigenvalue λ_0 is simple, with a strictly positive eigenfunction ψ_0, and every other eigenvalue is strictly smaller.[20]

Definition 4.1 (The lattice gap). *Let λ_1 be the largest eigenvalue of T below λ_0 on the gauge-invariant sector. The lattice mass gap is $m_{lat} = \log(\lambda_0/\lambda_1)$, in units of the lattice spacing; its value in physical units is m_{lat}/a.*

In the state defined by ψ_0, which is the limit of infinite time extent at fixed spatial volume, an observable A of the bond variables at positive times defines, by integration over the half-lattice with A inserted, a vector Φ_{A} on the time slice, and $\langle \vartheta \bar{A} \cdot \tau_{\mathrm{n}} A \rangle_0 = \langle \Phi_{\mathrm{A}}, (T/\lambda_0)^{\mathrm{n}} \, \Phi_{\mathrm{A}} \rangle$ for every number n of time steps. This is the lattice form of Theorem 3.1.

Lemma 4.2 (The gap governs reflection-positive decay). *Let $q = \lambda_1/\lambda_0$. For every gauge-invariant observable A at positive times and every $n \geq 0$,*

$$0 \;\leq\; \langle \vartheta \bar{A} \cdot \tau_n A \rangle_0 \;-\; |\langle A \rangle_0|^2 \;\leq\; q^n \, (\, \langle \vartheta \bar{A} \cdot A \rangle_0 \;-\; |\langle A \rangle_0|^2 \,). \tag{4.2}$$

Proof. Write $\Phi_{\mathrm{A}} = \langle \psi_0, \Phi_{\mathrm{A}} \rangle \psi_0 + \varphi$ with $\varphi \perp \psi_0$. Since $\langle \psi_0, \Phi_{\mathrm{A}} \rangle = \langle A \rangle_0$ and ψ_0 is fixed by T/λ_0, the left side of (4.2) equals $\langle \varphi, (T/\lambda_0)^{\mathrm{n}} \, \varphi \rangle$. The operator T/λ_0 is positive, and on the orthogonal complement of ψ_0 its norm is q. Hence $0 \leq \langle \varphi, (T/\lambda_0)^{\mathrm{n}} \, \varphi \rangle \leq q^{\mathrm{n}} \, \|\varphi\|^2$, and $\|\varphi\|^2 = \langle \vartheta \bar{A} \cdot A \rangle_0 - |\langle A \rangle_0|^2$ is the case $n = 0$. ∎

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[16] *YMM1*, p. 4.

[17] K. Osterwalder and E. Seiler, "Gauge field theories on a lattice", *Ann. Physics* 110 (1978), 440–471.

[18] Jaffe and Witten, *op. cit.*, §6.5, p. 11.

[19] M. Lüscher, "Construction of a selfadjoint, strictly positive transfer matrix for Euclidean lattice gauge theories", *Comm. Math. Phys.* 54 (1977), 283–292.

[20] R. Jentzsch, "Über Integralgleichungen mit positivem Kern", *J. Reine Angew. Math.* 141 (1912), 235–244.

The finite form of Lemma 4.2, that a gap places every connected two-point function of a positive symmetric transfer matrix between zero and q^{n} times its value at $n = 0$, is checked by the Lean kernel.[21]

On a torus of finite time extent N the correlations differ from those of the state ψ_0 by terms bounded, at a fixed lattice, by multiples of q^{N-n} and of $\mathrm{tr}(T/\lambda_0)^{N} - 1$, and these tend to zero as the time extent grows at fixed spatial volume. The constant on the right of (4.2) is a correlation between A and its mirror image, whose supports are separated by twice the distance of A from the reflection plane. It is therefore controlled by bounds on correlations at separated points, which is the only kind of bound a construction of a composite field can provide.

4.3 The renormalisation group of Bałaban

A continuum limit requires bounds on the lattice theory that are uniform as the spacing tends to zero along a sequence of bare couplings tuned by asymptotic freedom. Bałaban's renormalisation group provides such bounds on a torus. It integrates out the fluctuations of one length scale after another, each step followed by a rescaling, chooses gauges that vary from cell to cell and from scale to scale, and controls the effective actions uniformly in the spacing, with an analysis of asymptotic freedom and of the regions of large field.[22] The problem statement recorded that these estimates "need to be extended to the study of expectations of gauge-invariant functions of the fields."[23] The corpus is an attempt to carry out that extension and then to pass to infinite volume.

5. The Three-Dimensional Manuscripts

This section proves part (i) of the Ground Theorem.

5.1 The documents

The three-dimensional work consists of two manuscripts. The first, YMM1, is titled "Scalar fields from three-dimensional lattice Yang–Mills theory", carries no author or date line, and runs to 854 pages.[24] A revised edition, posted on 27 October 2026, runs to 856 pages. Its change list records twenty-four corrections and states: "The statements of the principal Theorems 0 and A–C are unchanged."[25] The second manuscript, YMM2, is titled "Curvature observables, dynamics, and conditional confinement in three-dimensional lattice Yang–Mills theory" and runs to 1,044 pages. It was posted with the note that it "has not been edited to even our standards, so we would like to continue cleaning it up and checking it", and its abstract states that "Confinement assumes a box inequality uniform along a fixed

[21] Edwin Boston III; listed in the Records under the lattice theorems in Lean.

[22] Bałaban, *op. cit.*; Jaffe and Witten, *op. cit.*, §6.5, p. 11.

[23] Jaffe and Witten, *op. cit.*, §6.5, p. 11.

[24] Page counts measured as stated in the Introduction.

[25] YM3/Revisions/CHANGES-2026-10-26.md in the repository.

physical scale and label."[26] Part (i) of the Ground Theorem concerns YMM1. YMM2 enters the residue of Section 7 through the curvature bilinears it adds.

5.2 The observable

YMM1 states its subject in its first sentences: "We study a scalar observable whose formal classical counterpart is $\mathrm{tr}(F^2)$, the Yang–Mills action density. Its quantum correlations are defined through centered Wilson plaquette energies."[27] On the lattice $a_K\mathbb{Z}^3$, with $a_K = L^{-K}$ and $L \geq 13$ odd, the smeared density of a compactly supported test function f is

$$\mathcal{O}_K(f) \;=\; a_K{}^{-1} \sum_p f(x(p))\, a_p, \tag{5.1}$$

the sum running over the positive plaquettes of all three orientations, $x(p)$ the barycentre of p and $a_p = 1 - \tfrac{1}{2}\,\mathrm{tr}\,U_p$ the plaquette energy; the centred density subtracts its expectation. The manuscript is explicit that "the notation $\mathrm{tr}(F^2)$ does not denote the pointwise square of a separately constructed continuum field strength."[28] The object of YMM1 is therefore a single gauge-invariant scalar field, the continuum limit of the plaquette energy.

5.3 The torus, the infinite volume and the components

Theorem A of YMM1 concerns a fixed torus of side $2L^m$. For $n \geq 2$ real Lipschitz test functions whose supports have empty common intersection, it asserts that the connected correlation of the smeared densities "exists along the full sequence of sufficiently fine lattices", locally uniformly in $b > 0$, and that "different positive labels b give different limiting families on the same torus."[29] On a fixed torus the manuscript therefore asserts a limit without passing to subsequences.

Theorem B passes to infinite volume, and there the limit is subsequential. For first-passage data (K_j, β_j) at a reference coupling g and periodic Gibbs states ν_j, "there is a common subsequence on which" the moments of the centred densities converge for every separated tuple, and the limit "has zero one-point function, is invariant under all Euclidean isometries, and is reflection positive at every plane."[30] Every such limit σ carries a canonical component measure $(X, \lambda, (\sigma_\xi)_{\xi\in X})$, with X compact metrizable, λ of full support and $\sigma = \int \sigma_\xi\, d\lambda(\xi)$, and each component obeys

$$|\sigma_\xi(X \cdot \tau_v Y) \;-\; \sigma_\xi(X)\,\sigma_\xi(Y)| \;\leq\; C_g(X,Y)\,|v|^{-1} \qquad (|v| \;\geq\; r(X,Y)), \tag{5.2}$$

uniformly in ξ.[31] The manuscript states the subsequential character of its conclusion in so many words: "The infinite-volume conclusion is subsequential. The fixed-torus estimate retains a power of the volume and does not give uniqueness of these infinite-volume limits."[32]

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[26] *YMM2*, "Curvature observables, dynamics, and conditional confinement in three-dimensional lattice Yang–Mills theory", YM3/YMM2.pdf in the repository, p. 1; commit message of 27 October 2026, 01:14:40 UTC.

[27] *YMM1*, p. 4.

[28] *YMM1*, p. 4.

[29] *YMM1*, p. 5.

[30] *YMM1*, p. 6.

[31] *YMM1*, p. 6; the display is taken from the source file YM3/YMM1.tex.

[32] *YMM1*, p. 7.

5.4 The axioms and the reconstruction

Theorem 0 of YMM1 is its Osterwalder–Schrader theorem. It asserts an odd integer $L \geq 13$, a physical inverse coupling $b > 0$, integers $K_j \to \infty$ and periodic Gibbs states at spacing L^{-K} and inverse coupling $L^K b$ along $K = K_j$, such that the limits of the moments of the centred densities exist for test functions with pairwise disjoint supports, extend uniquely to the test functions flat at the coincidence diagonals, and satisfy (E0) with the explicit bound $|S_n(\varphi)| \leq 2C^n(n!)^8|\varphi|_{8n}$, together with (E1)–(E4).[33] On their reflection Hilbert space these functionals "reconstruct a Hermitian scalar Wightman field on $\mathbb{R}^{1+2}$ with positive energy, a unique invariant vacuum, a dense common Poincaré-invariant polynomial domain, covariance, local commutativity and tempered vacuum distributions", and "the Hilbert space has dimension at least two."[34]

The proof beneath Theorem 0 selects "one of its canonical clustering components", and the clustering and the unique vacuum are properties of a component. Of the original limit Theorem B says: "The original limit also has this reconstruction, except that vacuum uniqueness and clustering are not asserted. Its zero-energy space is $L^2(X, \lambda)$, with the vacuum vector corresponding to the constant 1."[35] The original limit is thus a superposition of clustering theories, with as many vacua as the component measure has room for.

5.5 Non-triviality

Theorem C of YMM1 is a short-distance law for the two-point function. At a reference coupling below a further ceiling, for test pairs (f, h) whose supports lie at a distance of order d, every limit of Theorem B and every one of its components obeys

$$\kappa_2(f,h) \;=\; \frac{9}{16\pi^2}\, b^{-2}\, \mathcal{K}_{free}(f,h)\,(1 \,+\, e), \tag{5.3}$$

$$\mathcal{K}_{free}(f,h) \;=\; \iint\, f(x)\, h(y)\, |x \,-\, y|^{-6}\, dx\, dy,$$

with $|e| \leq \varepsilon(d) \to 0$ as $d \to 0$, uniformly in the limit, the component, the test pair and its position; and "the label is uniquely determined by the connected two-point function."[36] The kernel $|x - y|^{-6}$ is the two-point function of the free action density in three dimensions, and the factor b^{-2} carries the coupling. Non-triviality is thus a statement about short distances, as the problem statement requires in Section 3.3.

5.6 What the manuscript does not assert

YMM1 closes its statement of Theorem B with sentences that fix the boundary of its claim: "The observable throughout is the scalar plaquette-energy density defined above. The reconstructed field is a scalar field. The construction makes no identification with a tensor-valued field strength and

[33] *YMM1*, pp. 4–5.

[34] *YMM1*, p. 5.

[35] *YMM1*, pp. 5 and 7.

[36] *YMM1*, p. 7; the display is taken from YM3/YMM1.tex.

asserts no mass gap."[37] The decay (5.2) is a power, and an inverse first power. A mass gap, as Section 7 proves, is equivalent to decay at an exponential rate.

Proof of part (i). Each clause of part (i) is the content of the sentence cited for it: the convergence on a fixed torus, by Theorem A of YMM1 (§5.3); the subsequential infinite-volume limits and the components with the decay (5.2), by Theorem B (§5.3); the axioms and the reconstruction in a component, by Theorem 0 (§5.4); the non-triviality, by Theorem C (§5.5); the absence of vacuum uniqueness and of clustering for the original limit, by Theorem B (§5.4); and the absence of a mass gap, by the sentence quoted in §5.6. ■

6. The Four-Dimensional Draft and the Standing of the Corpus

This section proves parts (ii) and (iii) of the Ground Theorem.

6.1 The authors' account

The README describes the four-dimensional work in a note marked as Alpöge's:[38]

> "[LA: At the moment we do (appear to) have a YM4 (i.e. $\mathbb{R}^4$, not just in finite volume) existence argument / slop, but it is extremely long and basically not understandable as is. Thus in the meanwhile it is being written much more methodically (thus quite slowly!) and our intention with e.g. QED3 and then YM3 (both also $\mathbb{R}^3$) is to show some of the coolest YM4 arguments (e.g. winning O(4)-invariance via an RP trick) in a much simpler, and shorter to write, setting."

The same note asks that the field "not incentivize rushing out 10,000-page slop, which I think would be especially useless here!" The repository also holds a raw account, set in four dimensions and titled "The Yang--Mills continuum limit along Bałaban's renormalization group: a complete account of Theorem 0", in three LaTeX files totalling 26,610,818 bytes.[39] Woit described the four-dimensional material as "a 4d YM existence proof, but in the useless form of tens of thousands of pages of unchecked slop", and observed that "Alpoge/Sussman also claim they can do $\mathbb{R}^4$."[40]

6.2 The conditional theorem

The zeroth draft is titled "SU(2) Yang–Mills theory in four dimensions via Balaban's renormalisation group: construction, axioms and non-triviality", is dated 6 October 2026, and runs to 153 pages. Its theorem is conditional from its first sentence: "Assuming the proof inputs declared clause by clause in Section 9, running Balaban's renormalisation group with complex sources gives the following deductions".[41] Among its clauses are the following.[42]

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[37] *YMM1*, p. 7.

[38] README of the repository, *op. cit.*

[39] theorem_0_complete_account_raw_I.tex (10,554,720 bytes), raw_II.tex (9,092,940) and raw_III.tex (6,963,158), with a master file of 1,284 bytes, in the repository; sizes from the repository's file index.

[40] Woit, *op. cit.*

[41] *The zeroth draft*, p. 1.

[42] *The zeroth draft*, pp. 3–4.

(0) The running coupling exists at every scale between the cutoff and a reference length, its inverse square obeys two-sided linear flow bounds uniformly in the cutoff, and for SU(2) at small coupling it obeys the exact one-loop law.

(i) Bałaban's ultraviolet stability theorem holds, its one hypothesis supplied by (0).

(iii) The connected correlations of $n \geq 2$ smeared action densities with separated supports are bounded uniformly in the cutoff and, for test functions supported in balls, uniformly in the volume and the position.

(ii) In the draft's words, "they converge as the cutoff is removed—on each torus without passing to subsequences along bare couplings tuned at the reference length, on $\mathbb{R}^4$ along subsequences of cutoffs and volumes—to limits that satisfy, in the numbering of [21], the regularity condition E0 in a stated form, the invariance E1 under translations and the hypercubic group, the reflection positivity E2 and the symmetry E3, whence a Hilbert space with a vacuum vector, a non-negative Hamiltonian and spatial momenta".

(iv) At separation d the connected two-point function equals, up to a relative error at most one half, a fixed constant times a factor $\kappa \in [2/3, 4/3]$ times the fourth power of the running coupling at the scale d times an explicit free kernel, uniformly in the cutoff, the bare coupling, the volume and the position.

(vi) If the flux of the lattice rotation current through far spheres vanishes along one sequence of radii, uniformly in the cutoff, then every limit on $\mathbb{R}^4$ taken along cutoffs that grow fast enough compared with the volume is invariant under O(4).

The draft states that its conclusions up to non-triviality, except the convergence without subsequences on a fixed torus, "are complete deductions from Balaban's theorems as cited" and from a companion account.[43] It records with equal care what rests on what. Of a statement whose proof it defers it says, "we do not claim that a complete written proof exists, here or in the companion account", and of the uniqueness clause (v) it says that its supporting statements are proved "except five, which we do not claim as proved: three are written, with independent verification in progress, and two are not yet written".[44]

6.3 What the draft does not claim

The draft's introduction lists what it does not claim:[45]

> "We do not claim uniqueness of the infinite-volume limit (the limits on $\mathbb{R}^4$ are taken along subsequences of volumes), uniqueness of the vacuum, the cluster property, a mass gap, the growth condition of [22] in the number of points (hence field operators or Wightman functions), anything at coinciding points, invariance under all rotations without the hypothesis of clause (vi), statements about Wilson loops or about Balaban's block-averaged variables, or any statement for SU(N), $N \geq 3$."

[43] *The zeroth draft*, p. 1.

[44] *The zeroth draft*, p. 6.

[45] *The zeroth draft*, p. 5.

Its account of related work ends: "the present paper does not address the mass gap."[46] Its Remark 4.19 gives the reason that clustering and the gap are absent and states the subsequential character of the limit:[47]

> "(3) The cluster property (E4), uniqueness of the vacuum, a mass gap. The constants in our bounds do not decay as the separation of the supports grows, so no decay of the connected functions is available. … (5) Uniqueness. The limits on $\mathbb{R}^4$ are obtained along subsequences of cutoffs and volumes; independence of the subsequence, convergence without subsequences in the volume, independence of the regularisation, and convergence for merely Lipschitz test functions on $\mathbb{R}^4$ are not asserted."

6.4 The standing of the corpus

The repository opens with a disclaimer, which also states the authors' attribution of the work:[48]

> "Disclaimer. This repository contains work-in-progress, written largely by LLM. Its content has not been satisfactorily vetted. All claims should be understood as tentative, pending verification. Novel mathematical ideas should be attributed to Claude (under the guidance of LA)."

The phrase "LLM-approved" belongs to the README's paragraph on quantum electrodynamics: "While we have not verified the contents of the manuscript, we believe it is essentially correct and have done our best to reach a reasonable state of confidence. The manuscript is LLM-approved, but LLMs are fallible. A few parts have been checked by hand." Of YMM1 the README says: "We believe this manuscript is essentially correct also. [ES: I am less confident than for QED3, but I do think the manuscript is correct.]" It describes the manuscripts as "the result of cutting material from literally thousands of pages of slop", which "is then fed to Codex to extract a minimal manuscript", and it adds that "some definitions that are explicit in the slop must be inferred in the abridged version. We have not figured out how to fix this."[49] These sentences are the evidence for the words "LLM-approved" and "unverified" in the sentence this paper set out to prove.

6.5 The public reading

On Woit's thread a commenter observed that "the abstract of the 854-page slop "YMM1.pdf" states "Infinite-volume limits are subsequential."" Sussman replied the same evening:[50]

> "Of course subsequential results are weaker than results which do not require passing to a subsequence! But I must admit to not understanding the broader issue. The crux of the problem is to prove that the subsequential limit has good properties."

Section 7 takes this sentence at its word and identifies the property that suffices.

Proof of parts (ii) and (iii). Part (ii) is the content of the passages quoted in §§6.1–6.3: the conditional deductions and their clauses (§6.2), the two supporting statements not yet written (§6.2), the

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[46] *The zeroth draft*, p. 6.

[47] *The zeroth draft*, p. 75.

[48] README of the repository, *op. cit.*

[49] README of the repository, *op. cit.*

[50] Woit, *op. cit.* Comments of 27 October 2026, 6:30 pm and 8:35 pm ET.

properties not claimed (§6.3), and the authors' description of the four-dimensional argument (§6.1). Part (iii) is the content of the disclaimer and of the README's sentences on verification quoted in §6.4. ■

6.6 Further claims in public view

Six further claims stood in public view when this paper was written, and the ground is not fixed until each is placed beside the corpus. The first is an announcement. On 26 October 2026 OpenAI announced a mathematically rigorous existence proof of interacting quantum field theories in four dimensions, among them the Yang–Mills theory and quantum chromodynamics, stating that "We finished our formalization of Yang–Mills' existence on October 16th, 2026", and that a group led by David Gross, with Arthur Jaffe, Michael Douglas, Igor Klebanov and Nikita Nekrasov, would review the work before its release; Woit, who reproduces the announcement's sentences, gives its length as 1,147 pages, and the review group's own statement says that Bałaban's ideas "enabled OpenAI to follow the theory all the way to infinite distances and prove the existence of its proper continuum limit. But confinement and the mass gap remain very important open challenges."[51] If that proof is released and holds, it is a candidate for the ultraviolet half of the chain of §8.1, with the effective measure at a fixed physical scale as its output; by its own statement it is not a proof of the gap.

The second is a published claim of the whole. Mir Faizal and Arshid Shabir, "Reflection-Positive Construction of a Four-Dimensional SU(N) Yang–Mills Theory with Mass Gap and Confinement", posted to the arXiv on 9 June 2026 and published in *Fortschritte der Physik* 74 (2026), e70097, with the detailed calculations in four companion papers in the *International Journal of Geometric Methods in Modern Physics*, runs to 593 pages in its arXiv form.[52] Its abstract claims a reflection-positive lattice formulation with a transfer operator of uniform gap at strong coupling, a "finite-range, gauge-covariant multiscale analysis" that carries the gap and the area law "from one scale to the next with interlaced inequalities whose small defects can be summed", the Osterwalder–Schrader reconstruction, and universality: "a single-scale Lipschitz control and a telescoping argument bind all admissible reflection-positive slicings into a unique limiting measure", so that "the continuum reached from strong coupling meets the one approached along an asymptotically free trajectory, yielding one and the same theory". At the join the structure is this. Section 3 obtains the gap at strong coupling "at fixed lattice spacing" by the polymer expansion. Section 10 enters the weak-coupling regime under "Hypothesis 1 (Stated for convenience; proved as Theorem 10.4 below)", which asserts a one-step renormalisation map, analytic "in a neighborhood U of (g, K) = (0, 0)", obeying the one-loop recursion with a controlled remainder, Theorem 10.4 being the "one-step contraction near the Gaussian fixed point". Theorem 10.8, "Identification with the constructed continuum limit", then asserts that the continuum Schwinger functions "obtained by following an admissible, asymptotically free trajectory from the ultraviolet" and those "constructed by transporting the strong-coupling mass gap and Wilson-loop area law to the continuum within the admissible class" coincide, by the uniqueness of its Section 9 over that class. The passage between the two regimes is thus made by a

[51] Woit, *op. cit.*, the post of 26 October 2026, from which every sentence of the announcement and of the group's statement is quoted here; the work itself is unreleased, and this paper has read none of it.

[52] Faizal and Shabir, arXiv:2606.19362, doi 10.1002/prop.70097. This paper has read its abstract (p. 1), its table of contents (pp. 2–3) and the three statements at its join, Hypothesis 1 (p. 110), Theorem 10.4 (p. 116) and Theorem 10.8 (p. 128), and not its remaining pages; no expert assessment of it was found in print by 29 October 2026.

uniqueness statement over a class of constructions, where the chain of §8.1 makes it by a bound at the crossover scale. Read at the five places where such a construction must carry weight, it carries it as follows.[53] The multiscale analysis is a "scale-by-scale decomposition of Gaussian covariances", the resolvent of "a nonnegative, selfadjoint, gauge-covariant elliptic operator L on $\ell^2(a\mathbb{Z}^4)$ with nearest-neighbour structure and a spectral gap $m^2 \geq 0$", with "massive covariances $(A_h + \mu^2)^{-1}$" in the appendix: the mass of the fluctuation field is a parameter of the decomposition, not an output of the theory. The persistence of the gap across scales, Theorem 5.6, assumes "the interlacing/defect inequality $\gamma_{k+1} \geq \gamma_k - \delta_k$" with $\Sigma\delta_k < \infty$ and concludes $\inf \gamma_k \geq \gamma_0 - \Sigma\delta_j$ by the telescoping sum; the summability, Proposition 5.5, holds "Under the FRD locality hypotheses", which "guarantee", uniformly in the scale, that "every connected Ursell kernel $U_k(X)$ associated with a finite polymer X of diameter $\operatorname{diam}(X) = \ell$ obeys $|U_k(X)| \leq Ce^{-c\ell}$" — exponential clustering at every scale is the hypothesis. The continuum Schwinger functions of Theorem 7.1 are subsequential limits, by tightness, of "the continuum embedding of the corresponding lattice field as a random tempered distribution … via block-averaging followed by the usual scaling", at a fixed bare coupling. The weak-coupling half, Theorem 10.4, is carried on three pages that represent "the one-step effective action by integrating the fluctuation field with law $N(0, C_0)$", a Gaussian, and state that "the Brydges–Kennedy–Abdesselam–Rivasseau (BKAR) formula together with Kotecký–Preiss bounds guarantees absolute convergence of the cumulant expansion provided the a priori polymer norm is small", with the remainder bounded because "Power counting and analyticity give the bound $|R_k| \leq c_2|g_k|^5$"; the large-field regions, the gauge fixing of the fluctuation field and the non-linear averaging of a compact variable, which occupy Bałaban's papers, do not appear in them. And the uniqueness that identifies the two limits, Theorem 9.4, concludes that two constructions give the same Schwinger functions "If $\mu_k = \tilde{\mu}_k$ and $K_k = \tilde{K}_k$ on a core of local observables", that is, when they already share their marginal and their transition kernel; so Theorem 10.8 presupposes the identification it asserts. In this form the construction does not reach the object this paper names as open, Hypothesis I of §8.3, and does not discharge it. Its remaining pages are not examined here, and no expert assessment of it was found in print by 29 October 2026.

The third and fourth are manuscripts deposited on Zenodo, which no journal has refereed. Ramiro Fontes, "The SU(2) Yang–Mills Mass Gap via Curvature-Based Block Renormalization, Character Expansion, and Dobrushin Comparison", deposited on 7 March 2026 and revised on 11 March, states as its Main Theorem that for $\beta \geq \beta_0 = 9587$ the connected correlators of gauge-invariant observables decay exponentially, uniformly in the volume, and that every subsequential continuum limit carries a mass gap (p. 4).[54] Its renormalisation group runs while the effective coupling stays at least β_0, the first of its inductive hypotheses (p. 19), and stops after $(\beta - 9587)/0.258$ steps; the decay is then asserted for the effective measure at β_0 by a cluster expansion whose polymers lie inside one block of side two with its boundary held fixed, the rate being, in its words, "the inter-block decay of the effective action" (pp. 46, 85). Locality of an effective action is not clustering of its measure, and

[53] The same arXiv PDF: §2.3 (p. 17), Proposition 5.5 (pp. 55–56), Theorem 5.6 (p. 57), Theorem 7.1 (p. 80), Theorem 9.4 (p. 105), the proof of Theorem 10.4 (pp. 117–119) and the appendix on the decomposition (p. 174). The quotations are verbatim.

[54] Zenodo records 18906249 and 18969699; this paper has read, in the version of 11 March 2026 (ym_v65.pdf), the Main Theorem (p. 4), the inductive hypothesis (p. 19), the passage on the decay rate (p. 46), the note on the terminal coupling (p. 80) and the clustering step of the mass-gap theorem (p. 85). B. Lee, Zenodo record 19601454, version v13_r22: pp. 11–12, 19, 26, 72 and 83.

the manuscript's own note shows the consequence, stating that at $\beta = 9587$, its minimum, "no RG step is needed: the cluster expansion converges directly on the original lattice" (p. 80). The bare Wilson theory at $\beta = 9587$, a coupling at which the manuscript's own normalisation places the physical scale some $e^{25,800}$ lattice spacings out, would then have a correlation length of the order of the spacing. The mechanism is the spectral gap of the block's abelian Hessian, the non-abelian curvature entering as a small correction (p. 4), so the same argument would give the four-dimensional U(1) theory a gap at weak coupling, which Guth and Fröhlich and Spencer exclude: it fails the test of §8.3. Its one-block constants were re-derived on three roads on 29 October, and three of them are wrong. The 72-plaquette Hessian whose least eigenvalue 0.6893 it uses (p. 4) is an asymmetric truncation of the 120 plaquettes meeting a block of side two: over the block's 24 faces the full Hessian is $\text{faces}^{\mathrm{T}}\text{faces} + 3\mathrm{I}$, with spectrum 3, 7, 9, 11 of multiplicities 15, 6, 8, 3 exactly, so the least eigenvalue is 3 and the zero mode the manuscript removes is an artefact of its truncation. Its entropy inequality $(e\, c_{\mathrm{conn}})^{k-1} \le 20^{k}$ fails for animals of four plaquettes, where the tree-graph bound it cites gives 54.4 per plaquette. And its Peierls condition $20e^{-\beta(1-\cos\theta)} < 1$, which fixes $\beta_0 = 9587$ at its angle $\theta = 0.025$, bounds a ratio of integrals by a ratio of maxima: the probability that one plaquette's angle exceeds θ under the one-plaquette law at $\beta = 9587$ is 0.1118, not the 0.0500 the condition assumes, so the condition fails at the manuscript's own parameters by a factor 2.24, and corrected at that angle its threshold lies near 16,000.[55] Byoungwoo Lee, "Yang–Mills Mass Gap in Four Dimensions: Reduction and Closure for the BT–2 Route", deposited on 16 April 2026, obtains its gap $m_0 \ge \rho_0$ from its Assumption 2.1, a logarithmic Sobolev floor uniform in the spacing (pp. 11–12). The floor is carried by the Holley–Stroock lemma under the supposition $\mathrm{KL}(\mu_0 \| \mu) \le M$ (p. 72), with M among the quantities the manuscript holds fixed by choice (p. 11), and its appendix proves that the system of inequalities is solvable in its parameters (p. 83), not that the Yang–Mills measure satisfies it; for a Gibbs measure that relative entropy grows with the volume and as the spacing tends to zero. Neither manuscript reaches the crossover.

The fifth and sixth are repositories, read on 29 October at the places where their gaps are obtained. A GitHub repository announcing a resolution of the problem for SU(3), created on 16 October 2026 and last changed on 28 October, carries five volumes and a Lean corpus. Its master theorem is stated under two named hypotheses, a continuum bridge and a non-perturbative cluster decay, and concludes a gap of 1.552 ± 0.038 GeV, a simulation's number with its error bar; the gap enters twice by insertion, as the square of that number set equal to a Ricci lower bound on the orbit space, which no derivation supports, and as the decay rate of its cluster theorem, and once by assertion, a polymer contraction evaluated at the finest scale only and declared uniform toward the infrared, where asymptotic freedom denies the bound it needs; its own reviewers' guide names the first of these as a trap, and its own README classes the Lean corpus as a conditional reduction whose premise is the cluster decay. The polymer step is group-blind and would give the four-dimensional U(1) theory a gap: it fails the test of §8.3 where it is asserted.[56] A Zenodo record of 27 September

[55]The three roads are the plan seat's awk instruments, the fourth seat's blind re-derivation (the Hodge decomposition of the tesseract; Laplace for the marginal) and an integer and quadrature check, all under records/one-block/ of the repository with the ledger OPEN/ledger/one-block-the-unit-s90-s92.md and the D1 page beside it. The spectrum is an integer identity, $(H - 3)(H - 7)(H - 9)(H - 11) = 0$ with traces 192, 1440 and 12288; the marginal is $\int_0^{\pi} e^{\beta\cos\varphi} \sin^2\varphi \, d\varphi$ divided by the same integral from 0, the Laplace tail of a χ^2 with three degrees of freedom at $\beta\theta^2 = 6$.

[56]github.com/garygitton/yang-mills-millennium-problem: Volume V, p. 2 (Theorem 1.2) and p. 5 ((2.6), Theorem 2.1); Volume II, p. 14 (Theorem 4.5 and its proof, evaluated at k = 0) and pp. 23–24 (Theorem 7.1); the README's epistemic note and the

2026, written by a language model at its depositor's request and labelled as awaiting the depositor's review, proves a spectral gap for the SU(2) Kogut–Susskind Hamiltonian on open spatial boxes at strong coupling, uniform in the box and the spacing: unconditionally for $g^2 \geq 3.978$ with its order-2 inputs recomputed, conditionally to $g^2 \geq 3.684$ on coefficient bounds read and not re-implemented, by a vacuum equation, a Casimir inequality proved through a supply–demand theorem, and a spectral return. It states its own limit, that the continuum path leaves its domain after finitely many steps, and claims no crossing. It is a sibling of Theorem 8.7 in the Hamiltonian formulation, at a strong-coupling threshold that cannot be ranked against 0.2452 because no exact map joins the two couplings there, and not a claimant.[57]

7. The Residue and Its Completion

This section proves part (iv) of the Ground Theorem.

7.1 The residue

Table 1 sets each demand of the problem statement beside what the two manuscripts assert, with the section of this paper that carries the evidence.

The problem demands	YMM1, three dimensions	The zeroth draft, four dimensions
Every compact simple gauge group	SU(2) (§5.2)	SU(2); no statement for SU(N), $N \geq 3$ (§6.3)
All gauge-invariant local polynomials in the curvature	The scalar plaquette-energy density; curvature bilinears in YMM2 (§§5.1–5.2)	The action density (§6.2)
The infinite volume	Limits along subsequences (§5.3)	Limits along subsequences of cutoffs and volumes, under the declared inputs (§6.2)
Euclidean invariance (E1)	Asserted (§5.3)	Translations and the hypercubic group; O(4) under the flux hypothesis (§6.2)
Reflection positivity and symmetry (E2), (E3)	Asserted (§§5.3–5.4)	Asserted under the declared inputs (§6.2)
The cluster property (E4) and a unique vacuum	In each component, with decay $\lvert v\rvert^{-1}$; not for the original limit (§§5.3–5.4)	Not claimed (§6.3)
Field operators and the relativistic spectral condition	A scalar Wightman field in each component (§5.4)	Not claimed; a non-negative Hamiltonian (§§6.2–6.3)
Short-distance agreement with asymptotic freedom	The two-point law (5.3) (§5.5)	A two-point law in the running coupling, uniform in the volume (§6.2)
A mass gap $\Delta > 0$ with finite mass	Not asserted (§5.6)	Not addressed (§6.3)

REVIEWERS_GUIDE; the Lean theorems dimensional_transmutation_mass_gap_positive, canonical_balaban_polymer_activity_subunitary_evaluation (the inequality 582 < 1000) and scaleway_su3_measured_mass_gap_positive (a typed number); read by the fourth seat of this work on 29 October 2026.

[57] Zenodo record 23001803, version of 27 September 2026, the Yang–Mills record: §0.3, §1, Theorem 2.1, Lemma 2.2, Corollary 2.6 and Proposition 6.2; read by the fourth seat on 29 October 2026.

The problem demands	YMM1, three dimensions	The zeroth draft, four dimensions
A verified proof	Not satisfactorily vetted, by the authors' statement (§6.4)	Conditional; two supporting statements not yet written (§6.2)

Table 1. The demands of the problem statement and the assertions of the corpus.

7.2 The gap is exponential clustering

The decisive observation is that, for a reflection-positive theory, a mass gap and exponential decay of the reflection-positive correlations are one and the same property. The argument is the spectral theorem applied to the semigroup of Osterwalder and Schrader; its weight here lies in the identification it makes. Let S be reflection positive and invariant under time translations, and let H, Ω and $H \geq 0$ be the Hilbert space, the vacuum and the Hamiltonian that Osterwalder and Schrader construct from these two properties, with the relations of Theorem 3.1. This is exactly the structure that both manuscripts assert for their limits.

Theorem 7.1 (The gap is exponential clustering). *Let $m > 0$. The following conditions are equivalent.*

(a) Ω is, up to a phase, the only vector annihilated by H, and H has no spectrum in the interval $(0, m)$.

(b) For every finite family f and every $t \geq 0$, $S(\theta\bar{f} \cdot \tau_t f) - |S(f)|^2 \leq e^{-mt}\,(\,S(\theta\bar{f} \cdot f) - |S(f)|^2\,)$.

(c) For every finite family f there are a constant C_f and an unbounded set $T_f \subset [0, \infty)$ such that $S(\theta\bar{f} \cdot \tau_t f) - |S(f)|^2 \leq C_f e^{-mt}$ for every $t \in T_f$.

When they hold and H has dimension at least two, the mass $m^* = \inf \operatorname{spec}(H \text{ restricted to } \Omega^\perp)$ satisfies $m \leq m^* < \infty$.

Proof. For a finite family f put $\psi_f = [f] - S(f)\Omega$. Since $\langle \Omega, [f] \rangle = S(f)$, the vector ψ_f is the orthogonal projection of $[f]$ onto $\Omega^\perp$, and since $e^{-tH}\Omega = \Omega$,

$$\langle \psi_f, e^{-tH}\,\psi_f \rangle \;=\; S(\theta f \,\cdot\, \tau_t f) \;-\; |S(f)|^2, \quad t \;\geq\; 0. \tag{7.1}$$

The space $\Omega^\perp$ reduces H, and since the vectors $[f]$ are dense in H, the vectors ψ_f are dense in $\Omega^\perp$.

(a) ⇒ (b). By (a) the restriction of H to $\Omega^\perp$ has spectrum in $[m, \infty)$, so $e^{-tH} \leq e^{-mt}$ on $\Omega^\perp$ by the spectral theorem. Apply this to ψ_f in (7.1); the case $t = 0$ of (7.1) identifies $\|\psi_f\|^2$.

(b) ⇒ (c). Take $C_f = S(\theta\bar{f} \cdot f) - |S(f)|^2$ and $T_f = [0, \infty)$.

(c) ⇒ (a). Let μ_f be the spectral measure of H in the vector ψ_f, a finite positive measure on $[0, \infty)$. For $0 \leq m' < m$ and $t \in T_f$, identity (7.1) and condition (c) give

$$\mu_f([0, m'])\; e^{-tm'} \;\leq\; \int\; e^{-t\lambda}\, d\mu_f(\lambda) \;\leq\; C_f\; e^{-mt}, \tag{7.2}$$

so $\mu_f([0, m']) \leq C_f\, e^{-(m-m')t}$ for arbitrarily large t, and $\mu_f([0, m']) = 0$. Letting m' increase to m gives $\mu_f([0, m)) = 0$, which says that $E\psi_f = 0$ for the spectral projection E of H onto $[0, m)$. The ψ_f are dense in $\Omega^\perp$ and E is bounded, so E vanishes on $\Omega^\perp$. Hence $H \geq m$ on $\Omega^\perp$: no nonzero vector of $\Omega^\perp$ is annihilated by H, and H has no spectrum in $(0, m)$. This is (a).

Finally, if dim H ≥ 2 then $\Omega^{\perp} \neq \{0\}$, the spectrum of H on $\Omega^{\perp}$ is a nonempty closed subset of $[m, \infty)$, and its infimum m^* is finite. ■

The finite form of the implication (c) ⇒ (a), the gap read off the decay of a positive symmetric transfer matrix, is checked by the Lean kernel.[58]

Corollary 7.2 (Clustering). *Under the conditions of Theorem 7.1, for all finite families f and g and every $t \geq 0$, $|S(\theta\bar{g} \cdot \tau_t f) - S(\theta\bar{g})\, S(f)| \leq e^{-mt}\, \|\psi_g\|\, \|\psi_f\|$. If S also satisfies (E0)–(E3), the cluster property (E4) holds, and the same bound holds with τ_t replaced by the translation by t in any direction.*

Proof. The left side is $|\langle \psi_g, e^{-tH} \psi_f \rangle|$, and e^{-tH} has norm at most e^{-mt} on $\Omega^{\perp}$. Under (E0)–(E3) the vacuum is unique by Theorem 7.1, and Theorem 3.1 gives (E4). A translation in another direction is conjugate to τ_t by a Euclidean rotation, under which S is invariant by (E1), and the reflection Hilbert space built across the rotated hyperplane carries the same Hamiltonian spectrum, since the Schwinger functions it is built from are the same functions. ■

Measured against Theorem 7.1, the decay (5.2) asserted for the components of YMM1 is the decay of a theory whose mass gap has not been established: a bound by an inverse power of the separation is compatible with spectrum reaching down to zero. The distance between the power and the exponential is the mass gap.

7.3 From the lattice to the limit

Lemma 7.3 (Passage to the limit). *Let S_j be Schwinger functions converging to S, along a subsequence, on the test functions flat at the coincidence diagonals, and let $D \subset [0, \infty)$ be unbounded. Suppose that for every finite family f there are a constant C_f and numbers $r_j(t; f)$, with $r_j(t; f) \to 0$ as $j \to \infty$ for each $t \in D$, such that $S_j(\theta\bar{f} \cdot \tau_t f) - |S_j(f)|^2 \leq C_f\, e^{-mt} + r_j(t; f)$ for $t \in D$. Then S satisfies condition (c) of Theorem 7.1.*

Proof. For a finite family f and $t \geq 0$ the families $\theta\bar{f}$ and $\tau_t f$ are supported in opposite half-spaces, so every test function entering $S_j(\theta\bar{f} \cdot \tau_t f)$ is flat at the coincidence diagonals, and both sides of the hypothesis converge along the subsequence. The inequality passes to the limit, with $T_f = D$. ■

Theorem 7.4 (The completion). *Let a construction be given by Wilson measures on periodic lattices of spacings $a_j = a_0 L^{-K_j}$, with L a fixed integer and $K_j \to \infty$, whose sides tend to infinity, with bare couplings tuned along the sequence; and let the Schwinger functions S_j of smeared gauge-invariant observables converge along a subsequence to S. Let D be the set of times $n\, a_0 L^{-k}$ with n, k ≥ 0. Suppose that the lattice correlations cluster uniformly at a physical rate m > 0: for every finite family f there are a constant C_f and numbers $r_j(t; f) \to 0$ as $j \to \infty$ for each $t \in D$, such that*

$$S_j(\theta\bar{f} \cdot \tau_t f) - |S_j(f)|^2 \leq C_f\, e^{-mt} + r_j(t; f) \quad \text{for } t \in D.$$

Then S has a unique vacuum, its Hamiltonian has no spectrum in (0, m), and its connected reflection-positive correlations decay in time at the rate m. If S is non-trivial, its mass is finite. The hypothesis holds, with $r_j = 0$, when the S_j are the correlations of the states ψ_0 of spatial tori, the transfer matrices satisfy $m_{lat}(j) \geq m\, a_j$ on the sector generated by the observables, and the correlations at separated points are bounded uniformly in j.

[58] Edwin Boston III; listed in the Records under the lattice theorems in Lean.

Proof. The set D is unbounded and independent of f. Lemma 7.3 gives condition (c) of Theorem 7.1 for S, and Theorem 7.1 with Corollary 7.2 gives the conclusions; non-triviality means that H has dimension at least two. For the last sentence, each $t \in D$ is an integer multiple $n_j a_j$ of the spacing for all large j, and the smeared observable of $\tau_t f$ is then the lattice translate by n_j steps of the smeared observable of f. With $q_j = \lambda_1/\lambda_0$ for the transfer matrix of rung j, so that $-\log q_j = m_{\mathrm{lat}}(j) \geq m\, a_j$, Lemma 4.2 gives $S_j(\theta \bar{f} \cdot \tau_t f) - |S_j(f)|^2 \leq q_j^{n_j} \, (S_j(\theta \bar{f} \cdot f) - |S_j(f)|^2) \leq e^{-mt} \, (S_j(\theta \bar{f} \cdot f) - |S_j(f)|^2)$, and the bracket is a correlation at separated points, bounded uniformly in j by a constant depending only on f. ■

Corollary 7.5 (What the gap supplies against the footnote). *For a construction by compactness of the kind in Theorem 7.4, uniform clustering at a physical rate, and in particular a lattice gap uniform in physical units, supplies the mass gap and the uniqueness of the vacuum. With (E0)–(E3), (E1) meaning full Euclidean invariance, it supplies the properties that the footnote of Proposition 3.2 names: the mass gap and the axioms.*

Proof. Theorem 7.4 gives the gap and the unique vacuum; with (E0)–(E3), Corollary 7.2 gives the cluster property (E4), which completes the axioms. ■

7.4 The collapse of the components

Corollary 7.6 (The gap collapses the component measure). *If the original limit σ of Theorem B of YMM1 satisfies condition (c) of Theorem 7.1, then its component measure λ is a point mass, σ is itself a clustering theory, and its connected correlations decay exponentially in every direction.*

Proof. By Theorem B of YMM1 the zero-energy space of σ is $L^2(X, \lambda)$, with the vacuum the constant 1 (§5.4). By Theorem 7.1 the zero-energy space is spanned by Ω, so $L^2(X, \lambda)$ is one-dimensional. If λ were not a point mass, a Borel set B with $0 < \lambda(B) < 1$ would give a function 1_B independent of 1 in $L^2(X, \lambda)$. Hence λ is a point mass; since λ has full support, X is a single point and $\sigma = \sigma_\xi$. The decay is exponential in time by Corollary 7.2 and in every direction by the Euclidean invariance of σ. ■

A gap would therefore do more than add a property to the limit: it would remove the need for the conditioning by which the corpus obtains its clustering components.

Proof of part (iv). The residue is Table 1, each entry of which is proved in Sections 5 and 6. The completion is Theorem 7.4, and its equivalence with uniform exponential clustering is Theorem 7.1, carried along the sequence by Lemma 7.3. ■

8. The Ground Handed Over

8.1 The object that remains

The object that remains is the infinite-volume mass gap of four-dimensional SU(2) lattice Yang–Mills theory at every coupling, uniform in physical units along the scaling sequence: a constant $m > 0$ with $m_{\mathrm{lat}}(j) \geq m\, a_j$ at every rung of the construction's sequence, a lower bound on the lattice gap in physical units, uniform in the spacing and in the volume. By Theorem 7.4 it is the mass gap of every continuum limit point, and the same object for every compact simple gauge group completes the problem's demand for a gap. Two features of this inequality fix the work it demands.

The first is its scale. In lattice units the required gap tends to zero with the spacing, like $m\,a_{\mathrm{j}}$, so it cannot be read at a single coupling. The lattice gaps that are proved directly hold at strong coupling, where the transfer matrix mixes rapidly: for SU(2) this is the theorem of Osterwalder and Seiler, and for every finite gauge group it is among the theorems checked by machine in §8.2.[59] For SU(2) the constant of that gap is a theorem of Edwin Boston III: the conditional law of one link given its six staples, the heat-bath kernel, moves in total variation by at most $2\beta/(3\pi)$ times the change of the staple sum, because the mean absolute deviation of a coordinate of a von Mises–Fisher law on the three-sphere is at most its value $4/(3\pi)$ at the uniform law, with equality there alone; so Dobrushin's single-site condition holds on the full lattice for $\beta < 3\pi/72 = 0.1309$, and in the temporal gauge on lattices open in time, where every time link is the identity and a space link has fourteen neighbouring links rather than eighteen, four spatial plaquettes with three other space links each and two temporal plaquettes with one, for $\beta < 3\pi/56 = 0.1683$.[60] The construction's sequence runs into the scaling window of weak bare coupling, where asymptotic freedom ties the bare coupling to the spacing. The inequality must be carried from the one end of the sequence to the other.

The second is its carrier. The carrier is the renormalisation group itself: a chain of blocking steps from a fixed physical scale, at which the effective coupling is small and Bałaban's estimates hold, down to the scale of the gap, with the error of every step certified. Five links compose this carrier, and each has its status. The first is the ultraviolet half: the blocking steps from the cutoff to a fixed physical scale at which the effective coupling is still small, with the effective action held near Wilson's uniformly in the number of steps. This is Bałaban's theorem, and what the carrier needs of it beyond his published estimates is stated as an interface: the convergence of the correlation functions of the effective theory at that scale as the cutoff is removed, and the Euclidean invariance of the limit, the two pieces the zeroth draft deduces conditionally and names among its inputs. The second is a single bound at the crossover scale, where the effective coupling ceases to be small. Along the Gaussian scales above it the mixing coefficient of a fibre is exactly one, that of a massless Gaussian, so no contraction is to be certified there; the content lies at the crossover scale alone, and the object there is the influence functional of the blocked action: the covariance of a boundary-adjacent link's projection with every plaquette of the block under the worst boundary datum, equivalently the mixed second differences of the non-Wilson part of the renormalised action. This bound is the open mathematics of the route. The third is the infrared half, the gap of the blocked theory inside the strong-coupling disk, which is the theorem of Osterwalder and Seiler for SU(2) and a theorem checked by machine for every finite gauge group. The fourth is the uniformity of the second link's bound in the cutoff: the effective actions at the fixed scale differ across the sequence by tilts of small oscillation, and a bound stable under such tilts makes the decay rate and range of the third link independent of the cutoff, which is the inequality of Theorem 7.4 with one constant. The fifth is the assembly, Theorem 7.4 itself. Once the second link is in hand, two roads carry it to the gap: the transfer-matrix road of Propositions 8.1 and 8.2 below, and the Gibbs road of §8.3, where Theorem

[59] Osterwalder and Seiler, *op. cit.*, for the strong-coupling gap of the Wilson theory with a compact gauge group; the finite-group form with its explicit coefficients is Edwin Boston III's, listed in the Records.

[60] Edwin Boston III, the sharp one-link lemma: the bound on the mean absolute deviation is certified by an enclosure of 9,104 rectangles in the plane of the two parameters of the law, no cell undecided, with the axial case by Chebyshev's association inequality and the remainder by hand; the heat-bath kernel's constant follows by the path argument, each law along the path being a von Mises–Fisher law; its record and run of record are listed in the Records. The time-open count is the paper's.

8.5 states the passage from a Dobrushin bound on the blocked measure at one fixed physical scale, with the field inside the blocks clustering at that scale, to the gap of every limit point, the third and fourth links among its hypotheses and Proposition 8.4 the fourth link's exact form. The remaining entries of Table 1, namely every compact simple gauge group, the full algebra of curvature observables and the unconditional invariance under O(4), stand beside it as the rest of the existence that the problem demands.

8.2 The instruments in hand

The proof that follows begins with instruments in hand. Each reads a lattice gap exactly at fixed coupling, and each is stated here at the strength of its proof or of its record.

Proposition 8.1 (Dobrushin's coefficient bounds the gap). *Let T be a transfer operator with a continuous, strictly positive kernel on a compact space X, with largest eigenvalue λ_0 and positive eigenfunction ψ_0, and let $P(x, dy) = T(x, dy)\,\psi_0(y) / (\lambda_0\,\psi_0(x))$ be its Doob transform, a Markov kernel. For $k \geq 1$ let $\delta(P^k) = \sup_{x,y} \|P^k(x, \cdot) - P^k(y, \cdot)\|_{TV}$ be Dobrushin's coefficient. Then every real eigenvalue $\lambda \neq \lambda_0$ of T satisfies $|\lambda|/\lambda_0 \leq \delta(P^k)^{1/k}$, and when T is positive,*

$$m_{lat} \;\geq\; \frac{1}{k}\, log\, \frac{1}{\delta(P^k)} \quad for\ every\ k \;\geq\; 1. \tag{8.1}$$

When X is finite, m_{lat} equals the limit of the right side as $k \to \infty$.

Proof. The Doob transform has the eigenvalues λ/λ_0 of T, with eigenfunctions $u = \psi/\psi_0$, which are bounded because ψ_0 is bounded below on the compact space. For a real bounded function u, $P^k u(x) - P^k u(y)$ is the integral of u against the signed measure $P^k(x, \cdot) - P^k(y, \cdot)$, of total mass zero and total variation at most $2\delta(P^k)$, so $\operatorname{osc}(P^k u) \leq \delta(P^k)\operatorname{osc}(u)$. If $Pu = \mu u$ with μ real and u nonconstant, then $\operatorname{osc}(P^k u) = |\mu|^k \operatorname{osc}(u)$, whence $|\mu|^k \leq \delta(P^k)$. When T is positive every eigenvalue is non-negative, and the bound at λ_1 is (8.1). When X is finite, $\delta(P^k)$ is the norm of P^k on the signed measures of total mass zero, on which the spectrum of P is that of T/λ_0 with the simple eigenvalue 1 removed, and Gelfand's spectral-radius formula gives the limit. ∎

The coefficient is Dobrushin's and the limit is Gelfand's.[61] Proposition 8.1 turns the gap of Theorem 7.4 into a statement about mixing: a certified bound $\delta(P^k) \leq e^{-k\,m\,a_j}$ for the Doob chain of every rung's transfer matrix bounds every eigenvalue below the top of the whole spectrum, and therefore gives the gap $m_{\mathrm{lat}}(j) \geq m\,a_j$ on the sector generated by the observables.

A single coefficient $\delta(P^k)$ is a global quantity, and on a lattice of many sites it is close to one at every accessible k. The renormalisation group splits it into rung quantities that can each be certified.

Proposition 8.2 (The chain lemma; Edwin Boston III). *Let X be a finite state space, P the Doob chain of a transfer matrix on X with stationary law π, and $b : X \to Y$ a block map with fibres $F_y = b^{-1}(y)$ and conditional laws $\pi(\cdot \mid y)$. Let $(b_*\mu)(y') = \mu(F_{y'})$, and define the lumped chain $P'(y, \cdot) = \Sigma_{x \in b^{-1}(y)}\, \pi(x \mid y)\,(b_*P^2)(x, \cdot)$, so that one coarse step is two fine steps; for $k \geq 1$ put $Q = P'^k$. With $\|\mu - \nu\|_{TV} = \tfrac{1}{2}\,\Sigma\,|\mu - \nu|$, define the fibre forgetting η_k, the largest of $\|P^{2k}(x, \cdot) - P^{2k}(x', \cdot)\|_{TV}$ over pairs x, x' in one fibre; the lumping defect ε_k, the largest of $\|(b_*P^{2k})(x, \cdot) - Q(b(x), \cdot)\|_{TV}$ over x; and the fibre equilibration θ_k, the largest of*

[61] R. L. Dobrushin, "Central limit theorem for nonstationary Markov chains. I, II", *Theory Probab. Appl.* 1 (1956), 65–80 and 329–383; I. Gelfand, "Normierte Ringe", *Mat. Sbornik* 9 (51) (1941), 3–24.

$\|P^{2k}(x,\cdot\mid y')-\pi(\cdot\mid y')\|_{TV}$ over x and the fibres y' that $P^{2k}(x,\cdot)$ charges, where $P^{2k}(x,\cdot\mid y')$ is $P^{2k}(x,\cdot)$ conditioned on $F_{y'}$. Let M_k be the 2×2 matrix with rows $(\delta(Q)+2\varepsilon_k,\ 2\theta_k)$ and $(\eta_k,\ 2\eta_k)$. Then

$$\delta(P^{2k}) \le \delta(Q) + 2\varepsilon_k + 2\theta_k + 6\eta_k, \quad (\lambda_1/\lambda_0)^{2k} \le \rho(M_k), \quad m_{lat} \ge \frac{1}{2k}\, log\, \frac{1}{\rho(M_k)}, \quad (8.2)$$

where $\rho(M_k)=\tfrac12[(d+2\eta_k)+\sqrt{((d-2\eta_k)^2+8\theta_k\eta_k)}]$ is the spectral radius of M_k and $d=\delta(Q)+2\varepsilon_k$.

Proof. For $g:X\to\mathbb{R}$ let $\bar g(y)=\Sigma_{x\in b^{-1}(y)}\,\pi(x\mid y)\,g(x)$, $a=\operatorname{osc}(\bar g)$, $s=\max_y \operatorname{osc}(g \text{ restricted to } F_y)$ and $r=g-\bar g\circ b$, so that $|r|\le s$ and r has mean zero under every $\pi(\cdot\mid y)$. First, $P^{2k}(\bar g\circ b)(x)=\Sigma_{y'}\,(b_*P^{2k})(x,y')\,\bar g(y')=(Q\bar g)(b(x))+e_1(x)$, where $e_1(x)$ integrates $\bar g$ against a signed measure of total mass zero and total variation at most ε_k, so $|e_1|\le\varepsilon_k a$. Second, $P^{2k}r(x)=\Sigma_{y'}\,(b_*P^{2k})(x,y')\,\Sigma_{x'\in b^{-1}(y')}\,P^{2k}(x,x'\mid y')\,r(x')$, and because r has mean zero under $\pi(\cdot\mid y')$ the inner sum equals $\Sigma_{x'}\,[P^{2k}(x,x'\mid y')-\pi(x'\mid y')]\,r(x')$, of modulus at most $\theta_k s$; so $e_2=P^{2k}r$ satisfies $|e_2|\le\theta_k s$. Hence $P^{2k}g=(Q\bar g)\circ b+e_1+e_2$.

For $g'=P^{2k}g$ write a' and s' for its two oscillations. The fibre mean of g' is $Q\bar g$ plus a function bounded by $\varepsilon_k a+\theta_k s$, and $\operatorname{osc}(Q\bar g)\le\delta(Q)\operatorname{osc}(\bar g)$, so $a'\le(\delta(Q)+2\varepsilon_k)a+2\theta_k s$. For x,x' in one fibre, $|g'(x)-g'(x')|\le\eta_k\operatorname{osc}(g)\le\eta_k(a+2s)$, so $s'\le\eta_k(a+2s)$. Finally $\operatorname{osc}(g')\le a'+2s'$. Since $\delta(P^{2k})$ is the supremum of $\operatorname{osc}(P^{2k}g)$ over $\operatorname{osc}(g)\le 1$, and $\operatorname{osc}(g)\le 1$ gives $a,s\le 1$, the first inequality of (8.2) follows. The entries of M_k are non-negative, so the componentwise bounds $(a',s')\le M_k(a,s)$ chain: applying P^{2k} j times with the same k at every step, against the same Q, so that the rung quantities are fixed, gives $\operatorname{osc}(P^{2kj}g)\le(1,2)\,M_k^{\,j}\,(1,1)^{\mathrm T}$ for $\operatorname{osc}(g)\le 1$. Proposition 8.1 applied to P^{2kj} then gives $(\lambda_1/\lambda_0)^{2kj}\le\delta(P^{2kj})\le(1,2)\,M_k^{\,j}\,(1,1)^{\mathrm T}\le C\cdot j\cdot\rho(M_k)^j$, the factor j present only where M_k is not diagonalisable, and the j-th root removes it: this is the second inequality, of which the third is the logarithm. ∎

The lemma carries its own scope. The lumping defect accumulates over the coarse chain's mixing time, since $\varepsilon_k\le k\varepsilon_1$ by telescoping and $\varepsilon_1\le\eta_1$, so on a chain of n levels down to the mixing scale the level m must certify $\varepsilon_m 2^{n-m}<1$. The instrument is therefore finite: it reaches a gap through finitely many certified levels, and the continuum end of the sequence lies beyond it, with the renormalised trajectory of §8.1. Its place in Theorem 7.4 is exact: the inequality $m_{lat}(j)\ge m\,a_j$ at rung j holds as soon as one k and one block map certify $\rho(M_k)\le e^{-2k\,m\,a_j}$ there. Read in double precision on the decimation of the Z_2 transfer matrix of the 2×4 cross-section to 2×2, the lemma gives $m_{lat}\ge 1.538$ at $u=\tfrac13$ against the certified gap 3.831, and $m_{lat}\ge 0.867$ at $u=\tfrac12$ against the certified gap 1.763.[62]

Proposition 8.3 (The shadow ladder; Edwin Boston III). *Let G' be a finite subgroup of $SU(2)$ containing -1, weighted by Wilson's factor $w(g)=e^{b\chi(g)}$, $\chi=\operatorname{tr}$ the fundamental character, and let $c_\chi/c_0=\tfrac12\,\Sigma_g\,w(g)\,\chi(g)\,/\,\Sigma_g\,w(g)$ be its fundamental character coefficient; for $SU(2)$ itself the sums are Haar integrals. Let $m_n(G')$ be the moments of χ under the uniform measure on G' and t the largest integer with $m_k(G')=m_k(SU(2))$ for every $k\le t$. Then t is odd, and the functions $\Delta_{tube}(b)=-4\log(c_\chi/c_0)$ of G' and of $SU(2)$ satisfy*

$$\Delta_{tube}{}^{G'}(b) - \Delta_{tube}{}^{SU(2)}(b) = -\frac{4\delta}{t!}\,b^{t-1} + O(b^{t+1}), \quad \delta = m_{t+1}(G') - m_{t+1}(SU(2)). \quad (8.3)$$

The integer t is the design strength of G' as a set of points of the three-sphere $SU(2)$.

Proof. Since $-1\in G'$ and $\chi(-g)=-\chi(g)$, the odd moments vanish, so t is odd and every series below is even. Let $Z(b)=\Sigma_n\,m_n\,b^n/n!$ be the normalised partition function. Then $c_\chi/c_0=\tfrac12\,(\log Z)'(b)$, and since

[62] Edwin Boston III, the chain lemma; its record and run of record are listed in the Records.

$m_2 = 1$, $c_\chi/c_0 = (b/2)(1 + u(b))$ with $u(b) = O(b^2)$. The moments of G' and of SU(2) agree through order t, so $\log Z^{G'} - \log Z^{SU(2)} = \delta b^{t+1}/(t+1)! + O(b^{t+3})$. Since $u = (\log Z)'/b - 1$, differentiating and dividing by b gives $u^{G'} - u^{SU(2)} = \delta b^{t-1}/t! + O(b^{t+1})$, and since $\Delta_{\text{tube}} = -4\log(b/2) - 4\log(1 + u)$ with $u = O(b^2)$, (8.3) follows.

The moment $m_n(G')$ is the multiplicity of the trivial representation of G' in the n-th tensor power of the fundamental representation. By the Clebsch–Gordan rule the moments agree through order n exactly when no irreducible representation of SU(2) of spin j with $1 \le 2j \le n$ has a nonzero G'-invariant vector. The harmonic polynomials of degree $2j$ on the three-sphere are the matrix coefficients of spin j, and their average over G' is the projection onto the G'-invariant vectors, so the same condition says that G' is a design of strength n. ■

For the quaternion group Q_8 and the binary tetrahedral, octahedral and icosahedral groups 2T, 2O and 2I, Table 2 gives the exact values. The moments of SU(2) are the Catalan numbers 1, 2, 5, 14, 42, 132 at $n = 2, 4, \ldots, 12$; those of the four groups are 1, 4, 16, 64, 256, 1024; 1, 2, 6, 22, 86, 342; 1, 2, 5, 15, 51, 187; and 1, 2, 5, 14, 42, 133. The finite groups therefore approximate the strong-coupling gap of SU(2) to an order fixed by their design strength, with an exactly known first defect.[63]

Group	Order	Design strength t	$m_{t+1}(G')$ against SU(2)	δ	First defect $-4\delta/t!$
Q_8	8	3	4 against 2	2	−4/3
2T	24	5	6 against 5	1	−1/30
2O	48	7	15 against 14	1	−1/1260
2I	120	11	133 against 132	1	−1/9979200

Table 2. The shadow ladder: the order at which each finite subgroup's strong-coupling gap parts from that of SU(2), and the exact first defect.

The gap by counting; Edwin Boston III. For the Z_2 gauge theory in 2 + 1 dimensions on the periodic cross-sections 2×2, 2×3, 2×4 and 3×3 at couplings $e^{-2\beta}$ among ½, ⅓, ¼ and ⅕, fourteen cells in all, the two leading eigenvalues of the transfer matrix are bracketed by sign counts of integer matrices, by Sylvester's law of inertia, so that every gap is certified to thirteen or fourteen decimal places by counting alone, with no limit taken.[64] At all fourteen cells the brackets contain the eigenvalues computed to forty-five digits by a second engine that shares no computation with the first, and they meet a third, independently certified set of intervals. On the 3×3 cross-section at $e^{-2\beta} = ½$ the gap is 4.391458312669 in lattice units.

The gap through duality; Edwin Boston III. By Wegner's duality the gap of the Z_2 gauge theory in 2 + 1 dimensions at coupling β is the inverse correlation length of the three-dimensional Ising model at the dual coupling K^*, with $\tanh K^* = e^{-2\beta}$.[65] Edwin Boston III reads the first two coefficients of the strong-coupling expansion of the gap exactly on the 3×3 cross-section, meets through them the first

[63] Edwin Boston III, the shadow ladder; the record and its run of record are listed in the Records at the end of this paper.

[64] J. J. Sylvester, "A demonstration of the theorem that every homogeneous quadratic polynomial is reducible by real orthogonal substitutions to the form of a sum of positive and negative squares", *Philosophical Magazine* 4 (1852), 138–142. The computation is Edwin Boston III's; its record is listed in the Records.

[65] F. J. Wegner, "Duality in generalized Ising models and phase transitions without local order parameters", *J. Math. Phys.* 12 (1971), 2259–2272.

three coefficients of the low-temperature series of the Ising correlation length published by Arisue and Tabata, and, solving that published relation for the coefficients that no cross-section of side at most four can read, obtains the expansion on the infinite lattice[66]

$$\Delta(u) \;=\; -4\,\log u \;+\; u^2 \;-\; \tfrac{15}{2}u^4 \;+\; \tfrac{7}{3}u^6 \;-\; \tfrac{87}{4}u^8$$

$$+\; \tfrac{251}{5}u^{10} \;-\; \tfrac{331}{2}u^{12} \;+\; O(u^{14}), \quad u \;=\; \tanh\beta, \tag{8.4}$$

with all ten published coefficients met.

The transform on a tube; Edwin Boston III. For the SU(2) Wilson theory in 2 + 1 dimensions on the spatial torus of side two, in the temporal gauge, with the transfer matrix written in the spin-network basis truncated at spin $j \le 1$, a space of 2,018 states, and at $j \le 3/2$, a space of 33,342 states, Edwin Boston III reads the spectrum within the truncation and proves two facts about the operator. First, the inequality of Proposition 8.1 holds for its Doob chain, so that the gap is at least $(1/k)\log(1/\delta(P^k))$ for every k. Second, the centre flux through the two cycles of the torus commutes with the transfer matrix, so the chain of Proposition 8.2 lumps exactly by the flux, with lumping defect zero: the physical rate is the gap within the sector of zero flux, and the full gap is the least of the gaps between sectors. A tube is not $\mathbb{Z}^4$, and the theorem supplies no uniformity in the cross-section; it is the first exact non-abelian instance of the structure the transform must carry. At $j \le 1$ and $\beta_3 = 2.3$ the top eigenvalue is 3.86580, the full gap 1.01886 is the torelon of unit flux, the zero-flux gap is 2.771287, and the plaquette correlation along the tube decays at the rates 2.849, 2.843 and 2.837 at separations 6, 7 and 8, falling toward the sector's gap; across β_3 from 0.2 to 6 the correlation decays at the sector's own rate, and at $\beta_3 = 0.2$ the rate 11.983 meets the strong-coupling value $-4\log(c_\chi/c_0) = 11.99$ to 0.06 per cent. The truncation is not small, and every number is carried at its cutoff: read at $j \le 3/2$, the rate at $\beta_3 = 2$ falls by 6 per cent, and at $\beta_3 = 2.3$ the zero-flux gap reads 2.452280 against 2.771287, the torelon 0.931023 against 1.018863, and the correlation's rates 2.590, 2.563, 2.552 and 2.546 at separations 5 to 8 fall toward the sector's gap as before.[67]

The lattice theorems, checked by machine; Edwin Boston III. In the Lean proof assistant, above the Mathlib library and with no unproved step, Edwin Boston III has formalised the following lattice theorems, and the Lean kernel has checked them.[68] A Lean formalisation of the strong-coupling gap of the U(n) theories by Dobrushin's method, at a coupling of order $1/(32n(d-1))$, is Douglas and Rajasekaran's (§8.3); the theorems below are a separate development. For the transfer matrix of every finite-group lattice theory: every even-time quadratic form is non-negative; the spectral sum holds; the dominant root is simple and strictly dominant, with its residual law; the string energy is the exact limit of two Perron roots; and there is a mass gap on every finite cross-section, at every coupling. For every finite group acting faithfully by a real orthogonal representation, on every finite cross-section, the static potential of a Wilson loop is at least the mass gap of its sector. For the Z_2

[66] H. Arisue and K. Tabata, "Low-temperature series for the correlation length in d = 3 Ising model", *Nucl. Phys.* B435 (1995), 555–566. The reading is Edwin Boston III's; its record is listed in the Records.

[67] Edwin Boston III, the transform on a tube; the instrument, its records at both cutoffs and the run of record are listed in the Records. The spin-3/2 record at $\beta_3 = 2.3$ is posted with its digest in the repository; its gate run follows the run of record named there.

[68] Edwin Boston III, the lattice theorems in Lean; the files, the theorem names and the run of record are listed in the Records.

theory: Griffiths' inequalities hold, with the monotonicity in the couplings; the infinite-volume Wilson expectation exists; the string tension exists, with its strong-coupling bound; and on a slab of $T + 1$ cross-sections of $\mathbb{Z}^{d-1}$ the mass gap is uniform in the cross-section and in T whenever $(3m + 2)\tanh \beta < 1$, where $m = 2(d - 2)$ is the number of in-plane plaquettes on a link, that is, for $\beta < 0.1257$ when $d = 3$ and $\beta < 0.0716$ when $d = 4$. Beside these stand the finite forms of the arguments of Sections 4 and 7: the bound that a gap imposes on every connected two-point function, the gap read off the decay, the definition of a physical gap along a scaling sequence, and Dobrushin's comparison theorem.

8.3 The reduction

The passage from the second link to the gap is itself a theorem, and this section proves it at the strength of its hypotheses. Fix a physical length $a_c = L^{k_0} a_0$ with $k_0 \geq 0$, an integer multiple of the unit a_0 of Theorem 7.4, L being the fixed integer of its construction, and call it the coarse scale. Take the construction with lattices whose sides are multiples of a_c, the time extent an even multiple when L is odd so that the half period lies on the lattice, and whose reflection hyperplane is one of the hyperplanes $x_0 = n\, a_c$. Every spacing a_j then divides a_c; the half-open cubes of side a_c with corners on the coarse lattice $\mathbb{Z}^4$ of spacing a_c, reduced modulo the sides, partition the lattice of rung j into blocks, each link assigned to the block containing its midpoint. Write B_y for the block at the coarse point y and $|y - y'| = \max_\nu |y_\nu - y'_\nu|$ for the distance on the coarse lattice; the four links of a plaquette lie in blocks at mutual distance at most one. For a bounded observable f, a function of the links, let $S(f)$ be the set of blocks containing a link on which f depends, $N(f)$ its cardinality, $\|f\|$ its supremum norm, and $\delta_y(f)$ the oscillation of f under changes of the links of B_y alone, so that $\delta_y(f) \leq 2\|f\|$ and $\delta_y(f) = 0$ off $S(f)$; the distance between two sets of blocks is the least distance between their members. For a smeared Wilson loop, for a polynomial in such loops, and for every observable whose support and bound are fixed in physical units, $N(f)$ and $\|f\|$ are bounded uniformly in j.

A block map is a measurable bijection of the configurations of a block onto a product $V \times \Phi_j$, with measurable inverse, the same for every block, in which V is a fixed compact space, the space of the coarse variables, and Φ_j is compact. One SU(2) element per coarse link, the holonomy of the block along a fixed path with the remaining links in a fixed gauge, is such a map, and one on which the reduction below is empty. For the Gaussian field with covariance $g^2(-\Delta)^{-1}$ in Feynman gauge, the flux through an $n \times n$ square, the angle of the coarse plaquette this map defines, has variance $g^2(4G_3(0)n + o(n))$, where $4G_3(0) = 4\Sigma_{m\in\mathbb{Z}}\, G_4(m\, e_1) = 2W/3$, G_d is the inverse of the d-dimensional lattice Laplacian, and $W = \sqrt{6}\,\Gamma(1/24)\Gamma(5/24)\Gamma(7/24)\Gamma(11/24)/(32\pi^3)$ is the return sum of the three-dimensional walk. Indeed the variance is g^2 times the sum of G_4 over pairs of boundary links of the square, weighted by their orientations; perpendicular sides do not couple, each side with itself gives $S_n = \Sigma_{|m|<n}(n - |m|)G_4(m\, e_1)$, the opposite sides give $-T_n$ with $0 \leq T_n \leq n\, G_3(n\, e_2)$, which is bounded, and $S_n/n \to G_3(0)$ by dominated convergence, $G_4(m\, e_1)$ being summable. Along the asymptotically free sequence, where g^2_{YM} falls only as $1/\log n$, this variance diverges, so at one loop every character mean of the path holonomy tends to zero: its coarse variables tend to independent Haar variables, Hypothesis I below then holds trivially, Hypothesis II(a) becomes the clustering of the Wilson measure itself, and the theorem reduces nothing. That the same holds beyond one loop is the self-energy divergence of bare Wilson lines, which is standard and is not proved here for this map. The block average, in which each coarse link is the average over the n^3 parallel fine paths of its block, is

free of this: in the same Gaussian field the flux of its coarse plaquette has variance converging, as $n \to \infty$, to g^2 times the finite integral $2\int |\Pi_v\, \varphi(q_v)|^2\, |1 - e^{iq_z}|^2\, |q|^{-2}\, d^4q/(2\pi)^4$, with $\varphi(q) = (e^{iq} - 1)/(iq)$, the integrand being dominated by an integrable function on $\mathbb{R}^4$; along the sequence it falls like $1/\log n$, a weak-coupling variable, as the coarse variables of Bałaban's renormalisation group are. The proof of Theorem 8.5 below uses the block map only through the disintegration of μ_j along the coarse configuration and through versions of the fibre and blocked conditional laws defined for every coarse configuration, and it holds unchanged for any measurable map from the configurations of a rung to coarse configurations with values in V, with Hypotheses I and II read for such versions. This admits the averaged variables of Bałaban's renormalisation group, on blocks of odd side centred on the coarse points, whose coarse coordinate at a coarse point depends also on links of the adjacent blocks, once each is extended measurably to the configurations at which its average vanishes; so the reduction applies to the coarse variables of the ultraviolet analysis of §8.1 itself. Write $v_y \in V$ and $\varphi_y \in \Phi_j$ for the coordinates of the links of B_y. The coarse coordinate of a block is a function of that block's links alone, and a change of it at fixed fine coordinates changes the links of that block alone; so a change of one coarse variable alters the Wilson conditionals at that block and at the blocks adjacent to it, and nowhere else. The blocked measure λ_j is the law of the coarse configuration $v = (v_y)$ under the Wilson measure μ_j, and the fibre measure $\nu_j(\cdot \mid v)$ is the conditional law of the fine coordinates $\varphi = (\varphi_y)$ given v: the Gibbs measure on the fibres whose conditional at a block is the Wilson conditional of that block restricted to its fibre, defined for every v. The blocked measure has conditional laws $\lambda_j(dv_y \mid v)$ given the other coarse variables, and Dobrushin's interdependence matrix of the blocked measure is

$$C_j^{\lambda}(y, y') \;=\; sup\, \|\lambda_j(\cdot \mid v) \;-\; \lambda_j(\cdot \mid v')\|_{TV}, \tag{8.5}$$

the supremum over pairs v, v' of coarse configurations that differ only at y', and $\|\cdot\|_{TV}$ the total variation distance of the two conditional laws of v_y. The coarse variables are of one type at every rung, so the total variation in (8.5) is taken on one fixed space for every j.

Proposition 8.4 (The tilt lemmas). *Let p and p' be probability measures on a compact space X, ρ a bounded measurable function on X with oscillation $\operatorname{osc}\rho = \sup\rho - \inf\rho$, and $T_\rho p = e^\rho p \,/\, p(e^\rho)$ the tilt of p by ρ. Then*

$$\|T_\rho p \;-\; p\|_{TV} \;\le\; 1 \;-\; e^{-\operatorname{osc}\rho}, \qquad \|T_\rho p \;-\; T_\rho p'\|_{TV} \;\le\; e^{\operatorname{osc}\rho}\, \|p \;-\; p'\|_{TV}. \tag{8.6}$$

Consequently, let W be a positive weight on the configurations of finitely many sites with compact spin spaces, with interdependence matrix C, and δ a bounded function of the configuration; let $o_y(\delta)$ be the supremum, over configurations, of the oscillation of δ in the variable at y, and $\Delta_{yy'}(\delta)$ the supremum, over pairs of configurations differing only at y', of the oscillation in the variable at y of the difference of the two restrictions of δ. Then the weight We^δ has interdependence matrix at most

$$e^{o_y(\delta)}\, C(y, y') \;+\; 1 \;-\; e^{-\Delta_{yy'}(\delta)}. \tag{8.7}$$

Proof. Normalise ρ so that its infimum is 0 and its supremum is $o = \operatorname{osc}\rho$; the tilt is unchanged. Then e^ρ takes values in $[1, e^o]$ and $A = p(e^\rho)$ lies in the same interval, so $T_\rho p$ has density $e^\rho/A \ge e^{-o}$ with respect to p; for every measurable set B, $p(B) - (T_\rho p)(B) \le (1 - e^{-o})\, p(B) \le 1 - e^{-o}$, and the supremum over B is the first inequality. For the second, let $p_u = p + u(p' - p)$ for $0 \le u \le 1$ and $q_u = T_\rho p_u$, with $A_u = p_u(e^\rho) \ge 1$. For a measurable set B, $q_u(B) = p_u(e^\rho 1_B)/A_u$, and differentiating in u,

$$\frac{d}{du}\, q_u(B) \;=\; \frac{1}{A_u} \int\, (1_B \;-\; q_u(B))\, e^{\rho}\, d(p' \;-\; p).$$

The integrand $(1_B - q_u(B))\, e^{\rho}$ takes values in $[1 - q_u(B), (1 - q_u(B))e^{o}]$ on B and in $[-q_u(B)e^{o}, -q_u(B)]$ off B, so its oscillation is at most e^{o}; and the integral of a function of oscillation at most c against a signed measure of total mass zero is at most c times half the total variation of that measure, that is, $c\|p' - p\|_{TV}$. Hence $|dq_u(B)/du| \le e^{o}\|p' - p\|_{TV}$, and integrating from 0 to 1 gives $|(T_\rho p')(B) - (T_\rho p)(B)| \le e^{o}\|p' - p\|_{TV}$ for every B, which is the second inequality. For the last assertion, fix a site y and a configuration ω of the other sites, and let p_ω be the conditional law of the variable at y under W; the conditional law under $W e^{\delta}$ is $T_\rho p_\omega$ with ρ the restriction of δ to the variable at y at ω, of oscillation at most $o_y(\delta)$. If ω' differs from ω only at y', with restriction ρ', then $T_{\rho'} p_{\omega'} = T_{\rho' - \rho}(T_\rho p_{\omega'})$, so by the triangle inequality and the two inequalities just proved, $\|T_\rho p_\omega - T_{\rho'} p_{\omega'}\|_{TV} \le e^{o}\|p_\omega - p_{\omega'}\|_{TV} + 1 - e^{-\mathrm{osc}(\rho' - \rho)} \le e^{o} C(y, y') + 1 - e^{-\Delta}$, with $o = o_y(\delta)$ and $\Delta = \Delta_{yy'}(\delta)$. ■

Hypothesis I (The blocked bound). *There are numbers $\kappa > 0$ and $\alpha < 1$ and a rung j_0 such that at every rung $j \ge j_0$, on every lattice of the construction, $\Sigma_{y'} C_j(y, y')\, e^{\kappa|y - y'|} \le \alpha$ for every coarse point y.*

Hypothesis II (The fibre clustering). *There are a number $\rho' < 1$, a constant K' and a rung j_0 such that at every rung $j \ge j_0$, on every lattice of the construction, for every coarse configuration v and all bounded observables f and g, with d the distance between $S(f)$ and $S(g)$:*

(a) $|\nu_j(fg \mid v) - \nu_j(f \mid v)\, \nu_j(g \mid v)| \le K' N(f)\, \|f\|\, \|g\|\, \rho'^{d}$;

(b) the fibre mean $F(v) = \nu_j(f \mid v)$, as a function of the coarse configuration, satisfies $\delta_y(F) \le K' N(f)\, \|f\|\, \rho'^{r}$ for every coarse point y, where r is the distance from y to $S(f)$.

Hypothesis I is the second link of §8.1 in its finite form: a statement about the conditional law of one coarse variable at one fixed physical scale, with the exponential weight that fixes the range, and with constants independent of the cutoff and of the volume. Hypothesis II is the ultraviolet half of the chain at the scales below a_c: with the coarse variables held fixed, the field inside the blocks clusters at the block scale, and its means depend on distant coarse variables only through an exponentially small tail. It is demanded for every coarse configuration, the adversarial ones included, because a single-site criterion conditions on the worst case; this is where it exceeds the small-field expansions, which control the fluctuation field in typical backgrounds and the large fields by their probability, and it is named here as the demand it is. It cannot be weakened to typical coarse configurations with the frustrated ones priced by their probability. The Ising model at low temperature on the tori $(\mathbb{Z}/L\mathbb{Z})^{d}$, $d \ge 2$, satisfies every such form with both prices as small as one likes, and has long-range order. On the event that the $2(2d - 1)$ outer neighbours of an adjacent pair carry one sign, the maximal correlation of the pair given the rest is $\sinh 2\beta / (\cosh 2(2d - 1)\beta + \cosh 2\beta) \le e^{-4(d-1)\beta}$, every function of a two-valued variable being affine. The complementary event lies in the union of the $4d - 1$ events that one bond of the pair's star is unsatisfied, and by the chessboard estimate its probability is at most $(4d - 1) \cdot 2^{1/L} e^{-2\beta}$, with chessboard weight at most $2^{3^{\wedge}d} e^{-2\beta / c_d}$ for a constant c_d depending on d alone. Peierls' argument gives $\sigma_0 \sigma_x$ mean at least ½ for every x and L once β is large.[69] The chessboard

[69] R. Peierls, "On Ising's model of ferromagnetism", *Proc. Cambridge Philos. Soc.* 32 (1936), 477–481; R. B. Griffiths, "Peierls proof of spontaneous magnetization in a two-dimensional Ising ferromagnet", *Phys. Rev.* 136 (1964), A437–A439; J. Fröhlich, R. Israel, E. H. Lieb and B. Simon, "Phase transitions and reflection positivity. I", *Comm. Math. Phys.* 62 (1978), 1–34, for the chessboard estimate and its use on tori.

estimate that would price the frustrated configurations is the instrument that proves the order. More generally, no condition that depends on a measure only through the joint law of its single-pair conditional coefficients implies decay of correlations. In infinite volume μ^+ and $\tfrac{1}{2}(\mu^+ + \mu^-)$ are Gibbs measures of one specification and give that law the same distribution, a maximal correlation being invariant under a bijection of each variable's values, while only the first has decaying truncated correlations. The supremum in (8.5) runs over exactly the configurations that Dobrushin's comparison visits when it telescopes between two environments typical of different phases, every intermediate one frustrated. Neither hypothesis refers to a limit. Their consequence is the gap.

Theorem 8.5 (The reduction of the gap to the blocked bound). *Let a construction be as in Theorem 7.4, blocked at the coarse scale a_c with a block map as above, and suppose Hypotheses I and II. Put $\rho = \max(e^{-\kappa}, \rho')$. Then for every ρ_1 with $\rho < \rho_1 < 1$ there is a constant K, depending on ρ, ρ_1, α, K', $N(f)$, $N(g)$, $\|f\|$ and $\|g\|$ and on nothing else, such that at every rung $j \geq j_0$, on every lattice of the construction, for all bounded observables f and g, $|S_j(fg) - S_j(f)\,S_j(g)| \leq K\,\rho_1^{\,d}$, with d the distance between $S(f)$ and $S(g)$; and every limit point of the construction's Schwinger functions, on the sector generated by the observables that are bounded uniformly in j, has a unique vacuum and a Hamiltonian with no spectrum in $(0, m)$, where*

$$m \;=\; \frac{1}{a_c}\,\min\Big(\,\kappa, \log\,\frac{1}{\rho'}\,\Big). \tag{8.8}$$

Proof. Fix $j \geq j_0$, a lattice, and bounded observables f and g; write $F(\nu) = \nu_j(f \mid \nu)$ and $G(\nu) = \nu_j(g \mid \nu)$ for their fibre means, bounded by $\|f\|$ and $\|g\|$. The covariance splits along the block map: $\mu_j(fg) - \mu_j(f)\mu_j(g) = [\lambda_j(FG) - \lambda_j(F)\lambda_j(G)] + \lambda_j(\nu_j(fg \mid \cdot) - FG)$. The second term is the λ_j-mean of the covariance of f and g under the fibre measure, at most $K'N(f)\|f\|\|g\|\rho'^{d}$ by Hypothesis II(a).

For the first term put $G' = G + 2\|g\|$, with values between $\|g\|$ and $3\|g\|$, and $\lambda' = G'\lambda_j/\lambda_j(G')$. The blocked measure is a finite system of coarse variables with compact values, and Dobrushin's comparison theorem applies to it:[70] if λ' has conditional laws that differ from those of λ_j at y by at most b_y in total variation, for every conditioning, then for every bounded function F of the coarse configuration, $|\lambda'(F) - \lambda_j(F)| \leq \Sigma_{i,y}\,\delta_i(F)\,D(i,y)\,b_y$, where $D = \Sigma_{n\geq 0}\,(C^{\lambda_j})^n$. The conditional law of λ' at y is that of λ_j tilted by $\log G'$ as a function of ν_y, so by the first inequality of Proposition 8.4 the two differ by at most $b_y = 1 - e^{-\mathrm{osc}} \leq \mathrm{osc} \leq \delta_y(G')/\inf G' \leq \delta_y(G)/\|g\| \leq K'N(g)\,\rho'^{r'}$, where osc is the oscillation of $\log G'$ in ν_y and r' the distance from y to $S(g)$, by Hypothesis II(b). Under Hypothesis I the weighted matrix $\tilde{C}(y,y') = C^{\lambda_j}(y,y')\,e^{\kappa|y-y'|}$ has row sums at most α, and since $|y - y''| \leq |y - y'| + |y' - y''|$, induction on n gives $(C^{\lambda_j})^n(y,y')\,e^{\kappa|y-y'|} \leq \tilde{C}^n(y,y') \leq \alpha^n$; so $D(y,y') \leq e^{-\kappa|y-y'|}/(1-\alpha)$. With $\delta_i(F) \leq K'N(f)\|f\|\rho'^{r}$ by Hypothesis II(b), r the distance from i to $S(f)$, the comparison theorem gives

$$|\,\lambda'(F) \;-\; \lambda_j(F)\,| \;\leq\; \textstyle\sum_{i,y}\,\delta_i(F)\,D(i,y)\,b_y \;\leq\; \dfrac{K'^2\,N(f)\,N(g)\,\|f\|}{1-\alpha}\,\sum_{i,y}\,\rho^{\,e(i,y)},$$

where $e(i,y) = r_i + |i - y| + r'_y$, with r_i the distance from i to $S(f)$ and r'_y that from y to $S(g)$, and $\rho = \max(e^{-\kappa}, \rho')$ bounds the base of each of the three factors. By the triangle inequality $e(i,y) \geq d$, and the number of pairs (i,y) with $e(i,y) \leq n$ is at most $N(f)N(g)(2n+1)^8$, since i then lies within distance n of

[70] R. L. Dobrushin, "Prescribing a system of random variables by conditional distributions", *Theory Probab. Appl.* 15 (1970), 458–486, where the uniqueness theorem and the comparison theorem are proved for arbitrary single-site spaces; the comparison theorem is used here in the form of Theorem 8.20 of H.-O. Georgii, *Gibbs Measures and Phase Transitions*, second edition, de Gruyter, Berlin, 2011. Its finite form, for finite single-site spaces, is among the theorems checked by machine in §8.2.

$S(f)$ and y within distance n of $S(g)$. Hence the last sum is at most $N(f)N(g)\,\Sigma_{n\ge d}\,(2n+1)^8\,\rho^n$, which is ρ^d times a polynomial in d and is therefore at most $K_1\rho_1{}^d$ for every $\rho_1 \in (\rho, 1)$, with K_1 depending on ρ, ρ_1, $N(f)$ and $N(g)$. Since $\lambda_j(FG) - \lambda_j(F)\lambda_j(G) = \lambda_j(G')(\lambda'(F) - \lambda_j(F))$ and $\lambda_j(G') \le 3\|g\|$, the first term is at most $3K'^2N(f)N(g)\|f\|\,\|g\|K_1\rho_1{}^d/(1-\alpha)$. Adding the second term gives the covariance inequality, with $K = 3K'^2N(f)N(g)\|f\|\,\|g\|K_1/(1-\alpha) + K'N(f)\|f\|\,\|g\|$.

For the gap, let f be an observable, bounded uniformly in j, whose links lie in the time interval $[0, w]$. The links of $\theta\bar f$ lie in $[-w, 0]$ and those of $\tau_t f$ in $[t, t+w]$, and the block boundaries in time are the multiples of a_c, so the distance between $S(\theta\bar f)$ and $S(\tau_t f)$ is at least $\lfloor t/a_c \rfloor \ge t/a_c - 1$. The covariance inequality gives $S_j(\theta\bar f\cdot\tau_t f) - |S_j(f)|^2 \le (K/\rho_1)\,e^{-m_1 t}$ with $m_1 = \log(1/\rho_1)/a_c$, the constant independent of j and of the lattice because $N(f)$ and $\|f\|$ are. This is the hypothesis of Theorem 7.4 with $r_j = 0$ at every $t \in D$, so Theorem 7.4 leaves no spectrum in $(0, m_1)$ for every $\rho_1 \in (\rho, 1)$, hence none in $(0, m)$ with $m = \log(1/\rho)/a_c$, which is (8.8). ■

The reduction has a second form, in the currency in which the gap itself is an exact quantity. It replaces the supremum of Hypothesis II by an average over the blocked measure and the total variation of (8.5) by a maximal correlation, and its conclusion is that of Theorem 8.5.

Proposition 8.1′ (The gap is an L^2 norm). *Let T be a transfer operator on a compact space X with kernel $T(x, dy) = t(x, y)\,dy$, t continuous, strictly positive and symmetric, and T positive as an operator on $L^2(dy)$; let $\lambda_0 > \lambda_1$ be its two largest eigenvalues, ψ_0 the positive eigenfunction normalised in $L^2(dy)$, $\pi(dx) = \psi_0(x)^2\,dx$, and P the Doob transform of Proposition 8.1. Then P is self-adjoint on $L^2(\pi)$; on $L^2{}_0(\pi)$, the functions of π-mean zero, $\|P^k\| = (\lambda_1/\lambda_0)^k$ for every $k \ge 1$, so that $m_{lat} = -\log \|P\|_{L^2{}_0(\pi)}$; $\|P^k\|_{L^2{}_0(\pi)} \le \delta(P^k)$; and, with the eigenvalues λ_i of T counted with multiplicity,*

$$\int \chi^2(P^k(x,\cdot)\,\|\,\pi)\,\pi(dx) \;=\; Tr\,T^{2k}/\lambda_0{}^{2k} \;-\; 1 \;=\; \Sigma_{i\ge 1}\,(\lambda_i/\lambda_0)^{2k}. \tag{8.8a}$$

Proof. For bounded u and w, $\int w\cdot Pu\,d\pi = \lambda_0{}^{-1}\iint w(x)\psi_0(x)\,t(x,y)\,\psi_0(y)u(y)\,dx\,dy$, which is symmetric in u and w; so P is self-adjoint on $L^2(\pi)$. Multiplication by ψ_0 is a unitary map U of $L^2(\pi)$ onto $L^2(dx)$, ψ_0 being continuous and bounded below on the compact space X, and $(T/\lambda_0)(\psi_0 u) = \psi_0\,Pu$, that is $UPU^{-1} = T/\lambda_0$. The map U carries the constants to ψ_0 and $L^2{}_0(\pi)$ onto the orthogonal complement of ψ_0, so P on $L^2{}_0(\pi)$ is unitarily equivalent to T/λ_0 on that complement, a compact positive operator whose norm is its largest eigenvalue λ_1/λ_0; powers pass through the equivalence. For an eigenvalue $\lambda \ne 0$ of T with eigenfunction ψ, the eigenfunction ψ/ψ_0 of P is continuous, hence bounded, and it is nonconstant when $\lambda \ne \lambda_0$, so the oscillation argument of Proposition 8.1 gives $|\lambda/\lambda_0|^k \le \delta(P^k)$ for every eigenvalue $\lambda \ne \lambda_0$ of T; the norm of the self-adjoint P^k on $L^2{}_0(\pi)$ is the largest of these numbers. Finally $P^k(x, dy)$ has density $p_k(x,y) = t_k(x,y)/(\lambda_0{}^k\,\psi_0(x)\psi_0(y))$ with respect to π, t_k the kernel of T^k, so $\iint p_k{}^2\,d\pi\,d\pi = \lambda_0{}^{-2k}\iint t_k(x,y)^2\,dx\,dy = \lambda_0{}^{-2k}\,\mathrm{Tr}\,T^{2k} = \Sigma_i\,(\lambda_i/\lambda_0)^{2k}$, the operator T^k being Hilbert–Schmidt with continuous symmetric kernel; since $\int p_k(x,y)\,\pi(dy) = 1$ for every x, $\iint (p_k - 1)^2\,d\pi\,d\pi = \mathrm{Tr}\,T^{2k}/\lambda_0{}^{2k} - 1$, and the inner integral is $\chi^2(P^k(x,\cdot)\,\|\,\pi)$. ■[71]

[71] The Doob transform and the averaged χ^2 identity for reversible chains are classical: L. Saloff-Coste, "Lectures on finite Markov chains", *Lectures on Probability Theory and Statistics (Saint-Flour, 1996)*, Lecture Notes in Math. 1665, Springer, Berlin, 1997, 301–413; D. A. Levin, Y. Peres and E. L. Wilmer, *Markov Chains and Mixing Times*, second edition, American Mathematical Society, Providence, 2017, chapter 12. That the subdominant spectrum is bounded by Dobrushin's coefficient is classical as well: E. Seneta, *Non-negative Matrices and Markov Chains*, second edition, Springer, New York, 1981; R. L. Dobrushin, "Central limit theorem for nonstationary Markov chains. I, II", *Theory Probab. Appl.* 1 (1956), 65–80 and 329–383.

The average in (8.8a) is at least the multiplicity of λ_1 times $(\lambda_1/\lambda_0)^{2k}$, so it controls the gap only once the decay beats that multiplicity; the norm carries no multiplicity, and Proposition 8.1 dominates it by a supremum over pairs of starting points. The reduction below is stated in the same currency.

For the reduction in this currency the blocked measure is assumed to have the form

$$\lambda_j(dv) \;=\; e^{-A_j(v)} \prod_{y\in Y} m(dv_y), \tag{S}$$

with m a Borel probability measure on V, Y the set of coarse points of the lattice, and A_j a measurable function finite $\prod m$-almost everywhere: the effective action at the coarse scale. A block map is translation covariant when it commutes with the translations of the lattice by multiples of a_c; its blocked measure is then invariant under the translations of Y. For a rung j and a lattice of the construction, the inner product of $L^2(\lambda_j)$ is $\langle F, G\rangle = \lambda_j(\bar{F}G)$; in Lemmas 8.5A and 8.5B, Proposition 8.5C, Theorem 8.5D and Proposition 8.5E the rung and the lattice are fixed. For $y \in Y$ let H_y be the closed subspace of $L^2(\lambda_j)$ of functions of $v_{y^c} = (v_z)_{z\neq y}$, K_y the orthogonal projection onto it, which is the conditional expectation $K_yF = \lambda_j(F \mid v_{y^c})$, and $Q_y = I - K_y$. The L^2 local fluctuation of $F \in L^2(\lambda_j)$ at y is

$$\ell_y(F) \;=\; \|Q_yF\| \;=\; \lambda_j(Var_{\lambda_j}(F \mid v_{y^c}))^{1/2};$$

it vanishes unless F depends on v_y, it equals $\ell_y(\bar{F})$, and for bounded F it is at most $\delta_y(F)/2$ when F is real and at most $\delta_y(F)/\sqrt{2}$ in general, with $\delta_y(F)$ the oscillation of F under changes of v_y alone, a real variance being at most a quarter of the squared oscillation and the variance of a complex function the sum of the variances of its real and imaginary parts, each of oscillation at most $\delta_y(F)$. For distinct y, y' write $\xi = (v_z)_{z\neq y,y'}$ for the coarse environment of the pair, π_ξ for a version of the conditional law of $(v_y, v_{y'})$ given ξ, and

$$\rho_\xi(y,y') \;=\; sup\,\{\,|Cov_{\pi_\xi}(f(v_y), g(v_{y'}))| : Var_{\pi_\xi} f \;=\; Var_{\pi_\xi} g \;=\; 1\,\}$$

for its maximal correlation, zero when v_y or $v_{y'}$ is almost surely constant under π_ξ, the supremum over real or over complex functions being the same; it is a countable supremum over f and g in a uniformly dense subset of the continuous functions on V, hence measurable in ξ. The conditional maximal correlation matrix of the blocked measure is

$$R_j(y,y') \;=\; ess\ sup_\xi\ \rho_\xi(y,y'), \quad R_j(y,y) \;=\; 0, \tag{8.8b}$$

the essential supremum taken under λ_j. It is symmetric, its entries lie in $[0, 1]$, and it does not depend on the versions.[72]

Hypothesis I′ (The blocked bound in L^2). *There are numbers $\kappa > 0$ and $\alpha < 1$ and a rung j_0 such that at every rung $j \geq j_0$, on every lattice of the construction, $\Sigma_{y'} R_j(y,y')\, e^{\kappa|y-y'|} \leq \alpha$ for every coarse point y.*

[72]The maximal correlation is H. Gebelein's, "Das statistische Problem der Korrelation als Variations- und Eigenwertproblem und sein Zusammenhang mit der Ausgleichsrechnung", *Z. Angew. Math. Mech.* 21 (1941), 364–379, and A. Rényi's, "On measures of dependence", *Acta Math. Acad. Sci. Hungar.* 10 (1959), 441–451.

Hypothesis II′ (The fibre clustering, averaged over the blocked measure). *There are a number $\rho' < 1$, a constant K' and a rung j_0 such that at every rung $j \geq j_0$, on every lattice of the construction, for all bounded observables f and g, with d the distance between S(f) and S(g):*

(a) $|\lambda_j(\, \nu_j(fg \mid \cdot) - \nu_j(f \mid \cdot)\, \nu_j(g \mid \cdot)\,)| \leq K'\, N(f)\, \|f\|\, \|g\|\, \rho'^{d}$;

(b) the fibre mean $F = \nu_j(f \mid \cdot)$, as a function of the coarse configuration, satisfies $\ell_y(F) \leq K'\, N(f)\, \|f\|\, \rho'^{r}$ for every coarse point y, where r is the distance from y to S(f).

Both hypotheses involve the fibre measure and the conditional laws of λ_j only through λ_j-integrals and λ_j-essential suprema, so they do not depend on the versions. The one supremum in them is the essential supremum over the coarse environment in (8.8b): the conditional law of a pair of coarse variables is still taken at every environment, but the field inside the blocks enters it integrated with its true conditional weight, and in Hypothesis II′ an adversarial coarse configuration enters weighted by its probability. Neither hypothesis refers to a limit, and neither asks for a relation between rungs: each is used at each rung separately.

Hypothesis II implies Hypothesis II′ with the same ρ' and K', since the λ_j-mean of a function is at most its supremum and $\ell_y(F) \leq \delta_y(F)/\sqrt{2} \leq \delta_y(F)$. Hypothesis I implies Hypothesis I′ with the same κ and α when the block map is translation covariant, by Proposition 8.5E below; when it is not, Hypothesis I′ follows from Hypothesis I with $\tfrac{1}{2}(\alpha + \bar{\alpha})$ in place of α, provided the weighted column sums $\Sigma_{y'}\, C^{\lambda_j}(y', y) e^{\kappa|y-y'|}$ are at most some $\bar{\alpha}$ with $\alpha + \bar{\alpha} < 2$.

Lemma 8.5A (Two projections). *Let P and Q be orthogonal projections on a Hilbert space, N = ran P ∩ ran Q, P_N the orthogonal projection onto N, and $c = \|PQ - P_N\|$, the cosine of the Friedrichs angle between ran P and ran Q. Then for every vector F,*

$$\|(I - Q)PF\| \leq \|(I - Q)F\| + c\, \|(I - P)F\|.$$

Proof. By Halmos' theorem on two subspaces the space is the orthogonal sum of ran $P \cap$ ran Q, ran $P \cap$ ker Q, ker $P \cap$ ran Q, ker $P \cap$ ker Q and a generic part unitarily equivalent to $K \oplus K$, on which P acts as the matrix with rows $(I, 0)$, $(0, 0)$ and Q as the matrix with rows (C^2, CS), (CS, S^2), with $C = \cos\Theta$ and $S = \sin\Theta$ for a positive operator Θ on K with $0 < \Theta < \pi/2$ and trivial kernels of Θ and $\pi/2 - \Theta$; every summand reduces P and Q. On the four trivial summands $PQ - P_N$ vanishes, and on the generic part $P_N = 0$ and the norm of PQ is that of the row (C^2, CS), which is $\|C\|$; so $c = \|C\|$ when the generic part is not zero. On ran $P \cap$ ran Q, $(I - Q)PF = 0$; on ran $P \cap$ ker Q, $(I - Q)PF = F = (I - Q)F$; on the two summands inside ker P, $PF = 0$: on each the inequality holds with 0 in place of c. On the generic part write $F = (x, y)$. Then $I - Q$ is the matrix with rows $(S^2, -CS)$, $(-CS, C^2)$, so $(I - Q)PF = (S{\cdot}Sx, -C{\cdot}Sx)$, of norm $\|Sx\|$ since $C^2 + S^2 = I$, and $(I - Q)F = (S(Sx - Cy), -C(Sx - Cy))$, of norm $\|Sx - Cy\|$. Hence $\|(I - Q)PF\| = \|Sx\| \leq \|Sx - Cy\| + \|Cy\| \leq \|(I - Q)F\| + c\|y\|$, and $\|y\| = \|(I - P)F\|$. The summands are orthogonal and preserved by P and Q, so Minkowski's inequality over them gives the claim. ■[73]

Lemma 8.5B (The angle between two coarse variables). *For distinct y, y′ the cosine c(y, y′) of the Friedrichs angle between H_y and $H_{y'}$ is at most $R_j(y, y')$. Consequently, for every $F \in L^2(\lambda_j)$, $\ell_y(K_y F) = 0$ and*

[73] P. R. Halmos, "Two subspaces", *Trans. Amer. Math. Soc.* 144 (1969), 381–389; K. Friedrichs, "On certain inequalities and characteristic value problems for analytic functions and for functions of two variables", *Trans. Amer. Math. Soc.* 41 (1937), 321–364; for the angle and its equivalent forms, F. Deutsch, *Best Approximation in Inner Product Spaces*, Springer, New York, 2001, chapter 9.

$$\ell_{y\prime}(K_y F) \;\le\; \ell_{y\prime}(F) \;+\; R_j(y,y')\,\ell_y(F).$$

Proof. The intersection $H_y \cap H_{y'}$ contains every function of ξ. The cosine $c(y,y')$ is the supremum of $|\langle F, G\rangle|$ over unit vectors $F \in H_y$ orthogonal to $H_y \cap H_{y'}$ and $G \in H_{y'}$ orthogonal to $H_y \cap H_{y'}$. Such an F is a function of $(v_{y'}, \xi)$ orthogonal to every function of ξ, so $\lambda_j(F \mid \xi) = 0$: for λ_j-almost every ξ, $F(\cdot, \xi)$ has mean zero under the law of $v_{y'}$ under π_ξ. Likewise G is a function of (v_y, ξ) with $G(\cdot, \xi)$ of mean zero under the law of v_y under π_ξ. Disintegrating λ_j over ξ,

$$\langle F, G\rangle \;=\; \int \lambda_j(d\xi) \int \pi_\xi(dv_y\, dv_{y\prime})\, F(v_{y\prime}, \xi)\, G(v_y, \xi),$$

and the inner integral is at most $\rho_\xi(y,y')$ times the product of the norms of $F(\cdot,\xi)$ and $G(\cdot,\xi)$ in $L^2(\pi_\xi)$, by the definition of the maximal correlation. The Cauchy–Schwarz inequality over ξ gives $|\langle F, G\rangle| \le R_j(y,y')\|F\|\|G\|$, which is the first assertion. The second is $\|Q_y K_y F\| = 0$, and the third is Lemma 8.5A with $P = K_y$ and $Q = K_{y'}$. ∎

Proposition 8.5C (The fluctuation vector decays). *Let $\mathcal{L} = \Sigma_{y\in Y}\,(K_y - I)$, a bounded self-adjoint operator on $L^2(\lambda_j)$, and $P_t = e^{t\mathcal{L}}$ the semigroup of the heat-bath dynamics that resamples each coarse variable from its conditional law at rate one. Write $\ell(F) = (\ell_y(F))_{y\in Y}$. Then for every $F \in L^2(\lambda_j)$ and every $t \ge 0$, componentwise,*

$$\ell(P_t F) \;\le\; e^{-t(I\,-\,R_j)}\,\ell(F).$$

Proof. For $0 < h \le 1/|Y|$, $I + h\mathcal{L} = (1 - |Y|h)I + h\,\Sigma_y K_y$ with non-negative coefficients. Since ℓ_z is a seminorm, $\ell_z(K_z G) = 0$, and $\ell_z(K_y G) \le \ell_z(G) + R_j(y,z)\ell_y(G)$ for $y \ne z$ by Lemma 8.5B,

$$\ell_z((I \;+\; h\mathcal{L})G) \;\le\; (1 \;-\; |Y|h)\ell_z(G) \;+\; h(|Y| \;-\; 1)\ell_z(G) \;+\; h\,\Sigma_{y\neq z}\,R_j(y,z)\ell_y(G) \;=\; (((1 \;-\; h)I \;+\; hR_j)\ell(G))_z,$$

the matrix R_j being symmetric. The matrix $(1-h)I + hR_j$ has non-negative entries, so it preserves the componentwise order, and induction gives $\ell((I + t\mathcal{L}/n)^n F) \le ((1 - t/n)I + (t/n)R_j)^n\,\ell(F)$ for $n \ge t|Y|$. As $n \to \infty$, $(I + t\mathcal{L}/n)^n \to e^{t\mathcal{L}}$ in operator norm, $\mathcal{L}$ being bounded, each ℓ_z is continuous on $L^2(\lambda_j)$, and $((1 - t/n)I + (t/n)R_j)^n \to e^{-t(I-R_j)}$. ∎

Theorem 8.5D (The covariance theorem in L^2; Edwin Boston III). *Suppose that $\cap_{y\in Y} H_y$ consists of the constants and that the spectral radius $\rho(R_j)$ of R_j is less than one. Then for all F and G in $L^2(\lambda_j)$,*

$$|\lambda_j(FG) \;-\; \lambda_j(F)\lambda_j(G)| \;\le\; \Sigma_{x,y\in Y}\,\ell_x(F)\,D(x,y)\,\ell_y(G), \quad D \;=\; (I \;-\; R_j)^{-1} \;=\; \Sigma_{n\ge 0}\,{R_j}^n, \tag{8.8c}$$

with D entrywise non-negative, and

$$Var_{\lambda_j}(F) \;\le\; (1 \;-\; \rho(R_j))^{-1}\,\Sigma_{y\in Y}\,\lambda_j(\,Var_{\lambda_j}(F \mid v_{y^c})\,).$$

If moreover $\Sigma_{y'} R_j(y,y')\,e^{\kappa|y-y'|} \le \alpha < 1$ for every y, then $\rho(R_j) \le \alpha$ and $D(x,y) \le e^{-\kappa|x-y|}/(1-\alpha)$. Under (S) the intersection $\cap_y H_y$ consists of the constants.

Proof. The Dirichlet form of the dynamics is $E(F,G) = \langle F, -\mathcal{L}G\rangle = \Sigma_y \langle Q_y F, Q_y G\rangle$, so $|E(F,G)| \le \Sigma_y \ell_y(F)\ell_y(G)$. The kernel of $\mathcal{L}$ is $\cap_y H_y$, since $\mathcal{L}F = 0$ gives $\Sigma_y \|Q_y F\|^2 = \langle F, -\mathcal{L}F\rangle = 0$; it consists of the constants by assumption, so the spectral theorem for the bounded self-adjoint operator $\mathcal{L} \le 0$ gives $P_t G \to \lambda_j(G)$ in $L^2(\lambda_j)$ as $t \to \infty$. Since $\lambda_j(FG) = \langle \bar{F}, G\rangle$ and $(d/dt)\langle \bar{F}, P_t G\rangle = -E(\bar{F}, P_t G)$,

$$\lambda_j(FG) \;-\; \lambda_j(F)\lambda_j(G) \;=\; \int_0^\infty E(F, P_t G)\,dt,$$

and by Proposition 8.5C, with $\ell_y(\bar F) = \ell_y(F)$, $|E(\bar F,\ P_tG)| \le \Sigma_y\ \ell_y(F)(e^{-t(I-R_j)}\ell(G))_y$. The matrix R_j is symmetric with non-negative entries and spectral radius less than one, so $e^{-t(I-R_j)} = e^{-t}e^{tR_j}$ has non-negative entries and $\int_0^\infty e^{-t(I-R_j)}\,dt = (I - R_j)^{-1} = \Sigma_n\ R_j{}^n$; this is (8.8c). For the variance, $(d/dt)\|P_tF - \lambda_j(F)\|^2 = -2E(P_tF,\ P_tF) = -2|\ell(P_tF)|^2$, with $|\cdot|$ the euclidean norm on $\mathbb{R}^Y$; the componentwise domination of Proposition 8.5C by a non-negative vector gives $|\ell(P_tF)| \le |e^{-t(I-R_j)}\ell(F)| \le e^{-t(1-\rho(R_j))}|\ell(F)|$, the largest eigenvalue of the symmetric non-negative R_j being its spectral radius, and integration over $[0, \infty)$ gives $\mathrm{Var}(F) \le (1 - \rho(R_j))^{-1}|\ell(F)|^2$, which is the second inequality. Under the weighted row-sum bound put $\tilde R(y, y') = R_j(y, y')e^{\kappa|y-y'|}$, with row sums at most α; since $|y - y''| \le |y - y'| + |y' - y''|$, induction on n gives $R_j{}^n(y, y')e^{\kappa|y-y'|} \le \tilde R^n(y, y')$, and the row sums of $\tilde R^n$ are at most α^n, so $D(x, y)e^{\kappa|x-y|} \le \Sigma_n\ \alpha^n = 1/(1 - \alpha)$, while $\rho(R_j)$ is at most the largest row sum of R_j, hence at most α. Under (S), λ_j and $\prod m$ have the same null sets; a function in $\cap_y\ H_y$ agrees $\prod m$-almost everywhere, for each y, with a function not depending on v_y, and integrating out the coordinates one at a time with Fubini's theorem on the product measure shows that it is $\prod m$-almost everywhere constant. ■[74]

Proposition 8.5E (The currency inequality). *With versions of the conditional laws $\lambda_j(dv_y \mid v)$ defined for every coarse configuration, as in (8.5), for distinct y and y'*

$$R_j(y, y')^2 \ \le\ C_j{}^{\lambda}(y, y')\ C_j{}^{\lambda}(y', y), \quad \textit{so that} \quad R_j(y, y') \ \le\ \tfrac{1}{2}\,(\,C_j{}^{\lambda}(y, y') \ +\ C_j{}^{\lambda}(y', y)\,). \tag{8.8d}$$

For a translation-covariant block map, Hypothesis I therefore implies Hypothesis I′ with the same κ and α. When V has two points, at λ_j-almost every environment ξ at which both coordinates charge both points, $\rho_\xi(y, y')^2 = C_\xi(y \mid y')\ C_\xi(y' \mid y)$ in the notation of the proof: the inequality of the proof cannot be improved.

Proof. Fix $y \ne y'$ and an environment ξ, and write $X = v_y$, $X' = v_{y'}$. For λ_j-almost every ξ the single-site laws of (8.5) at the configurations (X, X', ξ) are conditional laws of π_ξ, and the operator M,

$$Mg(b) \ =\ \int\ \lambda_j(dv_y \mid X' \ =\ b, \xi) \int\ \lambda_j(dv_{y'} \mid X \ =\ v_y, \xi)\ g(v_{y'}),$$

which goes from X' to X and back, is a Markov kernel defined at every point, self-adjoint on $L^2(\pi_\xi')$, π_ξ' the law of X' under π_ξ, with norm $\rho_\xi(y, y')^2$ on the functions of mean zero: it is T^*T for the conditional expectation T from functions of X' to functions of X, whose norm on the functions of mean zero is the maximal correlation. Its kernel is the composition of two kernels whose Dobrushin coefficients are $C_\xi(y \mid y') = \sup_{b,b'} \|\lambda_j(dv_y \mid X' = b, \xi) - \lambda_j(dv_y \mid X' = b', \xi)\|_{TV} \le C\lambda_j(y, y')$ and $C_\xi(y' \mid y) \le$

[74]The variance inequality of Theorem 8.5D, in the sharper quadratic form $\mathrm{Var}(F) \le \ell(F)^T(I - R_j)^{-1}\ell(F)$, is M. Kassabov's theorem on arrangements of subspaces, "Subspace arrangements and property T", *Groups Geom. Dyn.* 5 (2011), 445–477, Theorem 5.1, read for the arrangement of the subspaces H_y, whose pairwise angles there are Friedrichs' angles, the intersections factored out (Lemma 8.5B). For the heat-bath dynamics of a Gibbs measure, R. Peyre, "Tensorizing maximal correlations", arXiv:1004.1602 (2010), Corollary 4.2.12, gives the spectral gap $(1 - \|\varepsilon\|)_+^2$ for a matrix ε bounding the maximal correlations of two sites under the conditional laws given every subset of the other sites. For two blocks the rate is the squared maximal correlation between them: J. S. Liu, W. H. Wong and A. Kong, "Covariance structure of the Gibbs sampler with applications to the comparisons of estimators and augmentation schemes", *Biometrika* 81 (1994), 27–40; Y. Amit, "On rates of convergence of stochastic relaxation for Gaussian and non-Gaussian distributions", *J. Multivariate Anal.* 38 (1991), 82–99, which computes such rates as products of projections through the angles between subspaces, and "Convergence properties of the Gibbs sampler for perturbations of Gaussians", *Ann. Statist.* 24 (1996), 122–140. The covariance bound in total variation, with Dobrushin's oscillations in place of the local fluctuations and his interdependence matrix in place of R_j, is H. Föllmer's, "A covariance estimate for Gibbs measures", *J. Funct. Anal.* 46 (1982), 387–395, and H. Künsch's, "Decay of correlations under Dobrushin's uniqueness condition and its applications", *Comm. Math. Phys.* 84 (1982), 207–222. The two-sided covariance bound (8.8c) in L^2, with the exponential decay of $(I - R_j)^{-1}$ under the weighted row sums, is this paper's; it does not follow from the quadratic form, which by polarisation loses the decay, and it is the form Theorem 8.5′ uses.

$C^{\lambda_j}(y', y)$, and Dobrushin's coefficient is submultiplicative, so $\delta(M) \le C^{\lambda_j}(y, y')C^{\lambda_j}(y', y)$. It remains to show that the norm of M on the functions of mean zero is at most $\delta(M)$. For a bounded g of mean zero and $n \ge 1$ the oscillation argument of Proposition 8.1 gives $\operatorname{osc}(M^n g) \le \delta(M)^n \operatorname{osc}(g)$, and $M^n g$ has mean zero, so its L^2 norm is at most $\sup|M^n g| \le \operatorname{osc}(M^n g)$. If the spectral projection E of the self-adjoint M, restricted to the functions of mean zero, onto $\{|\lambda| \ge \delta(M) + \varepsilon\}$ were not zero for some $\varepsilon > 0$, there would be a bounded g of mean zero with $Eg \ne 0$, the bounded functions being dense, and $\|M^n g\| \ge (\delta(M) + \varepsilon)^n \|Eg\|$ would contradict $\|M^n g\| \le \delta(M)^n \operatorname{osc}(g)$ for large n. Hence $\rho_\xi(y, y')^2 \le C^{\lambda_j}(y, y')C^{\lambda_j}(y', y)$ for λ_j-almost every ξ, which is the first assertion, and the arithmetic–geometric mean inequality gives the second. For a translation-covariant block map the blocked measure is invariant under the translations of Y, so the translates of the given versions at one coarse point y_0 are versions at every coarse point; read with them, $C^{\lambda_j}(y, y')$ is the given matrix's entry at $(y_0, y_0 + y' - y)$, its weighted row and column sums all equal the weighted row sum of the given matrix at y_0, at most α under Hypothesis I, and (8.8d) holds for these versions as for any, R_j not depending on them; then $\Sigma_{y'} R_j(y, y')e^{\kappa|y-y'|} \le \tfrac{1}{2} \Sigma_{y'} (C^{\lambda_j}(y, y') + C^{\lambda_j}(y', y))e^{\kappa|y-y'|} \le \alpha$. When $V = \{0, 1\}$, write p_{ab} for the probability of $X = a$, $X' = b$ under π_ξ, $p_{a\cdot}$ and $p_{\cdot b}$ for its marginals, and $\Delta = p_{11}p_{00} - p_{10}p_{01}$; every function of a two-point variable is affine in its indicator, so $\rho_\xi = |\Delta|/(p_{1\cdot}p_{0\cdot}p_{\cdot 1}p_{\cdot 0})^{1/2}$, while $C_\xi(y \mid y') = |p_{11}/p_{\cdot 1} - p_{10}/p_{\cdot 0}| = |\Delta|/(p_{\cdot 1}p_{\cdot 0})$ and $C_\xi(y' \mid y) = |\Delta|/(p_{1\cdot}p_{0\cdot})$, whose product is ρ_ξ^2. ■

Theorem 8.5′ (The reduction of the gap to the blocked bound, in L^2). *Let a construction be blocked at the coarse scale a_c as in Theorem 8.5, with a block map whose blocked measures have the form (S), and suppose Hypotheses I′ and II′. Put $\rho = \max(e^{-\kappa}, \rho')$. Then for every ρ_1 with $\rho < \rho_1 < 1$ there is a constant K, depending on ρ, ρ_1, α, K', $N(f)$, $N(g)$, $\|f\|$ and $\|g\|$ and on nothing else, such that at every rung $j \ge j_0$, on every lattice of the construction, for all bounded observables f and g, $|S_j(fg) - S_j(f)\,S_j(g)| \le K\,\rho_1{}^d$, with d the distance between $S(f)$ and $S(g)$; and every limit point of the construction's Schwinger functions, on the sector generated by the observables that are bounded uniformly in j, has a unique vacuum and a Hamiltonian with no spectrum in $(0, m)$, with $m = (1/a_c)\min(\kappa, \log(1/\rho'))$, the number of (8.8).*

Proof. Fix $j \ge j_0$, a lattice, and bounded observables f and g; write $F(v) = \nu_j(f \mid v)$ and $G(v) = \nu_j(g \mid v)$ for their fibre means, bounded by $\|f\|$ and $\|g\|$. The covariance splits along the block map, by the law of total covariance:

$$\mu_j(fg) - \mu_j(f)\mu_j(g) = [\lambda_j(FG) - \lambda_j(F)\lambda_j(G)] + \lambda_j(\,\nu_j(fg \mid \cdot) - FG\,).$$

The second term is at most $K'N(f)\|f\|\|g\|\rho'^{d}$ by Hypothesis II′(a). For the first, Hypothesis I′ gives $\rho(R_j) \le \alpha < 1$ and $D(i, k) \le e^{-\kappa|i-k|}/(1 - \alpha)$, (S) identifies $\cap_y H_y$ with the constants, and Theorem 8.5D gives $|\lambda_j(FG) - \lambda_j(F)\lambda_j(G)| \le \Sigma_{i,k}\, \ell_i(F)D(i, k)\ell_k(G)$. By Hypothesis II′(b), $\ell_i(F) \le K'N(f)\|f\|\rho'^{r_i}$ and $\ell_k(G) \le K'N(g)\|g\|\rho'^{r'_k}$, with r_i the distance from i to $S(f)$ and r'_k that from k to $S(g)$, so that

$$|\lambda_j(FG) - \lambda_j(F)\lambda_j(G)| \le \frac{K'^2\,N(f)\,N(g)\,\|f\|\,\|g\|}{1 - \alpha}\,\Sigma_{i,k}\,\rho^{e(i,k)}, \quad e(i, k) = r_i + |i - k| + r'_k,$$

$\rho = \max(e^{-\kappa}, \rho')$ bounding the base of each of the three factors. By the triangle inequality $e(i, k) \ge d$, and the number of pairs (i, k) with $e(i, k) \le n$ is at most $N(f)N(g)(2n + 1)^8$, since i then lies within distance n of $S(f)$ and k within distance n of $S(g)$. Hence the last sum is at most $N(f)N(g)\,\Sigma_{n \ge d}\,(2n + 1)^8\,\rho^n$, which is ρ^d times a polynomial in d and is therefore at most $K_1\rho_1{}^d$ for every $\rho_1 \in (\rho, 1)$, with K_1 depending on ρ, ρ_1, $N(f)$ and $N(g)$. Adding the second term gives the covariance inequality, with $K = K'^2N(f)N(g)\|f\|\|g\|K_1/(1 - \alpha) + K'N(f)\|f\|\|g\|$.

For the gap, let f be an observable, bounded uniformly in j, whose links lie in the time interval $[0, w]$. The links of $\theta\bar{f}$ lie in $[-w, 0]$ and those of $\tau_t f$ in $[t, t + w]$; the block boundaries in time are the multiples of a_c and a link belongs to the block of its midpoint, so $S(\theta\bar{f})$ lies in blocks of time index at most 0, $S(\tau_t f)$ in blocks of time index at least $\lfloor t/a_c \rfloor$, and the distance between them is at least $\lfloor t/a_c \rfloor \geq t/a_c - 1$. For $t \in D$ and all large j, t is an integer multiple of a_j; the measure μ_j is invariant under the reflection θ and under the translation τ_t, so $\mu_j(\theta\bar{f})$ is the complex conjugate of $\mu_j(f)$ and $\mu_j(\tau_t f) = \mu_j(f)$, and the covariance inequality gives $S_j(\theta\bar{f} \cdot \tau_t f) - |S_j(f)|^2 \leq (K/\rho_1)e^{-m_1 t}$ with $m_1 = \log(1/\rho_1)/a_c$, the constant independent of j and of the lattice because $N(f)$ and $\|f\|$ are. This is the hypothesis of Theorem 7.4 with $r_j = 0$ at every $t \in D$, so Theorem 7.4 leaves no spectrum in $(0, m_1)$ for every $\rho_1 \in (\rho, 1)$, hence none in $(0, m)$, $m = \log(1/\rho)/a_c$, the number of (8.8). ■

Theorem 8.5′ contains Theorem 8.5 for every translation-covariant block map whose blocked measures have the form (S), which Theorem 8.5 itself does not need: under Hypotheses I and II and (S), the hypotheses of Theorem 8.5′ hold with the same constants, by Proposition 8.5E and the remark after Hypothesis II′, and its conclusion is the same, the factor 3 that the tilt by G' puts into K in the proof of Theorem 8.5 being absent. Reflection positivity enters the proof only through Theorem 7.4, applied to the lattice measures μ_j; the blocked measure is not required to be reflection positive, and the step $\mu_j(\theta\bar{f})\mu_j(\tau_t f) = |\mu_j(f)|^2$ uses the invariance of μ_j under θ and τ_t alone, not its positivity. The fibre clause enters only through the λ_j-mean of the fibre covariance and the $L^2(\lambda_j)$ local fluctuations of the fibre means; that is why its average suffices, and why the matrix of the coarse variables must be read in the same currency, Dobrushin's comparison theorem consuming the supremum oscillations δ_y.

Proposition 8.4′ (The tilt lemma in L^2; Edwin Boston III). *Let π be the law of a pair (X, W) of random variables with values in compact metric spaces, δ a bounded measurable function of (x, w), and $\pi' = e^{\delta}\pi/\pi(e^{\delta})$ the tilt of π by δ. Put $o_X = \sup_w \operatorname{osc}_x \delta(x, w)$, $o_W = \sup_x \operatorname{osc}_w \delta(x, w)$, and $\Delta = \sup_{w,w'} \operatorname{osc}_x (\delta(x, w) - \delta(x, w'))$, the mixed second difference of δ in the sense of Proposition 8.4. Then the maximal correlations of the two laws satisfy*

$$\rho(\pi') \leq e^{(o_X + o_W)/2}\,\rho(\pi) + 1 - e^{-2\Delta}. \tag{8.8e}$$

Proof. If (X, W) is almost surely constant under π, both maximal correlations vanish and there is nothing to prove; otherwise three facts. First, for every law π of (X, W) under which (X, W) is not almost surely constant,

$$1 - \rho(\pi) = \inf \{\, [\, \pi(\mathrm{Var}_\pi(F \mid X)) + \pi(\mathrm{Var}_\pi(F \mid W))\,] \,/\, \mathrm{Var}_\pi(F) : F \in L^2(\pi) \text{ not constant} \,\}.$$

Let P_X and P_W be the conditional expectations onto the functions of X and of W, orthogonal projections on $L^2(\pi)$, so that the numerator is $\|F - P_X F\|^2 + \|F - P_W F\|^2 = \langle F, (2I - P_X - P_W)F\rangle$. If $\rho(\pi) < 1$, the ranges of P_X and P_W meet in the constants, a nonconstant function of X almost surely equal to a function of W having correlation one with it, and the cosine of the Friedrichs angle between them is $\rho(\pi)$; in the decomposition of Lemma 8.5A, $P_X + P_W$ has spectrum $\{1 \pm \cos\theta\}$ on the generic part, 1 on the two mixed summands and 0 on $\ker P_X \cap \ker P_W$, and since X or W is not almost surely constant, the generic part or a mixed summand is not zero; so the spectrum of $P_X + P_W$ off the constants has supremum $1 + \rho(\pi)$, and the infimum above is $1 - \rho(\pi)$. If $\rho(\pi) = 1$, either the two ranges share a nonconstant function, at which the quotient vanishes, or the cosine is one and the same decomposition makes the infimum zero.

Second, if $\pi'' = h\pi$ with $a \le h \le b$ for constants $0 < a \le b$, then $1 - \rho(\pi'') \ge (a/b)(1 - \rho(\pi))$: indeed $\pi''(\mathrm{Var}_{\pi''}(F \mid X)) = \inf_c \pi''((F - c(X))^2) \ge a \inf_c \pi((F - c(X))^2) = a\, \pi(\mathrm{Var}_\pi(F \mid X))$, the infimum over functions c of X, the same holds with W, and $\mathrm{Var}_{\pi''}(F) = \inf_c \pi''((F - c)^2) \le b\, \mathrm{Var}_\pi(F)$; the first fact finishes.

Third, if $\pi_1 = (u \otimes s)\pi/\pi(u \otimes s)$ for bounded functions u of x and s of w with positive infima, then $\rho(\pi_1) \le (\sup u \cdot \sup s/(\inf u \cdot \inf s))^{1/2}\, \rho(\pi)$. Write $\beta(x) = \pi(s(W) \mid X = x)$, $\zeta(w) = \pi(u(X) \mid W = w)$ and $Z = \pi(u \otimes s)$; the laws of X and W under π_1 have densities $u\beta/Z$ and $s\zeta/Z$ with respect to those under π, so the maps $f \mapsto f(u\beta/Z)^{1/2}$ and $g \mapsto g(s\zeta/Z)^{1/2}$ are unitary from $L^2(\pi_1{}^X)$ onto $L^2(\pi^X)$ and from $L^2(\pi_1{}^W)$ onto $L^2(\pi^W)$, and $\pi_1(f(X)g(W)) = \pi(a_1\tilde f(X) \cdot a_2\tilde g(W))$ with $\tilde f, \tilde g$ the images and $a_1 = (u/\beta)^{1/2}$, $a_2 = (s/\zeta)^{1/2}$. Hence the operator T_1 of π_1, $T_1 g = \pi_1(g(W) \mid X)$, is unitarily equivalent to $a_1 T a_2$, T the operator of π and a_1, a_2 acting by multiplication. For each of these laws the operator maps the constants to the constants and the functions of mean zero to the functions of mean zero, so it is the orthogonal sum of the identity on the constants and its restriction to mean zero, whose norm is the maximal correlation; the maximal correlation is therefore the second singular number $\inf\{\|A - E\| : \mathrm{rank}\, E \le 1\}$ of the operator. Writing $T = 1 \otimes 1 + T_0$ with $\|T_0\| = \rho(\pi)$, the operator $a_1 T a_2$ is a rank-one operator plus $a_1 T_0 a_2$, so its second singular number, which is $\rho(\pi_1)$, is at most $\|a_1\|_\infty \|a_2\|_\infty \rho(\pi)$; and $\beta \ge \inf s$, $\zeta \ge \inf u$ give $\|a_1\|_\infty \|a_2\|_\infty \le (\sup u \cdot \sup s/(\inf u \cdot \inf s))^{1/2}$.

Now fix (x_0, w_0) and write $\delta(x, w) = \delta(x, w_0) + (\delta(x_0, w) - \delta(x_0, w_0)) + \delta_{12}(x, w)$, where $\delta_{12}(x, w) = (\delta(x, w) - \delta(x, w_0)) - (\delta(x_0, w) - \delta(x_0, w_0))$ is at most Δ in modulus. The tilt by the first two terms is a tilt by $u \otimes s$ with $\sup u/\inf u \le \exp(o_X)$ and $\sup s/\inf s \le \exp(o_W)$, so the third fact gives $\rho(\pi_1) \le \exp((o_X + o_W)/2)\, \rho(\pi)$; the tilt of π_1 by δ_{12} has a density between two constants of ratio at most $e^{2\Delta}$, so the second fact gives $1 - \rho(\pi') \ge e^{-2\Delta}(1 - \rho(\pi_1))$. Hence $\rho(\pi') \le e^{-2\Delta}\rho(\pi_1) + 1 - e^{-2\Delta} \le \exp((o_X + o_W)/2)\, \rho(\pi) + 1 - e^{-2\Delta}$. ■[75]

The oscillations along each variable enter (8.8e) as a factor and the mixed difference as an addend, as in (8.7); an addend carrying the oscillations would not vanish for a pair on which δ has no joint term, while $1 - e^{-2\Delta}$ does, and it is the addends that the weighted row sums of Corollary 8.5F add up.

Corollary 8.5F (The effective action and the fourth link, in L^2). *(i) When the effective action A_j of (S) is bounded and the conditional laws $\lambda_j(dv_y \mid v) \propto \exp(-A_j(v))\, m(dv_y)$ are used in (8.5), $R_j(y, y') \le 1 - \exp(-\Delta_{yy'}(A_j))$ for distinct y and y', with $\Delta_{yy'}(A_j)$ the mixed second difference of the effective action in the sense of Proposition 8.4.*

(ii) Let the blocked measures of rungs $j \ge j_1$ on a common coarse torus be tilts $\lambda_j = e^{\delta_j}\lambda_{j_1}/\lambda_{j_1}(e^{\delta_j})$ of that of the rung j_1 by bounded functions δ_j. Then

$$R_j(y, y') \;\le\; e^{(o_y(\delta_j) + o_{y'}(\delta_j))/2}\, R_{j_1}(y, y') \;+\; 1 \;-\; e^{-2\Delta_{yy'}(\delta_j)},$$

with o_y and $\Delta_{yy'}$ as in Proposition 8.4; consequently, if $o_j = \sup_y o_y(\delta_j)$ and $\sup_y \Sigma_{y'}(1 - \exp(-2\Delta_{yy'}(\delta_j)))\, e^{\kappa|y-y'|}$ tend to zero as $j \to \infty$, uniformly in the lattice, Hypothesis I′ at the rung j_1 with the constant α_1 implies it at every sufficiently late rung with any $\alpha \in (\alpha_1, 1)$.

[75] The second fact is the perturbation argument of R. Holley and D. Stroock, "Logarithmic Sobolev inequalities and stochastic Ising models", *J. Stat. Phys.* 46 (1987), 1159–1194, read for the two-site heat bath.

Proof. (i) The conditional laws $\lambda_j(dv_y \mid v) \propto \exp(-A_j(v))m(dv_y)$ are defined for every v when A_j is bounded, the normalisations lying between $\exp(-\sup A_j)$ and $\exp(-\inf A_j)$, and (8.5) and Proposition 8.5E are read with them. The conditional law of v_y under one of two coarse configurations differing only at y' is the tilt of its conditional law under the other by the difference of the two actions, so the first inequality of Proposition 8.4 gives $C^{\lambda_j}(y, y') \le 1 - \exp(-\Delta_{yy'}(A_j))$. The mixed difference is the supremum of $|A_j(a, b) - A_j(a, b') - A_j(a', b) + A_j(a', b')|$ over the values a, a' of v_y, b, b' of $v_{y'}$ and the environment, so $\Delta_{yy'}(A_j) = \Delta_{y'y}(A_j)$, and Proposition 8.5E gives $R_j(y, y') \le (C^{\lambda_j}(y, y')C^{\lambda_j}(y', y))^{1/2} \le 1 - \exp(-\Delta_{yy'}(A_j))$.

(ii) The measures λ_j and λ_{j_1} are equivalent, with density proportional to e^{δ_j}, so for λ_j-almost every environment ξ the conditional law of $(v_y, v_{y'})$ under λ_j is the tilt of that under λ_{j_1} by $\delta_j(\cdot, \cdot, \xi)$, whose oscillations along v_y and $v_{y'}$ are at most $o_y(\delta_j)$ and $o_{y'}(\delta_j)$ and whose mixed difference is at most $\Delta_{yy'}(\delta_j)$; Proposition 8.4′ and the essential supremum over ξ give the bound. Weighting and summing, $\Sigma_{y'} R_j(y, y')e^{\kappa|y-y'|} \le e^{o_j}\alpha_1 + \Sigma_{y'}(1 - \exp(-2\Delta_{yy'}(\delta_j)))e^{\kappa|y-y'|}$, which is below any $\alpha \in (\alpha_1, 1)$ once the two quantities are small, at a rung that the uniformity makes independent of the lattice. ■

Corollary 8.5F (ii) is the fourth link of §8.1 in the L^2 currency, with the inputs that (8.7) asks of the tilts in total variation: their oscillations and their weighted mixed differences tending to zero.

Theorem 8.5′ and the statements it rests on cost two conditions beyond those of Theorem 8.5, each used at a named place. First, (S) is used in the proofs only to identify $\cap_y H_y$ with the constants, in Theorem 8.5D and through it in Theorem 8.5′, and to name the effective action that Corollary 8.5F (i) reads; for a block map whose blocked law is not of the form (S), Theorem 8.5D and Theorem 8.5′ hold verbatim with (S) replaced by that identification. Second, the passage from Hypothesis I to Hypothesis I′ uses translation covariance, through the equality of the weighted row and column sums of C^{λ_j}; without it, Hypothesis I′ holds with $\frac{1}{2}(\alpha + \bar{\alpha})$ in place of α, $\bar{\alpha}$ a bound on the weighted column sums, which must satisfy $\alpha + \bar{\alpha} < 2$. Nothing in the proofs of Theorem 8.5′, Theorem 8.5D, Propositions 8.4′, 8.5C and 8.5E, Corollary 8.5F, or Lemmas 8.5A and 8.5B uses that v_y is a function of the links of B_y alone, that the block map is a bijection of block configurations, or the value of L. Apart from Proposition 8.5E and Corollary 8.5F (i), whose subject is the matrix (8.5), none of them asks for a version of a conditional law beyond λ_j-almost every coarse configuration; Hypotheses I and II and the matrix (8.5) use versions defined for every coarse configuration.

Hypothesis I is read on the effective action of the coarse variables. Write the blocked measure as $\lambda_j(dv) = e^{-A}dv$ with respect to the product over the blocks of a fixed reference measure on V, so that $A = A_j(v)$ is the effective action at the coarse scale, and let $\Delta_{yy'}(A)$ be its mixed second difference in the sense of Proposition 8.4: the supremum, over coarse configurations differing only at y', of the oscillation in v_y of the difference of the two actions. The conditional law of v_y under one of the two configurations is the tilt of its conditional law under the other by that difference, so the first inequality of Proposition 8.4 gives $C^{\lambda_j}(y, y') \le 1 - e^{-\Delta}$ with $\Delta = \Delta_{yy'}(A)$. Hypothesis I is therefore a bound on the mixed second differences of the effective action between blocks, weighted by their distance: the influence functional of §8.1, and the number that a certificate of the second link must deliver at the coarse scale.

One might wish to state the hypothesis on the fine measure directly, taking the blocks as the sites of a Gibbs measure and the whole configuration of a block as its variable, so that Dobrushin's

condition for that measure would give every local observable's decay at once, with no fibre clause. That form is not uniform in the cutoff. Its interdependence coefficient between adjacent blocks is the total variation between two laws of all the links of a block under outside data that differ on a whole neighbouring face; as the spacing tends to zero that face carries ever more links, the two laws become distinguishable almost surely by an average over the face, and the coefficient tends to one at every coupling, so that the row sums tend to at least one. Whole-block total variation is the one norm that does not survive large blocks, and the hypothesis of the reduction is therefore stated on the coarse variables, whose type is fixed, with the field inside the blocks entering through its clustering alone.

Theorem 8.5 is the second link in the form the route attacks it. Hypothesis I is a bound on the blocked measure at the coarse scale, the object named in §8.1 as the influence functional of the blocked action. Hypothesis II is what the ultraviolet half must deliver below that scale; for the gauge group SU(2) the literature states nothing in this form, so it stands beside Hypothesis I as the second half of what the crossover must supply, and it is named as such. Proposition 8.4 is the fourth link in its exact form. The blocked measures λ_j have variables of one type at every rung, so it is meaningful to say that their conditional laws at the coarse scale are tilts of those of one rung j_1 by functions δ_j whose oscillations o_j and weighted mixed differences $\Sigma_{y'}(1 - e^{-\Delta})e^{\kappa|y-y'|}$ tend to zero; when they are, Hypothesis I at the rung j_1 with the constant α_1 implies it at every later rung with any $\alpha \in (\alpha_1, 1)$, by (8.7). That the measures stand in this relation is the ultraviolet interface of §8.1, and it is the place at which Bałaban's theorem enters the chain; the two statements the corpus leaves unwritten there, and the discharge of the theorem's own hypothesis on the sequence of couplings, are theorems on this work's path (§8.3).

The theorem places the four-dimensional gap at one remove from a finite statement. Hypotheses I and II concern the conditional law of one coarse variable and the field inside one block, at a fixed physical scale, with constants uniform in the cutoff, and nothing in them refers to a limit; their consequence, by Theorem 8.5, is the unique vacuum, the cluster property and the mass gap of every limit point, with the gap's value in (8.8). The proof that follows this paper is the proof of those two statements, and its measure of progress is a number: the distance, in blocking steps, between the scale at which the ultraviolet half delivers its interface and the scale at which the blocked bound is read. On the hierarchical model of the renormalisation group, in which each blocking step is the Migdal–Kadanoff recursion on a single-plaquette weight and is exact for the hierarchical lattice, that distance has been measured, and the crossover is crossed with certified numbers. Edwin Boston III has certified the flow of the SU(2) Wilson weight at 256 bits of precision with directed rounding, every character coefficient an interval and the truncation priced, and has certified a bound on the Dobrushin row of the flowed weight F_k placed on $\mathbb{Z}^4$ as the weight of every plaquette, at every step from $\beta = 2.2$ and $\beta = 2.3$ and from the eleventh step from $\beta = 5$ and the thirty-first from $\beta = 10$; the earlier steps from those two couplings the certificate refuses by name, the eight-term truncation leaving its lower bound on F_k non-positive near $\theta = \pi$. Write $F_k(U) = \exp(\beta_k \cos\theta + \delta_k(\theta))$ for U of angle θ, with β_k the Wilson coupling whose fundamental character coefficient is the step's own, so that δ_k is the weight's departure from Wilson's form. A weight tilted by e^{δ} keeps Dobrushin's interdependence up to the factor e^{o} and adds $1 - e^{-\Delta}$ for each neighbour, where o bounds the

oscillation of the tilt along the varied link and Δ its mixed difference between the two links;[76] each link lies in six plaquettes and has eighteen neighbouring links, two distinct links share at most one plaquette, and the Wilson part's row is $18 \cdot 4\beta_k/(3\pi)$ by §8.1, so the row is at most $B_k = e^{6 \operatorname{osc} \delta_k} \cdot (24/\pi)\,\beta_k + 18\,(1 - e^{-2 \operatorname{osc} \delta_k})$, which is the single link's disk $\beta_k < 3\pi/72$ of §8.1 when $\delta_k = 0$. From $\beta = 2.2$ and from $\beta = 2.3$ the budget closes at the eighth step, $B_8 \in [0.033548, 0.033599]$ and $B_8 \in [0.28150, 0.28192]$, after $B_7 \in [2.0760, 2.0791]$ and $B_7 \in [4.9347, 4.9420]$; from $\beta = 5$ it closes at the nineteenth, $B_{19} \in [0.026779, 0.026820]$ after $B_{18} \in [1.9154, 1.9182]$; and from $\beta = 10$ at the thirty-ninth, $B_{39} \in [0.0055722, 0.0055806]$ after $B_{38} \in [1.1355, 1.1372]$, where the coupling $\beta_{38} = 0.1177$ already lies inside the single link's disk and the departure from Wilson's form alone holds the budget above one. The eighth iterate from the crossover coupling therefore defines on $\mathbb{Z}^4$ a theory with a unique infinite-volume Gibbs measure and exponential clustering, and, its character coefficients being non-negative so that the weight is reflection positive, a lattice mass gap by Theorem 7.1; and eight is the first step at which the budget closes, the budget itself being certified above one at the step before. The hierarchical model's own confinement at every coupling, with lower bounds on its string tension and mass gap, is classical,[77] and what the certificate adds is the number of steps and the budget at each of them: on the hierarchical model eight blocking steps carry the crossover coupling into the disk, each one certified.[78] The record also says what the model is not. Its first step from $\beta = 2.2$ moves the coupling up where the running of $\mathbb{Z}^4$ moves it down, by an amount of the step's own size, so its first steps are not $\mathbb{Z}^4$'s: the distance Hypotheses I and II must cross is counted in the blocking steps of $\mathbb{Z}^4$, and the hierarchical count is the measured model of it. One more thing is known about the inequality that would carry the model's control to $\mathbb{Z}^4$. A comparison that bounded the mixed second differences of the effective action of $\mathbb{Z}^4$ by those of the hierarchical model, and held for every compact gauge group, would give the four-dimensional U(1) theory a gap wherever its hierarchical flow reaches the disk; the hierarchical U(1) model confines at every coupling by the same theorem, its flow descending at a rate exponentially small in the coupling, $7.3 \cdot 10^{-5}$ of β per step at an effective coupling of 1.39 and about $3 \cdot 10^{-13}$ at 2.83, where the SU(2) flow descends by 0.256 to 0.258 per step near $\beta = 5.5$, the rate of $\mathbb{Z}^4$'s one-loop running, 0.2575 per doubling; and the U(1) theory on $\mathbb{Z}^4$ has a massless phase at weak coupling.[79] The inequality the step needs therefore depends on the group, and that dependence

[76] The two tilt lemmas: a probability law tilted by e^{ρ} with ρ of oscillation at most o moves by at most $1 - e^{-o}$ in total variation, and two laws tilted by the same ρ keep their distance up to the factor e^{o}; the triangle inequality then gives the tilted interdependence $e^{o}C + (1 - e^{-\Delta})$. Both proofs are short and hold on any probability space; the finite form is checked by the Lean kernel in CrossoverCriterion.lean (Records).

[77] K. R. Ito, "Permanent quark confinement in four-dimensional hierarchical lattice gauge theories of Migdal–Kadanoff type", *Phys. Rev. Lett.* 55 (1985), 558; V. F. Müller and J. Schiemann, "Convergence of Migdal–Kadanoff iterations: a simple and general proof", *Lett. Math. Phys.* 15 (1988), 289–295, whose rate of convergence gives lower bounds on the string tension and the mass gap of the hierarchical four-dimensional gauge models. Ito's theorem covers the hierarchical U(N) models, U(1) among them.

[78] Edwin Boston III, the certified descent and the certified budget; the instruments, the records at the four starting couplings and the known answers are listed in the Records. The first step from $\beta = 2.2$ is read from the same record: the coefficient rises from 0.464 to 0.486, an effective coupling of 2.35, where the one-loop running of $\mathbb{Z}^4$ takes 2.2 to 1.94 under one doubling of the scale. Revision 30 counted instead the steps at which the coefficient c_1 itself falls below 0.131, the seventh from $\beta = 2.3$, the eighteenth from $\beta = 5$ and the thirty-seventh from $\beta = 10$, and compared them with a bound that §8.1 states on the coupling: the disk is $\beta < 3\pi/72$, $c_1 < 0.0327$ for a weight of Wilson's form, and the budget above, which also prices the departure from that form, closes one or two steps after the coefficient's crossing.

[79] A. H. Guth, "Existence proof of a nonconfining phase in four-dimensional U(1) lattice gauge theory", *Phys. Rev. D* 21 (1980), 2291–2307; J. Fröhlich and T. Spencer, "Massless phases and symmetry restoration in abelian gauge theories and spin systems", *Comm. Math. Phys.* 83 (1982), 411–454.

is the first property a proof of Hypothesis I must use; on the hierarchical model the two groups part in the rate of descent, which is where the running coupling lives.

Two things are known about the strong end of that distance. With the first-order transmission between links, the total variation moved by one neighbour's change, no block criterion of Dobrushin and Shlosman's kind gains on the single link at any block side, because a link deep inside a block branches to its eighteen neighbours exactly as a link of the full lattice does, and the criterion's series has the single link's ratio. A block gains only through the transmission across an integrated face, which carries the product of the two pinned means of the face's other edges, an average over the interior: its one-face constant is an explicit function of the moments of the von Mises–Fisher law on the three-sphere, the covariance of the transmitted variable being a multiple of the identity plus one rank-one term whose size is a closed form in two Bessel ratios, certified in a run of record. What is not in hand is the theorem that composes that transmission along paths inside a block: every composition that is proved, Dobrushin's comparison theorem among them, takes a supremum over configurations at each step and so composes the first-order sensitivities only, which return the single link's disk; a convergent expansion of the block measure would compose the averages, and its radius has no proof beyond that disk.[80] The paper carries no number for that widening until both are theorems.

One averaged composition is rigorous, and it is an instance of Theorem 8.5 in which the fibre clause holds exactly. Call a set of links of $\mathbb{Z}^4$ a *perfect cover* when every plaquette contains exactly one of its links. One exists with period two in every direction: reading the coordinates of a site x modulo 2, take the link from x in direction 1 when $x_2 = x_3 = x_4$, in direction 2 when $x_4 = x_1 \neq x_3$, in direction 3 when $x_2 = x_1 \neq x_4$, and in direction 4 when $x_3 = x_1 \neq x_2$. The four conditions partition the sixteen residues, so the set has one link at every site and density one quarter, and a direct check of the ninety-six plaquettes of the torus of side two shows that each contains exactly one of its links; since the condition is local and the set periodic, the same holds on $\mathbb{Z}^4$ and on every torus of even side.[81] Two consequences are used below: no plaquette contains two links of the cover, and a link outside the cover lies in six plaquettes whose six cover links are distinct, since two links share at most one plaquette.

Lemma 8.6 (The constants of the von Mises–Fisher law on the three-sphere). *For w in $\mathbb{R}^4$ let ν_w be the law on the unit sphere S^3 of $\mathbb{R}^4$, identified with SU(2), whose density with respect to the uniform measure is proportional to $e^{\langle u,\, w\rangle}$. Write $r = |w|$, $Z(r) = 2I_1(r)/r$ for its normalising constant, $\varphi = \log Z$ and $m = \varphi' = I_2/I_1$, with I_ν the modified Bessel functions. The mean of ν_w is $m(r)\hat{w}$; its covariance has the*

[80] Edwin Boston III; the block criterion's accounting, the one-face law and the closed form of its covariance are entered on the ledger of the work that follows this paper, with their readings on the exact cube.

[81] Edwin Boston III; the cover was found by an exhaustive search on the torus of side two and checked again, on a road sharing no line with the search, on every plaquette of the tori of side two, four and six, where a single bent site leaves ten plaquettes of the smallest torus with zero or two links of the set; Records R7.

eigenvalue $\varphi''(r)$ on $\hat{w}$ and $m(r)/r$ on the orthogonal complement; and its third cumulant along a unit vector u with $\langle u, \hat{w}\rangle = c$ is $\varphi'''(r)c^3 + 3(D(r)/r)c(1-c^2)$, where $D = \varphi'' - m/r$. For every $r \geq 0$:

(i) $m(r)/r \leq 1/4$ and $\varphi''(r) \leq 1/4$;

(ii) $-m(r)^2/3 \leq D(r) \leq 0$;

(iii) the third cumulant of ν_w along every unit vector is at most $m(r)/2$ in absolute value.

Proof. The cumulants of ν_w are the derivatives at w of $x \mapsto \varphi(|x|) - \varphi(|w|)$, the logarithm of the normalised Laplace transform of ν_w. Its gradient at w is $m\hat{w}$ and its Hessian has the stated eigenvalues; along the line $x + tu$ the radius $\rho(t) = |x + tu|$ has $\rho' = c$, $\rho'' = (1 - c^2)/r$ and $\rho''' = -3c(1 - c^2)/r^2$ at $t = 0$, so the third derivative $\varphi'''\rho'^3 + 3\varphi''\rho'\rho'' + \varphi'\rho'''$ is the stated form. The recurrences $I_1' = I_2 + I_1/r$ and $I_2' = I_1 - 2I_2/r$ give $\varphi' = m$ and the Riccati identity

$$m' \;=\; 1 \;-\; \frac{3m}{r} \;-\; m^2. \tag{8.9}$$

Put $z = r^2/4$ and $A = \Sigma_k\, z^k/(k!(k+2)!)$, $B = \Sigma_k\, z^k/(k!(k+1)!)$, $C = \Sigma_k\, z^k/(k!(k+3)!)$, so that $A = (2/r)^2 I_2$, $B = (2/r)I_1$ and $C = (2/r)^3 I_3$, all three positive, with $m/r = A/(2B)$ and $m^2 = zA^2/B^2$. Comparing coefficients gives $B - 2A = zC \geq 0$, which is the first half of (i). By (8.9), $\varphi'' = 1 - 3A/(2B) - zA^2/B^2$, and the substitution $B = 2A + zC$ gives

$$\frac{1}{4} \;-\; \varphi'' \;=\; \frac{z\,(A^2 - \tfrac{3}{4}\,BC)}{B^2}, \qquad D \;=\; \varphi'' \;-\; \frac{m}{r} \;=\; -\,\frac{z\,(A^2 - BC)}{B^2}. \tag{8.10}$$

Now $A^2 - BC = (2/r)^4(I_2^2 - I_1 I_3)$. The product formula $I_\mu I_\nu = \Sigma_k\, (r/2)^{\mu+\nu+2k}\,(\mu+\nu+2k)!/(k!(\mu+k)!(\nu+k)!(\mu+\nu+k)!)$[82] makes the k-th coefficient of $I_2^2 - I_1 I_3$ equal to $(4+2k)!/(k!(4+k)!(2+k)!(3+k)!)$, which is positive, and that of $3I_1 I_3 - 2I_2^2$ equal to $k/(2+k)$ times the k-th coefficient of $I_1 I_3$, which is not negative. Hence $0 \leq A^2 - BC \leq A^2/3$. The left inequality is Turán's inequality for the modified Bessel functions.[83] By (8.10) it gives $D \leq 0$, and, since then $\tfrac{3}{4}BC \leq \tfrac{3}{4}A^2 \leq A^2$, it gives $\varphi'' \leq 1/4$, the second half of (i). The right inequality gives $|D| \leq zA^2/(3B^2) = m^2/3$, which is (ii). For (iii), differentiate (8.9): $\varphi''' = 3|D|/r - 2m\varphi''$, where $0 \leq 3|D|/r \leq m^2/r \leq m/4$ by (ii) and (i), and $0 \leq 2m\varphi'' \leq m/2$ because φ'' is a variance at most $1/4$. With $a = 3|D|/r$ and $b = 2m\varphi''$, the third cumulant at c in $[0,1]$ is $ac(2c^2 - 1) - bc^3$, and it is odd in c. When $2c^2 \geq 1$ it lies between $-bc^3 \geq -(m/2)c^3$ and $ac(2c^2-1) \leq (m/4)c(2c^2-1)$; when $2c^2 < 1$ it equals $-(ac(1-2c^2) + bc^3)$, whose size is at most $(m/4)c(1-2c^2) + (m/2)c^3 = (m/4)c$. In every case its size is at most $(m/4)c(1+c^2) \leq m/2$. ■

Theorem 8.7 (A lattice mass gap at strong coupling; Edwin Boston III). *Consider the SU(2) lattice gauge theory on $\mathbb{Z}^4$ with Wilson's action, each plaquette weighted by $\exp(\tfrac{1}{2}\,\beta\,\mathrm{tr}\,U_p)$, at $0 < \beta \leq 0.241$. Its infinite-volume Gibbs measure is unique. For every $m_0 < \log(1/\alpha(\beta))/2$, with $\alpha(\beta) \leq 0.9963$ the number of (8.12), there is a constant K, depending on β and m_0 alone, such that for all bounded observables f and g depending on finitely many links, $|\langle fg\rangle - \langle f\rangle\langle g\rangle| \leq K\,N(f)\,N(g)\,\|f\|\,\|g\|\,e^{-m_0 d}$, with $N(f)$ the number of links on which f depends and d the least distance, in the maximum norm, between the midpoints of a link of f and a link of g; the same bound holds on every torus of even side. The Hamiltonian $H = -\log T$ of the transfer matrix T annihilates a unique vacuum and has no spectrum in the interval $(0, m_0)$.*

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[82] G. N. Watson, *A Treatise on the Theory of Bessel Functions*, second edition, Cambridge University Press, 1944, §5.41.

[83] K. V. Thiruvenkatachar and T. S. Nanjundiah, "Inequalities concerning Bessel functions and orthogonal polynomials", *Proc. Indian Acad. Sci. A* 33 (1951), 373–384.

Proof. Let L_1 be the perfect cover and L_0 its complement, and read each element of SU(2) as a unit vector of $\mathbb{R}^4$, so that ½ tr $U_p = \langle U_a, S_p(a)\rangle$ for the cover link a of a plaquette p, the staple $S_p(a)$ being the product of the other three links of p, or of their inverses, read as a unit vector. Each plaquette contains exactly one link of L_1, so the Wilson weight is the product over a in L_1 of $\exp\langle U_a, \eta(a)\rangle$, with $\eta(a) = \beta \Sigma_{p \ni a} S_p(a)$ and $|\eta(a)| \le 6\beta$. Given the links of L_0, the links of L_1 are therefore independent, U_a distributed by ν_w at $w = \eta(a)$, and the marginal law of the links of L_0 has density proportional to $\prod_{a \in L_1} Z(|\eta(a)|)$ with respect to the product of the Haar measures: a Gibbs measure with a potential of finite range. For x in L_0 and a plaquette $p \ni x$, let a_p be the cover link of p, $\theta_p(U_x) = \beta S_p(a_p)$, which is β times an isometry of $\mathbb{R}^4$ applied to U_x, and $w_p = \eta(a_p) - \theta_p(U_x)$, the contribution of the five other plaquettes of a_p, so that $|w_p| \le 5\beta$. The conditional law of U_x given the other links of L_0 is proportional to $\prod_{p \ni x} Z(|w_p + \theta_p(U_x)|)$, the six factors coming from six distinct cover links.

By Taylor's theorem along the segment from w_p to $w_p + \theta_p$, which stays in the ball of radius 6β, and since m is increasing, $\log Z(|w_p + \theta_p|) = \varphi(|w_p|) + \langle m(|w_p|)\hat{w}_p, \theta_p\rangle + ½\langle \theta_p, \Sigma_p \theta_p\rangle + R_p$, with Σ_p the covariance of ν_w at $w = w_p$ and $|R_p| \le (\beta^3/6)T_6$, $T_6 = m(6\beta)/2$, by Lemma 8.6(iii). The linear terms add to $\langle U_x, v\rangle$ for a vector v to which p contributes a vector of length at most $\beta m(5\beta)$; the quadratic term of p equals a constant plus $½\beta^2 D(|w_p|)\langle \theta_p/\beta, \hat{w}_p\rangle^2$, whose oscillation in U_x is at most $½\beta^2 D_5$, with $D_5 = m(5\beta)^2/3$, by Lemma 8.6(ii). The conditional law of U_x is therefore the tilt of ν_v by a function $h = \Sigma_p h_p$ of oscillation at most

$$\omega \;=\; 3\beta^2 D_5 \;+\; 2\beta^3 T_6. \tag{8.11}$$

Two facts bound the influence of one link on that law. First, if p is a probability measure and g a bounded function of oscillation o, the tilt $T_g p$ differs from p in total variation by at most $\tanh(o/4) \le o/4$: with g normalised to take values in $[0, o]$, for a set B of p-measure s on which e^g averages a and off which it averages a', $(T_g p)(B) - p(B) = s(1-s)(a-a')/(sa + (1-s)a')$, which is largest at $a = e^o$, $a' = 1$ and $s = 1/(1 + e^{o/2})$, where it equals $\tanh(o/4)$. Second, by the sharp one-link lemma of §8.1, the mean absolute deviation of $\langle U, e\rangle$ under any law ν_v is at most $4/(3\pi)$ for every unit vector e, and under a tilt of ν_v by a function of oscillation at most ω it is at most $(3e^\omega - 1)/2$ times as large: if the tilt has density g with values in $[c, ce^\omega]$ with respect to ν_v, where $c \le 1$, and $Y = \langle U, e\rangle - \nu_v(\langle U, e\rangle)$ with $M = \nu_v(|Y|)$, the deviation is at most $\nu_v(g|Y|) + |\nu_v(gY)| \le ce^\omega M + c(e^\omega - 1)M/2$, because $\nu_v(Y^+) = M/2$. Differentiating along the segment from v to v', two tilts of ν_v and $\nu_{v'}$ by one function of oscillation at most ω differ in total variation by at most $\rho(\omega)(2/(3\pi))|v - v'|$, with $\rho(\omega) = (3e^\omega - 1)/2$.

Now let a link y of L_0 change, the others fixed. If y is one of the two other links of a plaquette p of x, as twelve links are, then v moves by at most $2\beta m(5\beta)$, because a factor of the isometry moves by at most 2, and h_p moves by a function of oscillation at most $\beta^2 D_5 + (2/3)\beta^3 T_6$. If y lies in one of the five other plaquettes of a cover link a_p, as ninety links do, fifteen for each of the six, then w_p moves by at most 2β, so v moves by at most $\beta \cdot ¼ \cdot 2\beta = \beta^2/2$, the Jacobian of the mean being the covariance, whose norm is at most ¼ by Lemma 8.6(i); the quadratic part of h_p moves by a function of oscillation at most $\min(\beta^2 D_5, 2\beta^3 T_6)$, the second bound because the covariance moves in norm by at most 2β times the bound on the third cumulant, the norm of a symmetric trilinear form being its supremum on the diagonal; and the remainder moves by at most $(2/3)\beta^3 T_6$. By the two facts the interdependence coefficients are at most

$$C_{dir} \;=\; \rho(\omega)\,\frac{4\beta}{3\pi}\,m(5\beta) \;+\; \frac{1}{4}\Big(\,\beta^2 D_5 \;+\; \frac{2}{3}\beta^3 T_6\,\Big),$$

$$C_{med} \;=\; \rho(\omega)\,\frac{\beta^{2}}{3\pi} \;+\; \frac{1}{4}\big(\,min(\beta^{2}D_{5},\,2\beta^{3}T_{6}) \;+\; \frac{2}{3}\beta^{3}T_{6}\,\big), \qquad \alpha(\beta) \;=\; 12\,C_{dir} \;+\; 90\,C_{med}. \tag{8.12}$$

A link y that occupies several of these positions is counted once for each, which bounds its coefficient because total variation is subadditive along a chain of single changes, and no other link of L_0 affects the law of U_{x}; the column sums obey the same count. Every term of α is increasing in β, and Table 3 gives its value: α(0.241) = 0.996297 < 1. By Dobrushin's uniqueness theorem the Gibbs measure of the links of L_0 is unique,[84] and, since the links of L_1 are conditionally independent given them with a law that is fixed, so is the Gibbs measure of the theory. The coefficients vanish between links at distance more than two, so the weighted row sums of Hypothesis I are at most $\alpha e^{2\kappa}$, which is less than one for κ < log(1/α)/2; Hypothesis II holds with zero fibre covariance for observables with disjoint supports, whose links in L_1 are distinct and conditionally independent, and with fibre means that depend on the links of L_0 within distance one of their support. The argument of the proof of Theorem 8.5, applied on one lattice with the links of L_0 as the coarse variables, gives the clustering bound with any m_0 < log(1/α)/2, uniformly over the tori of even side and in the infinite volume. The Wilson action is reflection positive and its transfer matrix is strictly positive,[85] the bound applies to $\theta\bar{f}$ and $\tau_t f$ at every integer time t, and Theorem 7.1, applied to the reconstruction at integer times, gives the vacuum and the gap. ■

β	12 C_{dir}	90 C_{med}	α(β)	ω	ρ(ω)
0.200	0.250918	0.418916	0.669834	0.004576	1.006879
0.220	0.303206	0.516348	0.819554	0.006609	1.009946
0.230	0.331235	0.569908	0.901143	0.007838	1.011804
0.240	0.360539	0.626868	0.987407	0.009224	1.013901
0.241	0.363540	0.632756	0.996297	0.009372	1.014124
0.2415	0.365046	0.635714	1.000760	0.009446	1.014237

Table 3. The interdependence row sum (8.12) of the decimated measure, with the constants of Lemma 8.6, evaluated at 256 bits with outward rounding; it crosses one between 0.241 and 0.2415.

The numbers of Table 3 are certified on three roads that share no line; the derivation that they evaluate is the proof above, and its one test from outside the proof is a search from below over the decimated specification's own data, which found no configuration whose coefficient exceeds the constants (the direct coefficient reaching 97 per cent of its bound at β = 0.245, the mediated 89 per cent). The certificate evaluates m at 5β and 6β from its series with a bound on the tail, in integer arithmetic at 256 bits with every quantity rounded outward; a second evaluation at sixty digits agrees with it to 5×10^{-7} at every tabulated coupling and places the root of α = 1 at β = 0.241415; and a third encloses α with directed rounding over the whole interval and certifies it below one on [0, 0.2414]. With the two constants replaced by their exact values, $|D(5\beta)|$ for $m(5\beta)^{2}/3$ and $|\varphi'''(6\beta)|$ for $m(6\beta)/2$, the second being the largest third cumulant along any direction at radius at most 6β, each

[84] R. L. Dobrushin, "Prescribing a system of random variables by conditional distributions", *Theory Probab. Appl.* 15 (1970), 458–486; Sources [17].

[85] K. Osterwalder and E. Seiler, Sources [5]; M. Lüscher, Sources [7].

enclosed with certified error in a run of record, the same row sum stays below one on [0, 0.2452].[86] Theorem 8.7 is a theorem at strong coupling, 1.84 times the single link's disk $3\pi/72$ of §8.1, and it reaches no crossover: it claims nothing above $\beta = 0.2452$, the test of the abelian group in this section does not bear on it, and the interval between it and the crossover near $\beta = 2.3$ is the part of the second link that Hypotheses I and II must still supply. That no decimation of $\mathbb{Z}^4$ places both integrated links of every plaquette opposite one another, which a two-colouring argument shows in every dimension above two, is the reason the product of two pinned means of the block criterion cannot be composed in this way, and the reason that wider widening remains the conjecture above.[87]

The same construction runs in three dimensions, where every link lies in four plaquettes.

Theorem 8.8 (A lattice mass gap at strong coupling in three dimensions; Edwin Boston III). *Consider the SU(2) lattice gauge theory on $\mathbb{Z}^3$ with Wilson's action, each plaquette weighted by $\exp(\tfrac{1}{2}\beta\,\mathrm{tr}\,U_p)$, at $0 < \beta \le 0.3739$. Its infinite-volume Gibbs measure is unique. For every $m_0 < \log(1/\alpha_3(\beta))/2$, with $\alpha_3(\beta) \le 0.9996$ the row sum of the proof below, there is a constant K, depending on β and m_0 alone, such that for all bounded observables f and g depending on finitely many links, $|\langle fg\rangle - \langle f\rangle\langle g\rangle| \le K\,N(f)\,N(g)\,\|f\|\,\|g\|\,e^{-m_0 d}$, with $N(f)$ and d as in Theorem 8.7; the same bound holds on every torus of even side. The Hamiltonian of the transfer matrix annihilates a unique vacuum and has no spectrum in the interval $(0, m_0)$.*

Proof. The proof of Theorem 8.7, with the three-dimensional cover and its counts. The links (x, e_1) with x_2 and x_3 even, (x, e_2) with x_1 even and x_3 odd, and (x, e_3) with x_1 and x_2 odd, three families of parallel lines, meet every plaquette of $\mathbb{Z}^3$ exactly once. In the (1, 2)-plane at base point x the two e_1-links sit at second coordinates x_2 and $x_2 + 1$ with the same x_3, so exactly one of them belongs to the set when x_3 is even and neither when it is odd, while the two e_2-links sit at first coordinates x_1 and $x_1 + 1$ with the same x_3, so exactly one belongs when x_3 is odd and neither when it is even; in the (1, 3)-plane the e_1-links give one exactly when x_2 is even and the e_3-links one exactly when x_2 is odd; in the (2, 3)-plane the e_2-links give one exactly when x_1 is even and the e_3-links one exactly when x_1 is odd. The set has period two, so the same holds on every torus of even side.

Each link lies in $P = 4$ plaquettes. The integrated links are therefore independent von Mises–Fisher variables given the kept ones, each with an argument of norm at most 4β; the conditional law of a kept link is a product of four ratios of the function Z; its row has $2P = 8$ direct entries and $3P(P - 1) = 36$ mediated ones; the constants of Lemma 8.6 enter at 3β and 4β in place of 5β and 6β, as $D_3 = m(3\beta)^2/3$ and $T_4 = m(4\beta)/2$; and the oscillation of the reweighting is $\omega = 2\beta^2 D_3 + (4/3)\beta^3 T_4$, the sum over the link's four plaquettes of the per-plaquette term $\beta^2 D/2 + \beta^3 T/3$ of Theorem 8.7's proof. The row sum $\alpha_3(\beta) = 8C_{\mathrm{dir}} + 36C_{\mathrm{med}}$, with C_{dir} and C_{med} the coefficients of that proof at these arguments, is 0.976276 at $\beta = 0.37$ and 0.999547 at $\beta = 0.3739$, and crosses one at $\beta = 0.373975$. The rest of the proof is unchanged. ∎

The numbers are certified on two roads that share no line: an evaluation in 256-bit arithmetic with every quantity rounded outward, which reproduces Table 3's value at $\beta = 0.2410$ to nine digits,

[86] Edwin Boston III; the certificate and its sharp pass, the second evaluation, the third road's enclosure and the enclosures of the two constants are Records R7.

[87] Edwin Boston III; the two-colouring argument, the range of the decimated specification and its row sum are on the ledger of the work that follows this paper, with their records.

and an evaluation at sixty digits; they agree at every coupling read. With the exact constants $|D(3\beta)|$ and $|\varphi'''(4\beta)|$ in place of the two bounds the row crosses one at $\beta = 0.3841$, and in two dimensions, where a link lies in two plaquettes, the same construction closes at $\beta = 0.8252$. Three dimensions is where the ultraviolet half of a construction is in print without a hypothesis on the sequence of couplings, so there the interval between Theorem 8.8 and the small-field regime is the whole of what remains, the rehearsal of the four-dimensional crossing on ground whose two ends are both published.

The open part. Theorem 8.7 certifies the gap for $\beta \le 0.2452$ by Dobrushin's comparison, on proved lemmas alone; Theorem 8.9 below extends it to $\beta \le 0.4764$ by the flux-surface gas, conditional on the one lemma its statement names; nothing above 0.2452 is proved unconditionally, and nothing above 0.4816, where the same inequality holds in the tree dressing (0.4764 in the dressing of the page), is proved at all. The inequality that produces it, Dobrushin's comparison theorem, is group-blind, and correctly so: it holds for every compact single-spin space, and the four-dimensional U(1) theory at weak coupling, whose photon is massless, satisfies its unweighted finite-volume form as readily as a massive theory does, since a boundary's influence there decays as the fourth power of the distance and the unweighted block sum is of order $L^3 \log L$ against the block's volume L^4. The group enters only through the evaluation of Hypothesis I, the weighted row sums $\Sigma_y\, C(x, y)e^{\kappa|x-y|}$ of the blocked measure, which diverge for every $\kappa > 0$ in a Coulomb phase and converge in a massive one; what is open is a certified evaluation of those sums at a coupling where neither the strong-coupling nor the weak-coupling expansion converges. The weak end has no published number. Bałaban's ultraviolet stability theorem in four dimensions, Theorem 1 of his paper of record, is proved under the hypothesis that every effective coupling of the sequence, from the bare one at the cutoff to the one at the unit lattice, is at most a constant γ that exists and is never computed ("the constant γ depends on all other constants"); the coupling grows toward coarse scales, so the condition binds at the unit lattice, where his action weighs a plaquette exactly as (8.12)'s does with $\beta = 1/g^2$, and the distance is the interval $[0.2452, 1/\gamma^2]$ on proved lemmas, $[0.4764, 1/\gamma^2]$ under Theorem 8.9's condition, with γ unread. The hypothesis's own proof, his Theorem 2 of 1987 with its two-sided bound on the couplings, was announced there, stated in the 1989 paper to be unpublished ("so we have Theorem 1 with the assumption"), and never appeared: his complete list of publications after 1989 contains no paper on four-dimensional gauge theory,[88] so the four-dimensional ultraviolet half of every construction of the corpus's kind has stood on it as an assumption. It is not an assumption here but a theorem on this work's path. Its content is exact: his equations (0.18) and (0.20) define the effective coupling at each step by subtracting from the inverse coupling the coefficient β_{k+1} that his extraction (0.28) reads off the step's output as the marginal term, and Theorem 2 is the statement that this coefficient stays between two positive multiples of $\log L$ at every step, from which the existence of the bare coupling, the bound (0.31) and asymptotic freedom follow by summation and the intermediate value theorem. The coefficient has two parts per step, the background dependence of the fluctuation determinant of his Gaussian normalisation — the one-loop coefficient of the block scheme, whose continuum value for SU(2) is 0.2575 per doubling in Wilson's normalisation of the coupling — and a second-order part bounded through Theorem 1's own estimates; both are stated

[88] The zbMATH author listing for T. Bałaban, 1989–2022, read on 29 October 2026: the Euclidean fermion block renormalisation (1989), the two large-field papers (1989), the classical N-vector series (1995–1999), and the bosonic many-body series with Feldman, Knörrer and Trubowitz (2008–2022); no paper on four-dimensional gauge theory.

with their pages on the open ledger, with the falsifier beside them: for U(1) the one-loop coefficient vanishes, so the lower bound and the theorem fail, and the chain separates the groups at the ultraviolet end as Theorem 8.7 separates them at the infrared end. When the bound is proved here, γ becomes a number this work produces and the interval has two ends of its own. Below that ceiling the first rung of the renormalisation step, the small-field split of one block of side two with its Peierls and Kotecký–Preiss conditions, has its constants re-derived on the open ledger (the block Hessian's least eigenvalue 3, the entropy 54.4 per plaquette, the large-field weight the exact one-plaquette marginal): the coupling at which that split converges is about 70, conditional on the activity bound of the small-field expansion and on the chessboard constant, neither proved there, and, proved, it would mark where a Bałaban-type analysis has a verified foothold and not where decay begins, since the same expansion converges for the free massless field, which has no gap. The lower end of the interval is Theorem 8.7's. Osterwalder and Seiler proved a gap at every sufficiently small β with no coupling named; Shen, Zhu and Zhu named one, which for SU(2) on $\mathbb{Z}^4$ is $\beta < 1/12$ in Wilson's normalisation;[89] Douglas and Rajasekaran formalised in Lean, for the U(n) theories, the Dobrushin route of Chatterjee's lecture notes, at $\beta < 1/(32n(d-1))$ for the action $\beta \Sigma_p (n - \mathrm{Re\ Tr}\ U_p)$, which for n = 2 on $\mathbb{Z}^4$ is $\beta < 1/96$ in Wilson's normalisation.[90] Theorem 8.7 reaches 0.2452, about three times the first of these thresholds and more than twenty times the second, and Theorem 8.8 reaches 0.3739 on $\mathbb{Z}^3$, against 1/8 and 1/64 there, with the rate and the certified constants.[91] Dimock's expositions treat the abelian and scalar cases and state every threshold the same way. Within the certified form the next widening is the same theorem in the Wasserstein metric of the three-sphere, whose one-link constant is 1/4, exact in the two symmetric directions and supported to one per cent between them, against total variation's $4/(3\pi)$; with the constants proved so far it gains nothing over Theorem 8.7, it is predicted to close near $\beta = 0.30$ under two constants not yet proved, and it is unguessed to the digit until they are. The tools that would cross the interval are named and not claimed: a finite-volume criterion with typical rather than extremal boundary data, for which no theorem exists in a non-abelian gauge theory; a correlation inequality, Griffiths' second inequality for untruncated SU(2) Wilson loops, which holds on the exact tube of §8.2 at β = 0.5 and 1.0, is undecided at 2.0 and 2.3, and from which, by the r^{-8} plaquette correlations of the U(1) Coulomb phase, no finite-volume criterion for truncated correlations can follow; and the centre of the group, whose one published every-

[89] H. Shen, R. Zhu and X. Zhu, "A stochastic analysis approach to lattice Yang–Mills at strong coupling", *Comm. Math. Phys.* (2023), doi:10.1007/s00220-022-04609-1, arXiv:2204.12737, (1.2)–(1.3): their action is $N\beta\ \mathrm{Re}\ \Sigma\ \mathrm{Tr}\ Q_p$, so their β is one quarter of Wilson's for SU(2), and their condition $|\beta| < 1/(16(d-1))$ is $\beta < 1/12$ on $\mathbb{Z}^4$ and $\beta < 1/8$ on $\mathbb{Z}^3$ in Wilson's normalisation.

[90] M. R. Douglas and F. Rajasekaran, *lgt — Lattice Gauge Theory in Lean 4*, github.com/mrdouglasny/lgt, the headline theorems of LGT/MassGap/StrongCoupling.lean proved on 4 May 2026 (version 0.1.0, 6 September 2026): for U(n) Wilson theory on (ℤ/Nℤ)^d, d ≥ 2, N ≥ 3, and $\beta < 1/(32n(d-1))$, the connected correlation of two plaquette traces is at most $32n^2/(1-\alpha)\ \alpha^{(\mathrm{dist}-1)/2}$ with α < 1, uniformly in N; other compact groups in U(n) by an instance. Their action is $\beta\ \Sigma(n - \mathrm{Re\ Tr}\ U_p)$, so their β is one half of Wilson's for n = 2: $\beta < 1/96$ on $\mathbb{Z}^4$ and $\beta < 1/64$ on $\mathbb{Z}^3$ in Wilson's normalisation. They cite S. Chatterjee, *Gauge Theory Lecture Notes* (2026), Chapters 15–16, for the route.

[91] T. Bałaban, *Comm. Math. Phys.* 109 (1987), 249–301: Theorem 1 (p. 259) and Theorem 3 (p. 267), the recursion (0.20) (p. 256), the action (0.2) with the normalised trace (p. 252), read from the pages on 29 October 2026. The statement that the sequence can be kept small in four dimensions, his Theorem 2 (pp. 259–260), is deferred there to "a separate paper"; the four-dimensional stability theorem itself is Theorem 1 of *Large field renormalization II*, *Comm. Math. Phys.* 122 (1989), 355–392, p. 355, read on 29 October 2026: "If the sequence of the effective coupling constants is contained in an interval]0, γ] with a sufficiently small positive γ, then the effective densities ρ_k have the form, and satisfy all the conditions and bounds, described in Sect. 2", followed by "Theorem 2 of [I] allows us to remove the assumption … The proof of Theorem 2 … has not been published yet, so we have Theorem 1 with the assumption"; its bibliography names no paper proving Theorem 2.

coupling argument rests on a matching inequality that its examiners found established only where the high-temperature expansion converges, and which, as structured, would confine U(1) as well.[92]

Theorem 8.9 (The lattice mass gap at strong coupling, extended by the flux-surface gas; Edwin Boston III). *Consider the SU(2) lattice gauge theory on $\mathbb{Z}^4$ with Wilson's action, each plaquette weighted by exp(½ β tr U_p), at 0 < β ≤ β*, where β = *0.4764 is the largest coupling on the grid of step 2 × 10^{-4} at which the certified Kotecký–Preiss inequality of the flux-surface gas holds in the dressing of its page (in the tree dressing of §20 of that page the same inequality holds to β* = 0.4816). Then, CONDITIONAL ON LEMMA S12'S REMAINDER — the activity bound $4u^F$ for those of the tail's contact leaves that lie outside the classes proved on the companion page. The bound is proved where every component of the contact preimage on the sheet has cycle rank at most one; for a sphere sheet whose contact preimage is a forest, or whose face words, one dropped, form a primitive system of the contact graph's fundamental group, which holds in particular for at most three faces over at most two independent contact cycles of links of multiplicity four; for a contact graph with one cycle, trees attached, when a link of the cycle has multiplicity four, or when every link of the cycle has multiplicity six and the cycle is a skew hexagon or a rectangular cycle whose interior links at its straight vertices have multiplicity two, which every unit square, 1 × n rectangle and 2 × 2 square is; and for every leaf of the census to F = 22. It is open on the rest, among them contact preimages of higher cycle rank beyond those cases and single six-fold cycles of other shapes, the first a skew octagon, where 8 is proved: the one input of this theorem not yet proved, named again in the proof and on the companion page's §25 (l) and its entries (p) — the polymer expansion of the theory converges absolutely in every finite volume with bounds uniform in the volume, the infinite-volume Gibbs measure is unique, and for all bounded gauge-invariant observables A and B depending on finitely many links, with d the least distance between their supports, $|\langle AB\rangle - \langle A\rangle\langle B\rangle| \le K_A K_B \|A\| \|B\| e^{-\alpha d}$, where α = α(β) is any value at which the inequality is certified (at β = β *the certificate holds for α in [0.0041, 0.0047]) and K_A depends on A's support alone. On the gauge-invariant sector the Hamiltonian H of the transfer matrix has spectrum in {0} ∪ [α, ∞) in lattice units: a mass gap of at least α(β). Without that condition the same conclusions hold for 0 < β ≤ 0.241 on proved constants and to 0.2452 on the certified ones (Theorem 8.7); no coupling above 0.2452 in this paper rests on proved lemmas alone.*

Proof. The proof is the flux-surface gas of the companion page (*The flux-surface gas*, §§2–25 of the OPEN ledger, every section dated before the run it reports and every pricing lemma read by a second seat), and this paragraph is its chain. The polymer representation is classical: expanding each plaquette weight in characters, $f = \Sigma_j d_j a_j \chi_j$ with $a_j = I_{2j+1}(\beta)/I_1(\beta)$, the partition function is a gas of closed sets of plaquettes whose links all carry an even multiplicity, incompatible when they share a link; a closed branch-free surface at spin ½ weighs $u^F 2^\chi$ (Osterwalder–Seiler, Drouffe–Zuber). What the page proves beyond that is a pricing of every decoration of such a surface in exact representation theory rather than in supremum norms: the cut identity (S4′), the lone wall at exactly $3a_1$ (S7), the acyclic contacts free (S6), and THE LOCAL THEOREM (Theorem L, §24): the weight of any set of spin-1 plaquettes Y attached to a sheet S, divided by $(3a_1)^{|Y|}$ times the sheet's weight, is at most $\Pi_v r_v$ over

[92] E. T. Tomboulis, "Confinement for all values of the coupling in four-dimensional SU(2) gauge theory", arXiv:0707.2179 (2007); K. R. Ito and E. Seiler, "On the recent paper on quark confinement by Tomboulis", arXiv:0711.4930 (2007), and "Further discussion of Tomboulis' approach to the confinement problem", arXiv:0803.3019 (2008); Tomboulis's reply, arXiv:0712.2620 (2007). Sources [22]–[23].

the vertices Y touches, with $r_v^2 = 2^{m_{v}-2}\cdot S(C^{½}_{m_{v}} \cup \Gamma_v)$ a Haar integral of the vertex's own star — the sheet's cycle of spin-½ edges and Y's spin-1 chords at v. The stars form a FINITE family, the simple subgraphs of $K_{2,2,2,2}$ on the eight directions at a vertex, so the rate $\gamma(s) = 3\theta(s)^4$ with $\theta(s) = \max_{k\le s} r2max(k)^{1/2k}$ is a finite table: r2max(k) is the exact maximum of the star integral over every star of k chords where every row of the enumeration at k has been read by the exact intertwiner engine ($k \le 12$ and the rows below the engine's cap elsewhere), and on the rows not read it is the lesser of two certified bounds — the Frobenius sieve $|S| \le \Pi_l \sqrt{R_l}$, R_l the dimension of the invariants at link l, and THE REALIGNED OPERATOR-NORM BOUND (Lemma R, §25): contracting the star's link projectors in any order, $|S| \le \|P_1\|_F \Pi_t \|M_t\|_{op} \|P_n\|_F$ with M_t the projector realigned along the split, and $\|M_t\|^2_{op} \le \lambda_{max}(Tr_N P_t)\lambda_{max}(Tr_S P_t)$ by the block Cauchy–Schwarz inequality, every eigenvalue bound certified by an exact LDL^T decomposition in rational arithmetic. Every exact value read (3464 of them, at every k) lies under its own realigned bound, and the bound's planted defect is refused by name. With the closing-tree count $N(F) \le 5^{F-1}$ of Z_2-minimal surfaces, the dressing of the contacts — the contact leaves' activity $4u^F$, a theorem on the classes the statement lists, and on the rest a lemma still owed: the one input of this theorem not yet proved, named on the page's §25 (l) with its closing road in the companion register OPEN/the-far-shore.md §4.0 (the vertex-norm theorem at spin ½ with the finite table of even stars, and the sparsity of a Z_2-minimal leaf's stars or a sharper dense-leaf count) — and the composite tree, the Kotecký–Preiss condition $\Sigma_{X\ni\ell}|W(X)|e^{\alpha|L(X)|} \le \alpha$ is a finite inequality in the Bessel ratios a_j, the table and α, and the certificate (aperanton_kp_cert.awk in its TABLE mode, gawk at 256 bits with every quantity an interval of width 2^{-128} rounded outward) decides it on the grid: it holds at β *and fails one step above it. The gap then follows as in §16 of the page (D1): an observable's support enters the gas as a boundary, the relative leaves are counted by the same closing tree, a cluster joining the two supports is link-connected across the distance between them, and the Kotecký–Preiss tree bound in its two-weight form supplies* $e^{-\alpha d}$*; Theorem 7.1 turns the clustering of the reflection-positive correlations into the gap of the transfer matrix, and the uniform bounds pass to the infinite volume. The certificate's run of record is aperanton_table_gate.2026-11-01d (eight arms, five mutant tests, verdict GREEN; R9).*

What is new in Theorem 8.9 and what is classical. Classical: the polymer representation and the weight of a closed surface (Osterwalder and Seiler 1978; Drouffe and Zuber 1983); the Kotecký–Preiss criterion (1986) and its tree bound; the inequalities $\|AB\|_F \le \|A\|_F\|B\|_{op}$, the block Cauchy–Schwarz inequality and Sylvester's criterion; Theorem 7.1. New, and only in this work: the pricing of the branched decorations by the exact representation theory of the vertex star rather than by supremum norms — the cut identity, the lone wall, the acyclic contacts, the local theorem with its finite family of stars — the exact table through $k = 12$ with the kill-switch read on every row (no star above the per-corner rate 1.113533 of the four walls on a link), the certified realigned bound that closes the unread rows, and the number β *itself, certified rather than estimated. Theorem 8.7's* $\beta \le 0.2452$ *was Dobrushin's comparison on a decimated measure; Theorem 8.9 reaches* $\beta = 0.4764$ by the gas's own geometry, conditional on Lemma S12's remainder. Both are lattice gaps at strong coupling, not the continuum gap that is the object of the Clay problem; the distance to Bałaban's unread γ is the interval $[0.2452, 1/\gamma^2]$ on proved lemmas and $[0.4764, 1/\gamma^2]$ under that condition.

8.4 Coda

The ground, then, is this. A construction of the corpus's kind, as its manuscripts state it, delivers a limit that is controlled at short distances and silent at long ones: a theory whose correlations obey the axioms of positivity and symmetry, whose coupling is seen to run, and whose large-distance behaviour is not controlled. Theorem 7.1 shows that the missing control is a single spectral fact. Theorem 7.4 shows that this fact is a property of the rungs: an inequality on the lattice, where reflection positivity has made a quantum mechanics of the theory before any limit is taken. The proof that follows takes up that inequality on that sequence.

Sources

Every source was read on 28 October 2026. The repository sources are cited by file name; the repository is at https://github.com/ethanwsussman/Yang--Mills-LLM-experiment.

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[2] L. Alpöge and E. Sussman, repository *Yang--Mills-LLM-experiment*: README.md (last changed 26 October 2026, 16:18:10 UTC); YM3/YMM1.pdf and YM3/YMM1.tex, "Scalar fields from three-dimensional lattice Yang–Mills theory", 854 pages; YM3/Revisions/YMM1-2026-10-26.pdf, 856 pages, and YM3/Revisions/CHANGES-2026-10-26.md; YM3/YMM2.pdf, "Curvature observables, dynamics, and conditional confinement in three-dimensional lattice Yang–Mills theory", 1,044 pages; ym4_manuscript_zeroth_draft.pdf, "SU(2) Yang–Mills theory in four dimensions via Balaban's renormalisation group: construction, axioms and non-triviality", 6 October 2026, 153 pages; QED3/qed3.pdf, 263 pages; RPM1.pdf, 30 pages; theorem_0_complete_account_raw_I–III.tex. Repository created 6 October 2026, 09:51:17 UTC; most recent commit 27 October 2026, 10:45:28 UTC.

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Records of the computations of Edwin Boston III

Each record is held, with its instrument and its run of record, in the repository of Edwin Boston III.

[R1] *The shadow ladder* (Proposition 8.3): record g6_tube.2026-10-28a, with its instrument g6_tube.pl; landed in commit aea98a3af; re-derived in the run of record pentapylon_gate.2026-10-28r, nine arms, nineteen mutant tests, verdict GREEN.

[R2] *The gap by counting* (§8.2): record g6_inertia.2026-10-28b; run of record pentapylon_gate.2026-10-28r, verdict GREEN; landed in commit e11e1269c. The independently certified intervals it meets: run of record tomeus_gate.2026-10-27e, ten arms, ten mutant tests, verdict GREEN.

[R3] *The gap through duality* (§8.2): the dual-road records of 28 October 2026 and record g6_ising_anchor.2026-10-28a; landed in commit 44be883de with the run pentapylon_gate.2026-10-28q, eight arms, seventeen mutant tests, verdict GREEN; re-derived in run r.

[R5] *The chain lemma* (Proposition 8.2): instrument g9_chain.pl and record g9_chain.2026-10-29a; landed in commit 103e8425a; run of record pentapylon_gate.2026-10-28s, twelve arms, twenty-five mutant tests, verdict GREEN.

[R6] *The sharp one-link lemma, the certified descent and the transform on a tube* (§§8.1, 8.2, 8.3): instruments g18_mad_lemma.pl with g17_mixed_survey.pl, mkflow.awk and g20_tube.pl; records g18_m1, g18_axis, g18_mono and g18_direct_b0_1 … b8_20 of 2026-10-29b, mkflow_b2.2, b2.3, b5 and b10 of 2026-10-29d and 2026-10-29e, and g20_tube_j1.2026-10-29b with the spin-3/2 readings g20_tube_j32_b2 and b2.3 of 2026-10-29a; run of record pentapylon_gate.2026-10-29w, fifteen arms, thirty-two mutant tests, verdict GREEN at 02:20 EDT on 29 October 2026. The certified budget of §8.3: instrument aperanton_tilt_cert.awk (gawk -M, 256 bits, every quantity an interval under directed rounding) on the four records of 2026-10-29e, with the records tilt_cert.2026-10-29a at a grid of 20,000 points, reproduced byte for byte on 2 November 2026, and tilt_cert.2026-10-29b at 2,000 points, whose coarser grid refuses by name a few more of the early steps (the first from 2.2 and 2.3, two more from 5 and 10) and which agrees with the first at every step both records read, the closing step from every start among them; its known answer, the Wilson weight at $\beta = 3/10$ read to every printed digit ($B = 2.2918311805232928$) and a bent weight read off the line, run beside it; the finite form of the criterion, CrossoverCriterion.lean, nineteen theorems, interdep_tilt_sharp and rowsum_tilt_le_sharp among them, accepted by the kernel at arm 115 of the run aperanton_gate.2026-11-01d, in which the budget's own arm, 120, also passed and whose one RED is a surviving mutant test of arm 114 (the file of that run is posted under records/aperanton/runs, beside the other gate runs of 30 October to 1 November); the U(1) control, instruments aperanton_mk_flow.awk and aperanton_mk_flow_u1.awk with

the record mk_flow.2026-10-28a. The instruments, the records and the run's own file are posted with their digests in the repository, under records/pentapylon.

[R7] *The lattice mass gap at strong coupling* (Theorem 8.7): the cover, instrument aperanton_perfect_cover.pl and record perfect_cover.2026-10-29a; the certificate, instrument aperanton_decimation_cert.awk and records decimation_cert.2026-10-29a and, with the exact constants, 2026-10-29b; the enclosures of the two constants, record g19_vmf.2026-10-29b with its instrument; the third road, instrument g24_rowsum.awk and record g24_rowsum.2026-10-29a; and the second evaluation, verify73ter.py with its output, which shares no line with the certificate and checks the cover on the tori of side two and four, the coefficients of Turán's inequality exactly, and the row sum to sixty digits. The enclosures of the two constants and the one-face constants are of record: the run pentapylon_gate.2026-10-29x, seventeen arms, thirty-eight mutant tests, verdict GREEN at 04:06 EDT on 29 October 2026, re-runs them live and compares them byte for byte; the certificate, its second evaluation and the third road await their own gate run. The derivation is the proof of §8.3; its test from outside is the search from below, instrument g23_decim.pl with its records of 2026-10-29b, which found no configuration exceeding the constants; and it was read independently, line by line, by a second reader on 29 October 2026, who found no defect, reproduced Table 3 to six digits on a fourth road, and checked Lemma 8.6 and the one-link bound by deterministic quadrature at 240 radii and seventy law-direction pairs. The files and their digests are posted in the repository, under records/aperanton, records/pentapylon and records/second-road.

[R8] *The lattice mass gap at strong coupling in three dimensions* (Theorem 8.8): the cover, instruments g37_cover_dim.pl and g37_tile_check.pl with the record g37_cover_dim.2026-10-29a; the row in dimension D, instrument g38_row_dim.awk with the record g38_row_dim.2026-10-29a; and the second evaluation, rowdim.py and coverrule3d.py with their outputs, which share no line with them; posted with their SHA-256 under the repository's records/pentapylon and records/second-road, the theorem stated with both roads on the page OPEN/ledger/c3-the-three-dimensional-rehearsal.md.

[R9] *The lattice mass gap at strong coupling, extended* (Theorem 8.9): the flux-surface gas page OPEN/ledger/s1-the-flux-surface-gas.md, §§2–25, each section dated before its run; the exact star table and its records local_tail_table.2026-11-01d.txt, kp_cert_table_scan_page.2026-11-01d.txt and _tree; the certified realigned bound, instrument realign_cert.py, records realign_bound.2026-10-31a–d and the census realign_census.2026-10-31b (3464 exact values under their bounds, 0 discords); the class record class_exact.2026-10-31a; the certificate aperanton_kp_cert.awk in its TABLE mode; the run of record aperanton_table_gate.2026-11-01d, eight arms, five mutant tests, verdict GREEN; landed in commit c928db955.

[R4] *The lattice theorems in Lean* (§§4.2, 7.2, 8.2): the files EvenTime.lean, Spectral.lean, Perron.lean, StringEnergy.lean and MassGap.lean (the transfer theorem for every finite group); Nonabelian.lean and Separation.lean (the static potential); Griffiths.lean, Rectangle.lean and StrongCoupling.lean (the Z_2 inequalities, expectation and string tension); Slab.lean and Z2Slab.lean, theorem transfer_gap_sharp (the Z_2 slab); Scaling.lean, theorem twopoint_le_of_gap and definition PhysicalGapAlong; Decay.lean, theorems gap_of_decay and eigenvalue_le_of_decay; Dobrushin.lean, theorem cov_le. Run of record aperanton_gate.2026-10-27u, forty-eight arms, twenty-nine mutant tests, verdict GREEN; landed in commit 7dd4c66b2.